
Course Outline

1. Introduction
 2. Neuronal Signaling
 3. Visual System
 4. Computational Methods I
 5. Decision making
 6. Multi-modal Sensory Systems
 7. Hippocampus, Spatial Cognition & Memory
 8. Working memory
 9. Motor Systems
 10. Computational Methods II
 11. Learning & Plasticity
 12. Value and Reward
 13. Project Presentation
 14. Final Q&A
-

系统和计算神经科学导论

五、抉择

Decision Making

中国科学院脑科学与智能技术卓越创新中心（神经所） 杨天明 TYANG@ION.AC.CN

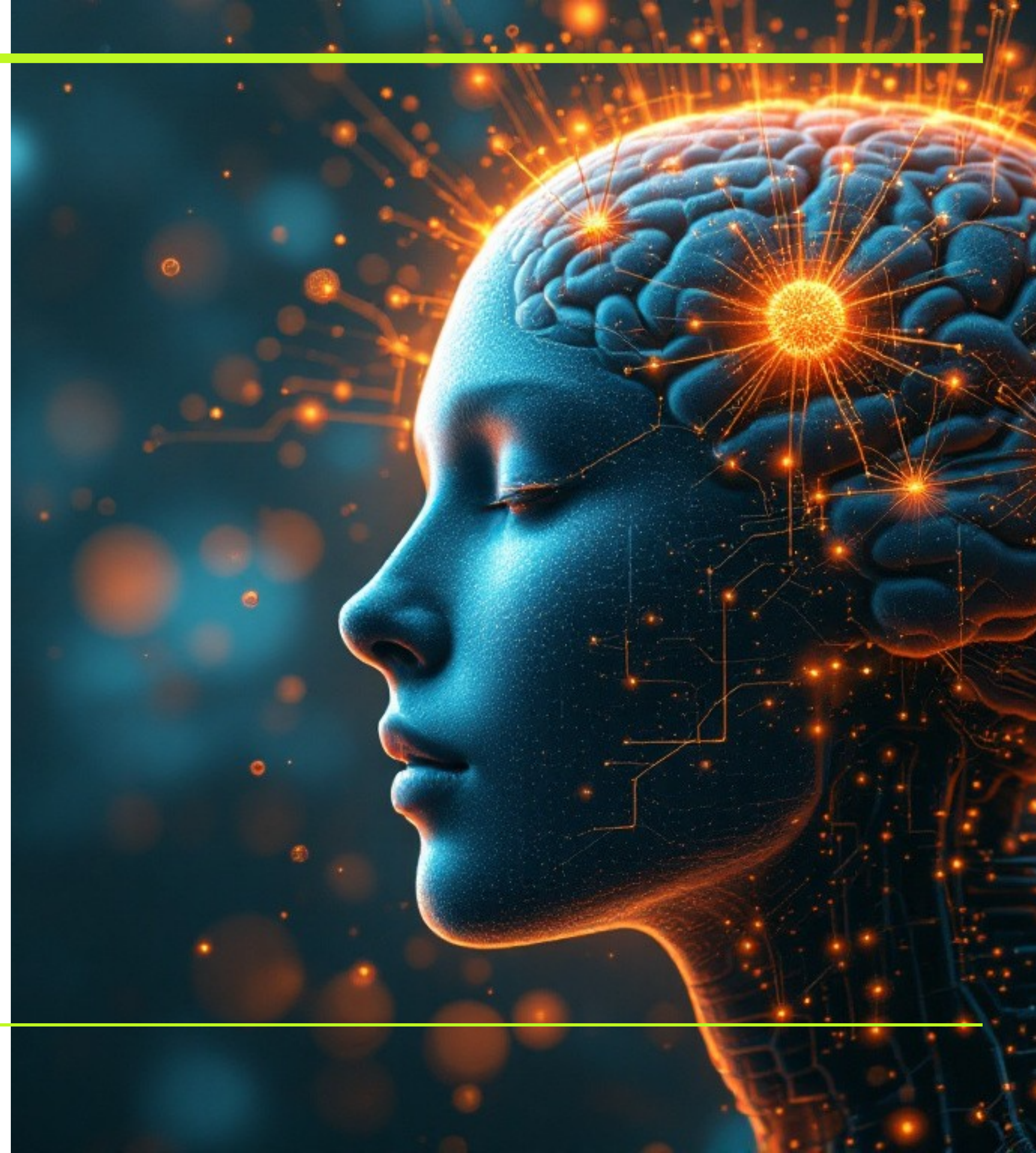
❖ **Overview**

❖ **DDM**

❖ **SPRT**

❖ **Confidence**

❖ **Attractor Network**



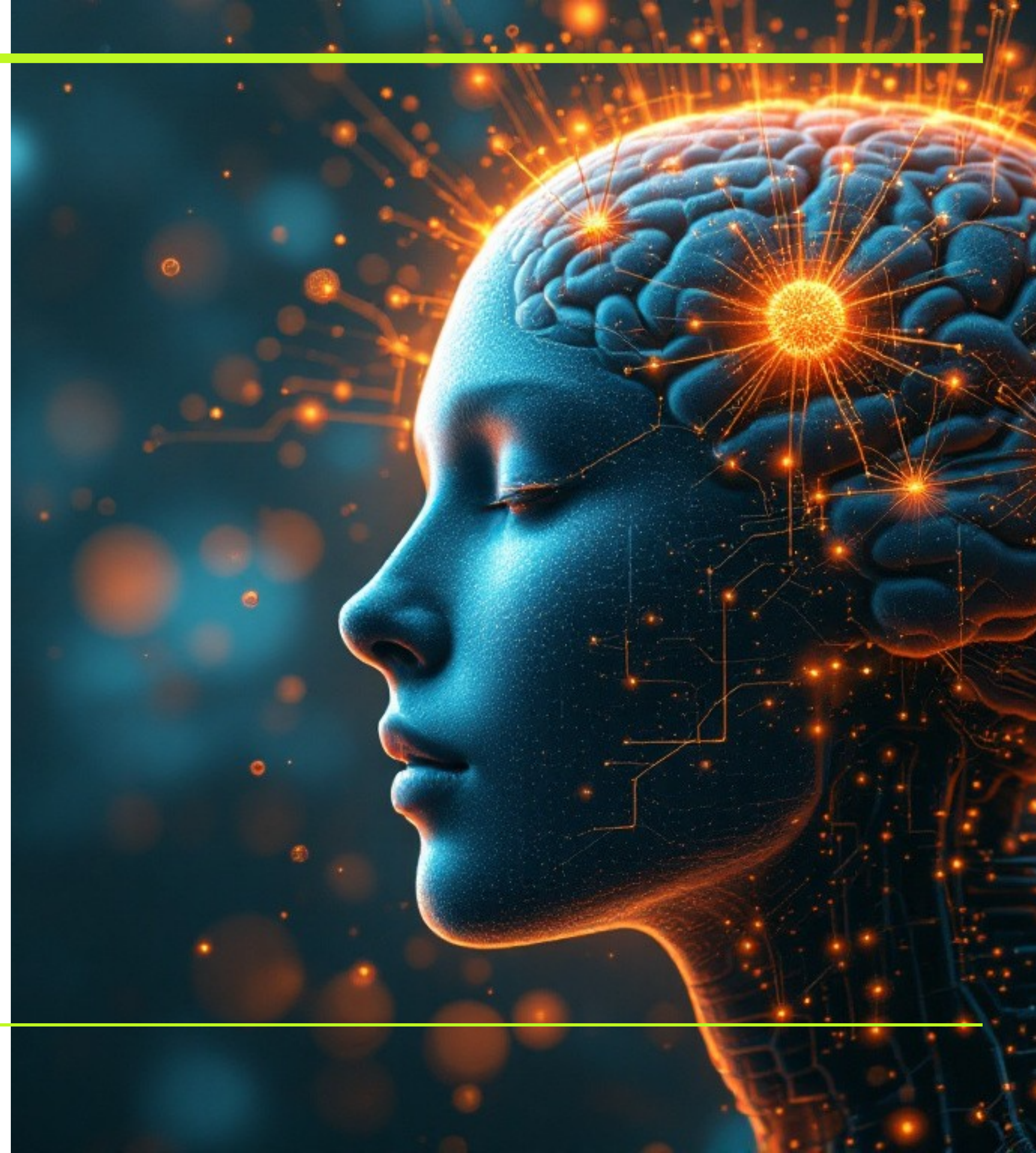
❖ Overview

❖ DDM

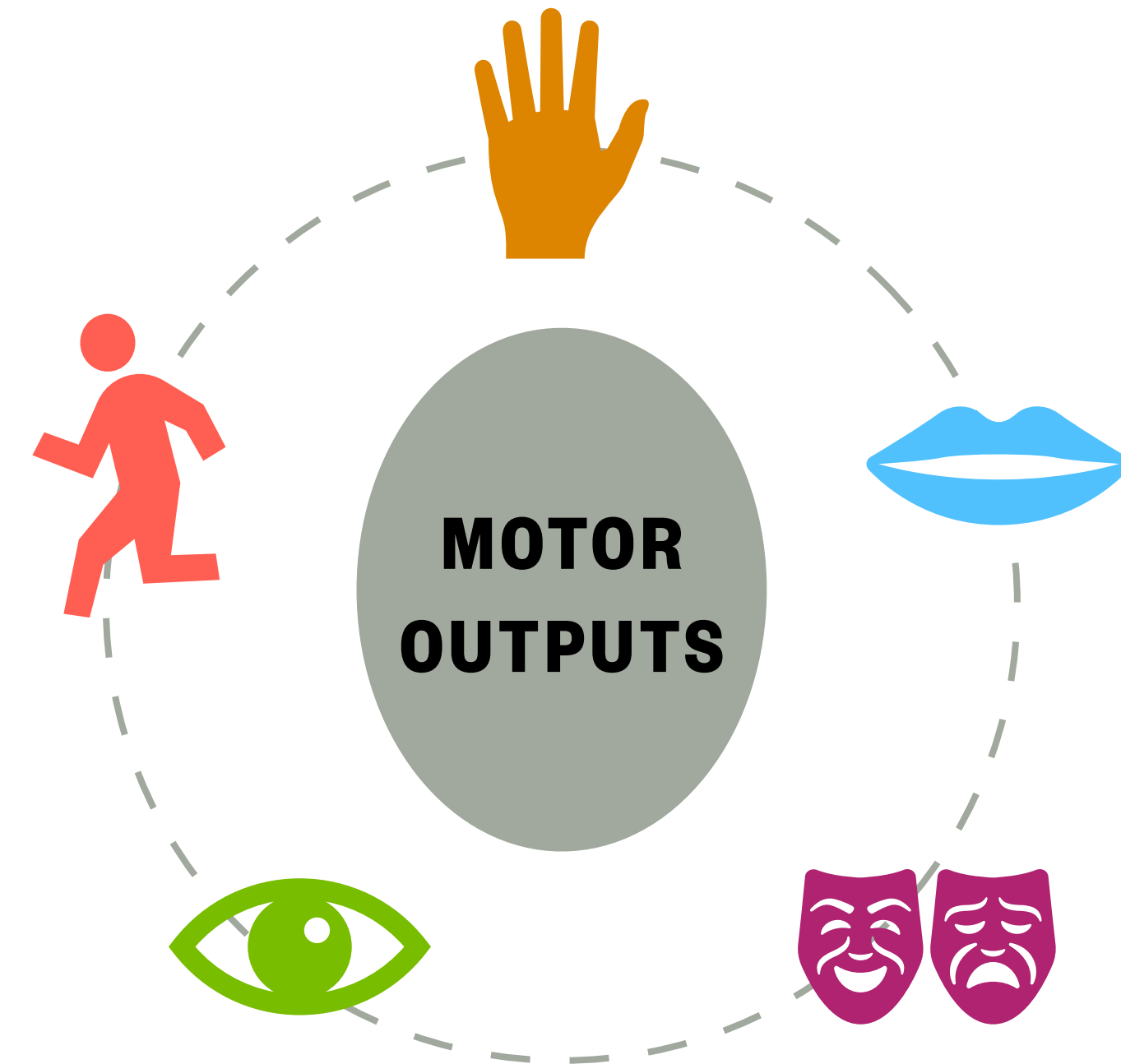
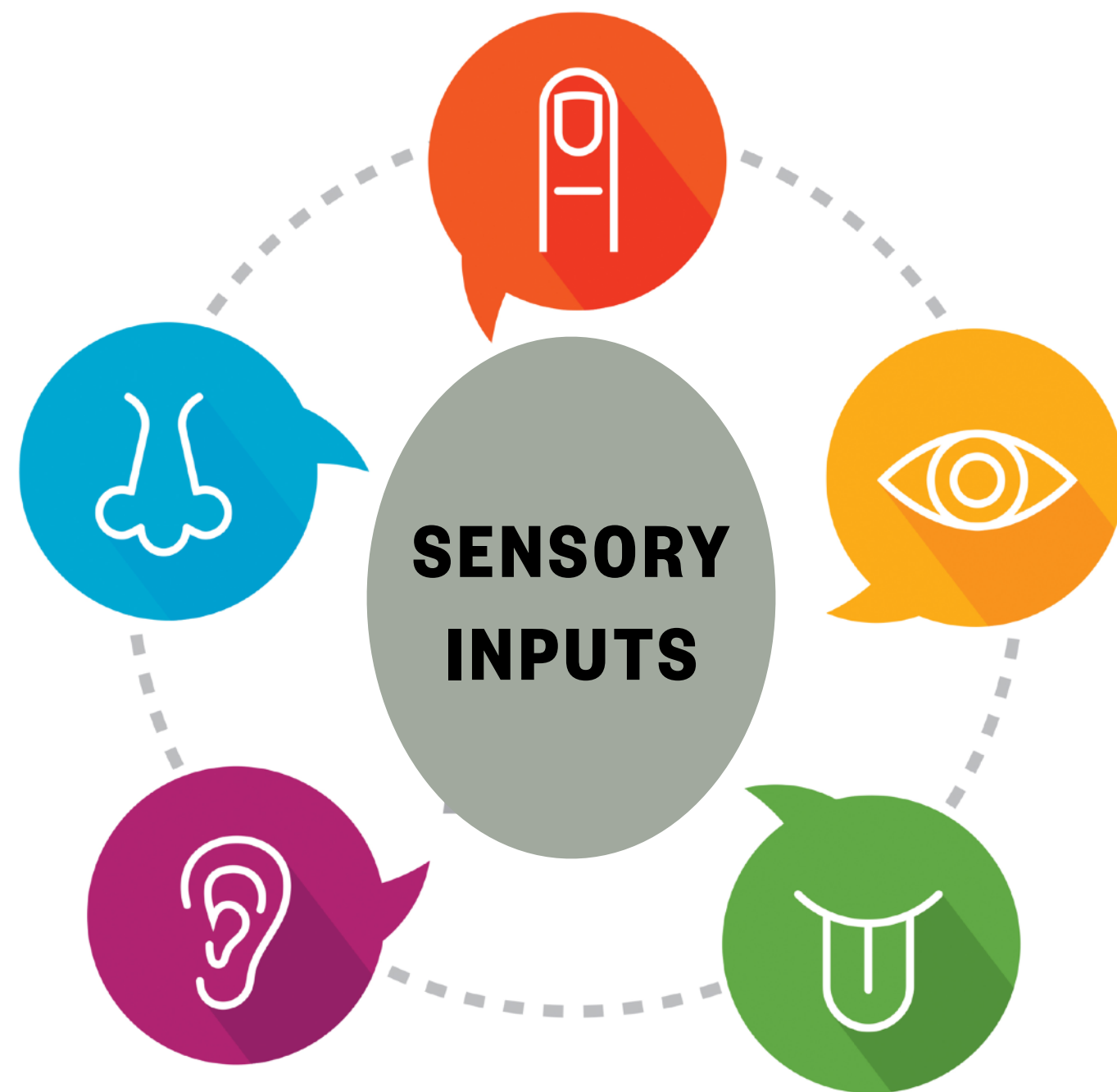
❖ SPRT

❖ Confidence

❖ Attractor Network



Definition



Different Types

- Perceptual decision making
 - Value-based decision making
 - Social decision making
 - Other complex decision making
-

Different Types

- Perceptual decision making
 - Value-based decision making
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 - Other complex decision making
-

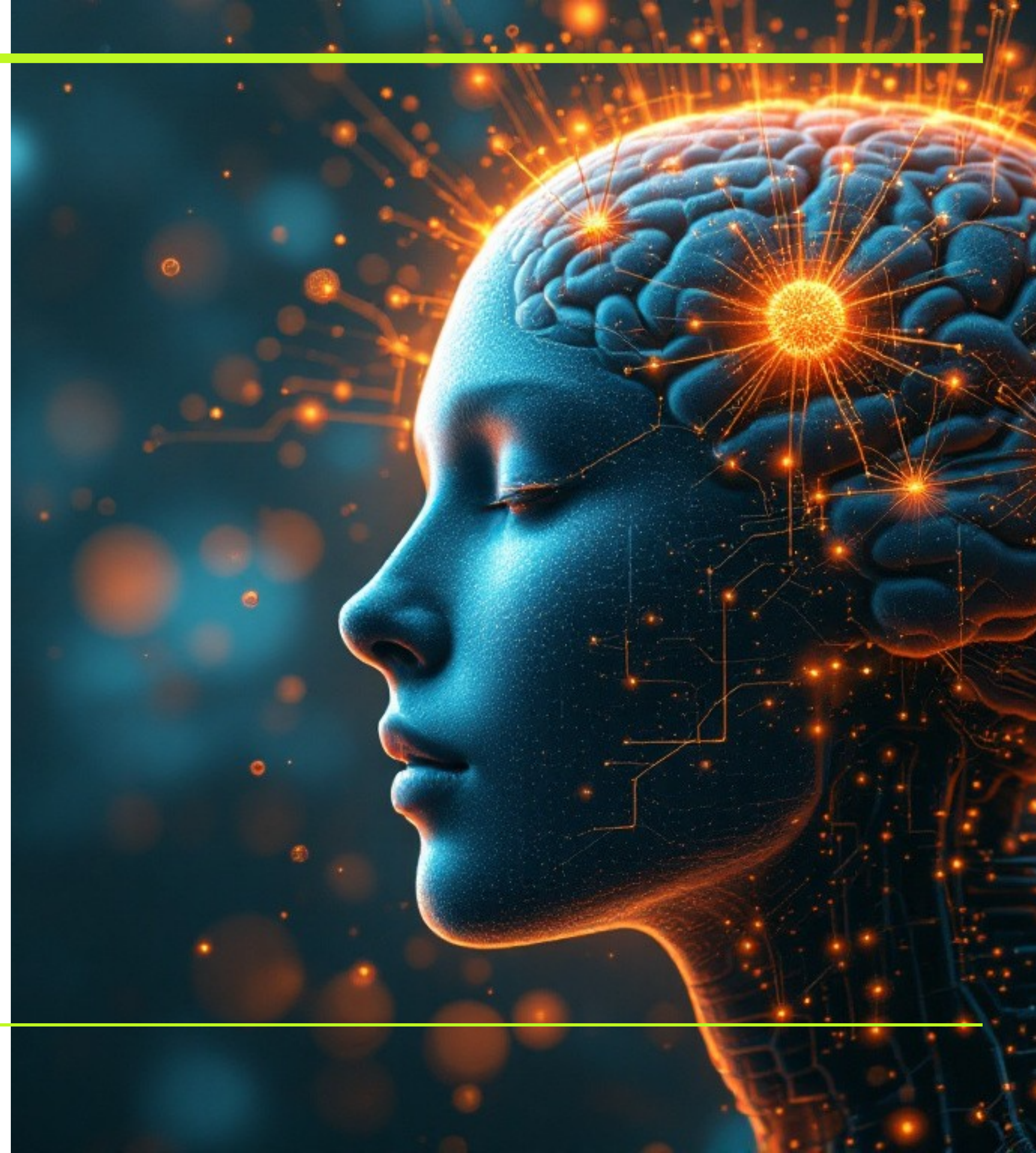
❖ Overview

❖ **DDM**

❖ Probabilistic Reasoning

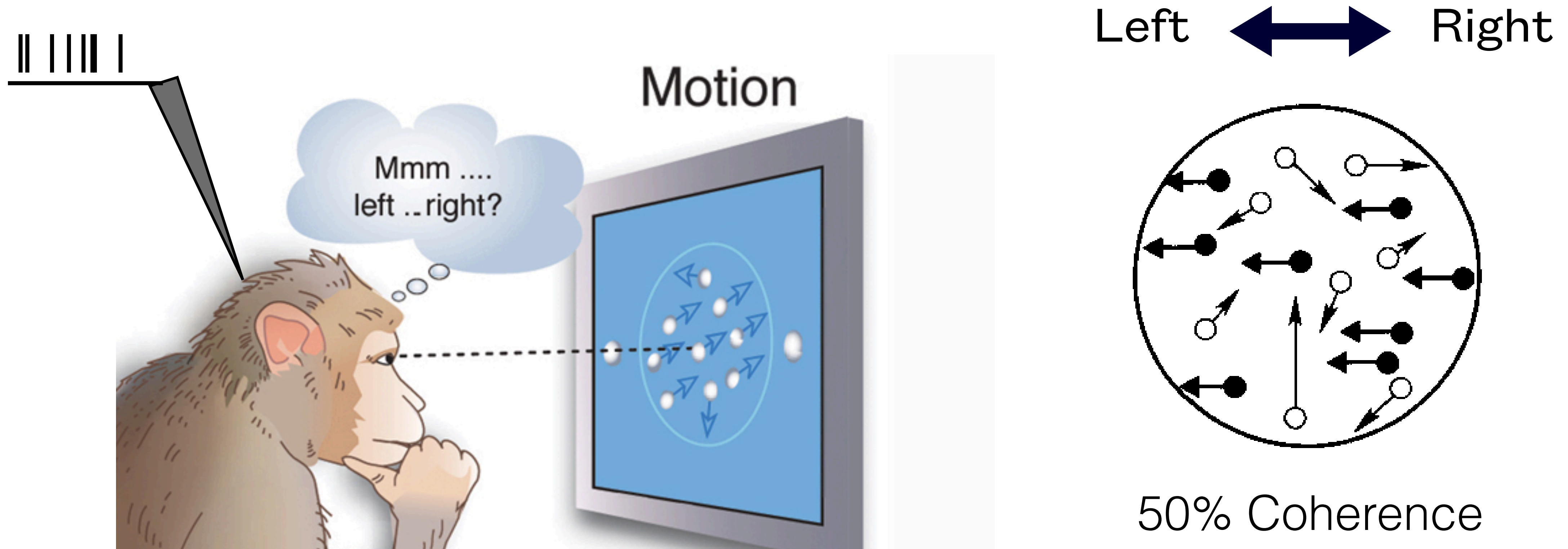
❖ Confidence

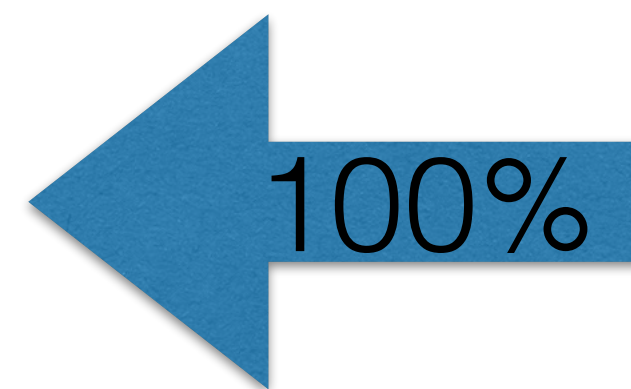
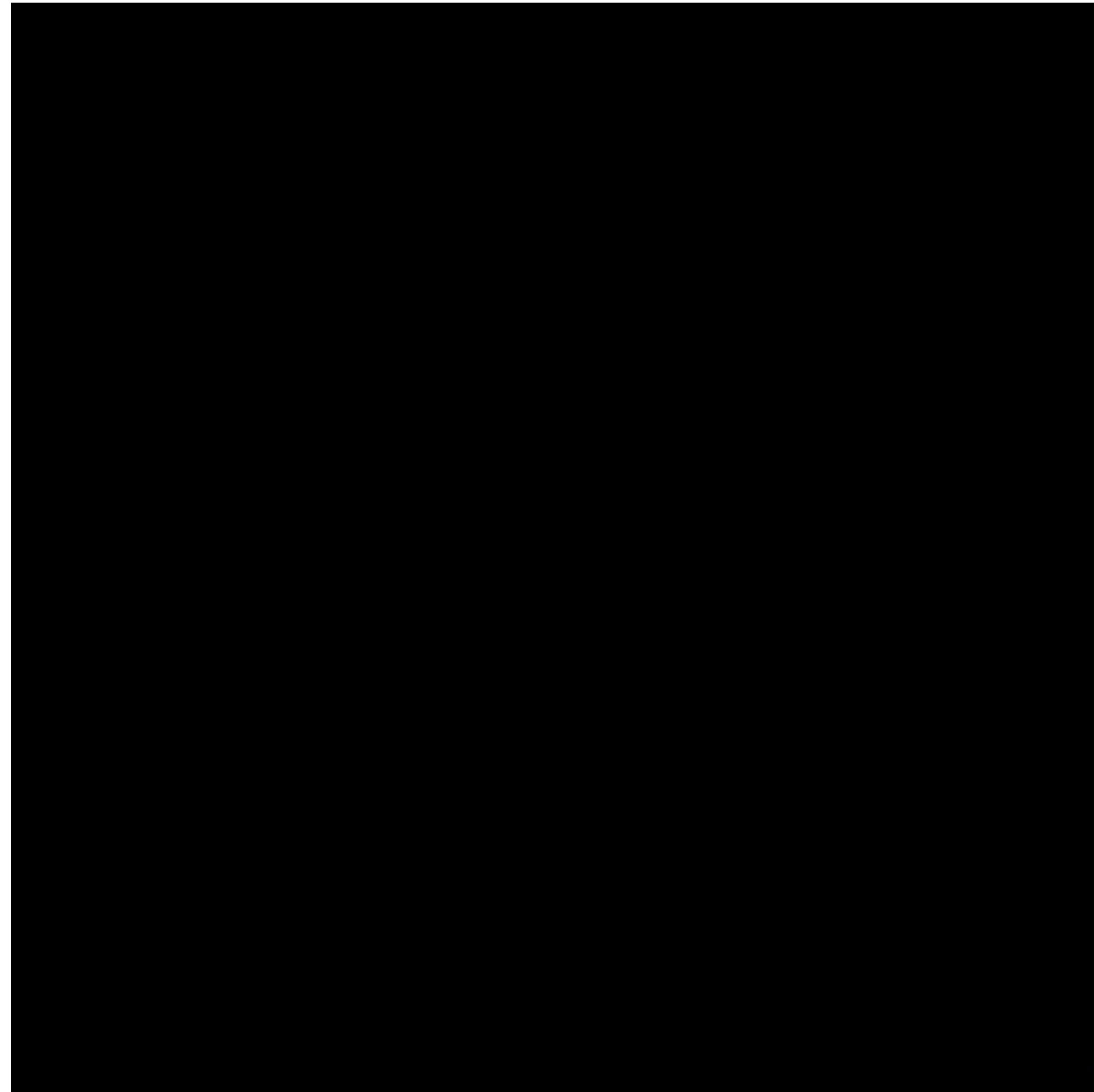
❖ Attractor Network

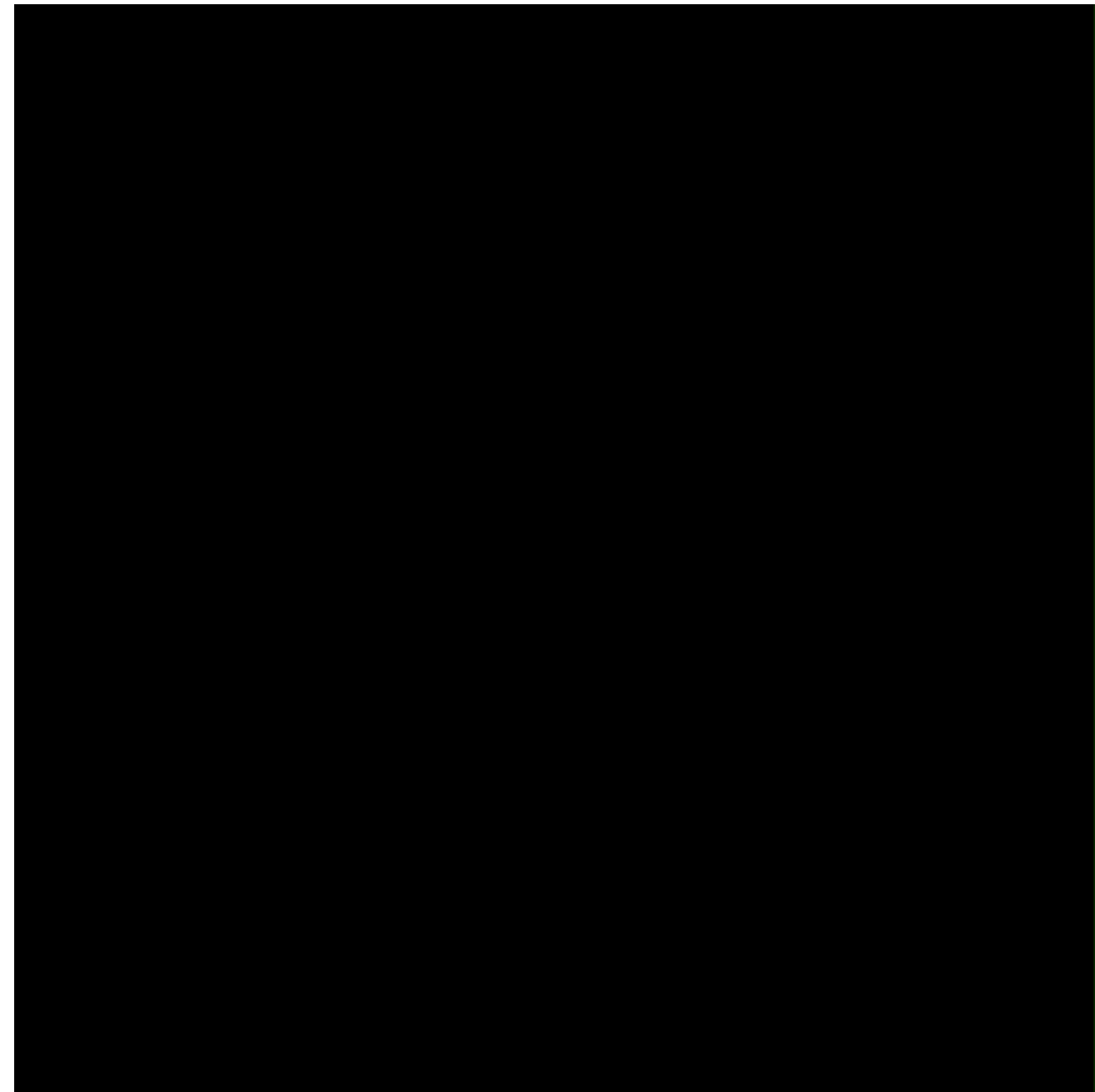


Random Dots Motion Direction Discrimination

2-Alternative Forced Choice



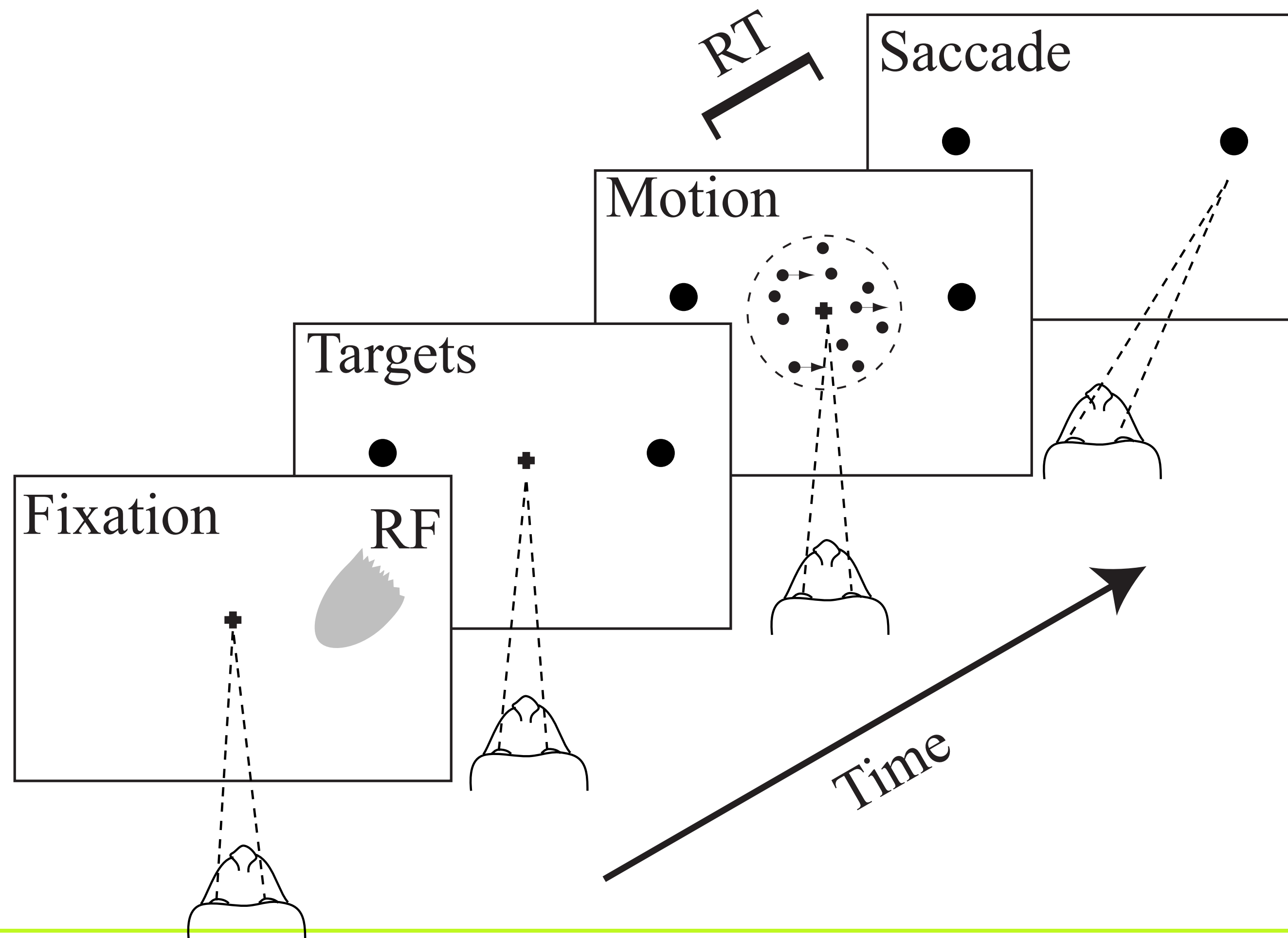




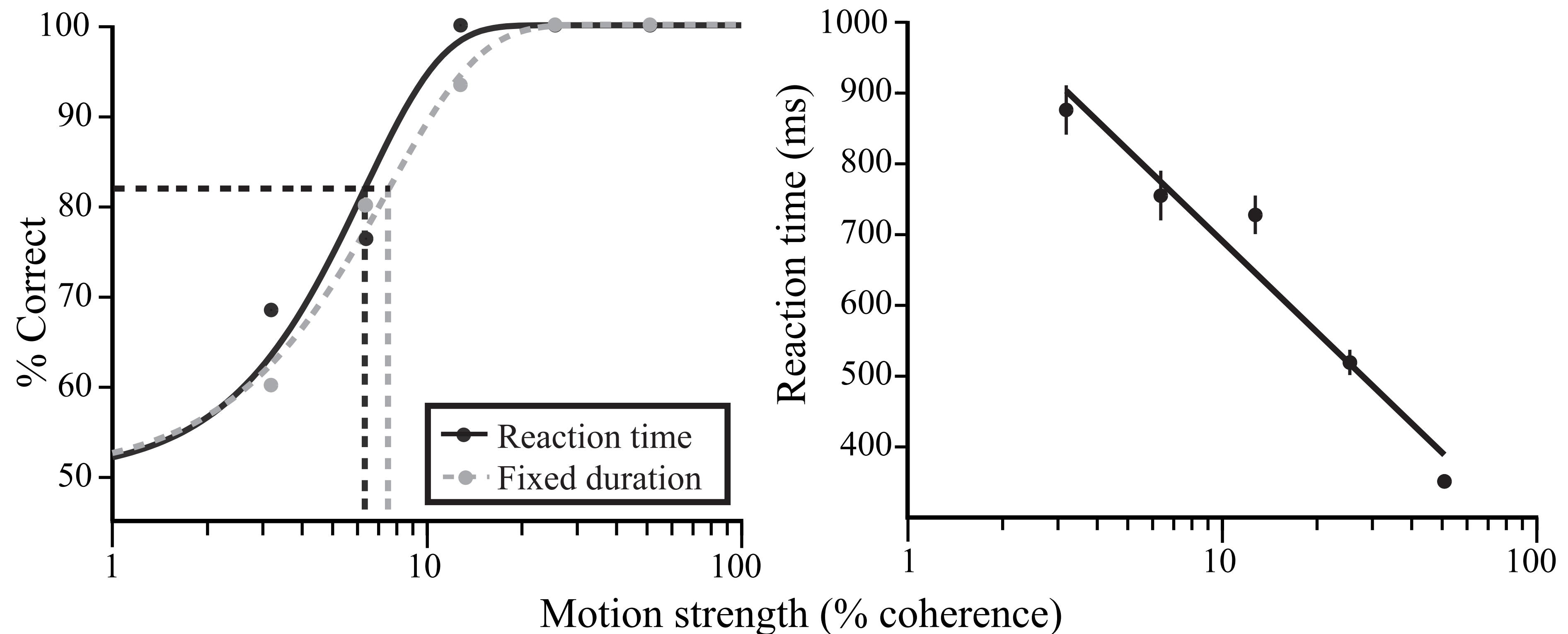
12.8%

A blue arrow pointing to the right, positioned below the text '12.8%'. The arrow's tail is on the left, and its head points towards the right edge of the image.

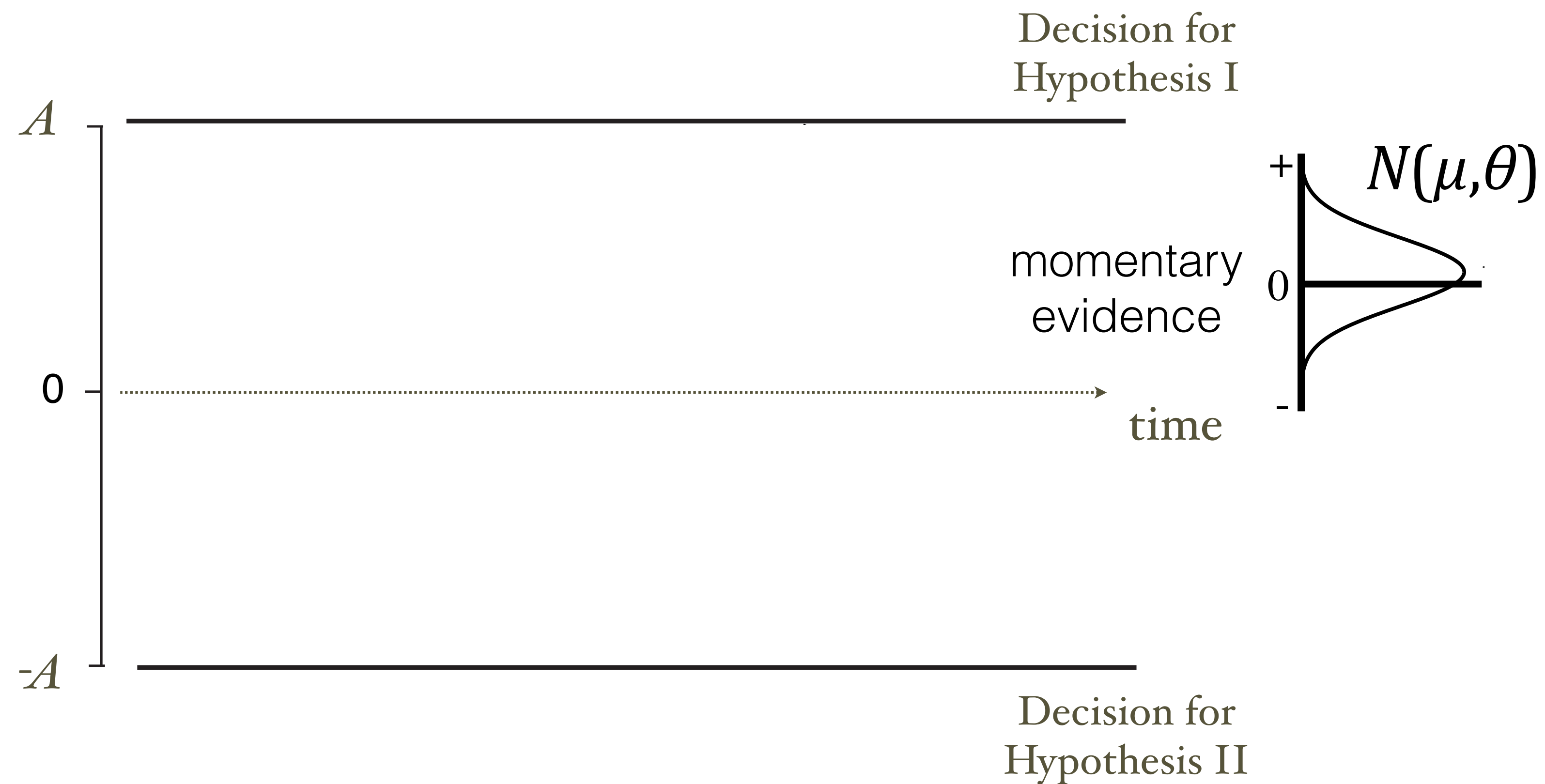
Reaction Time Task



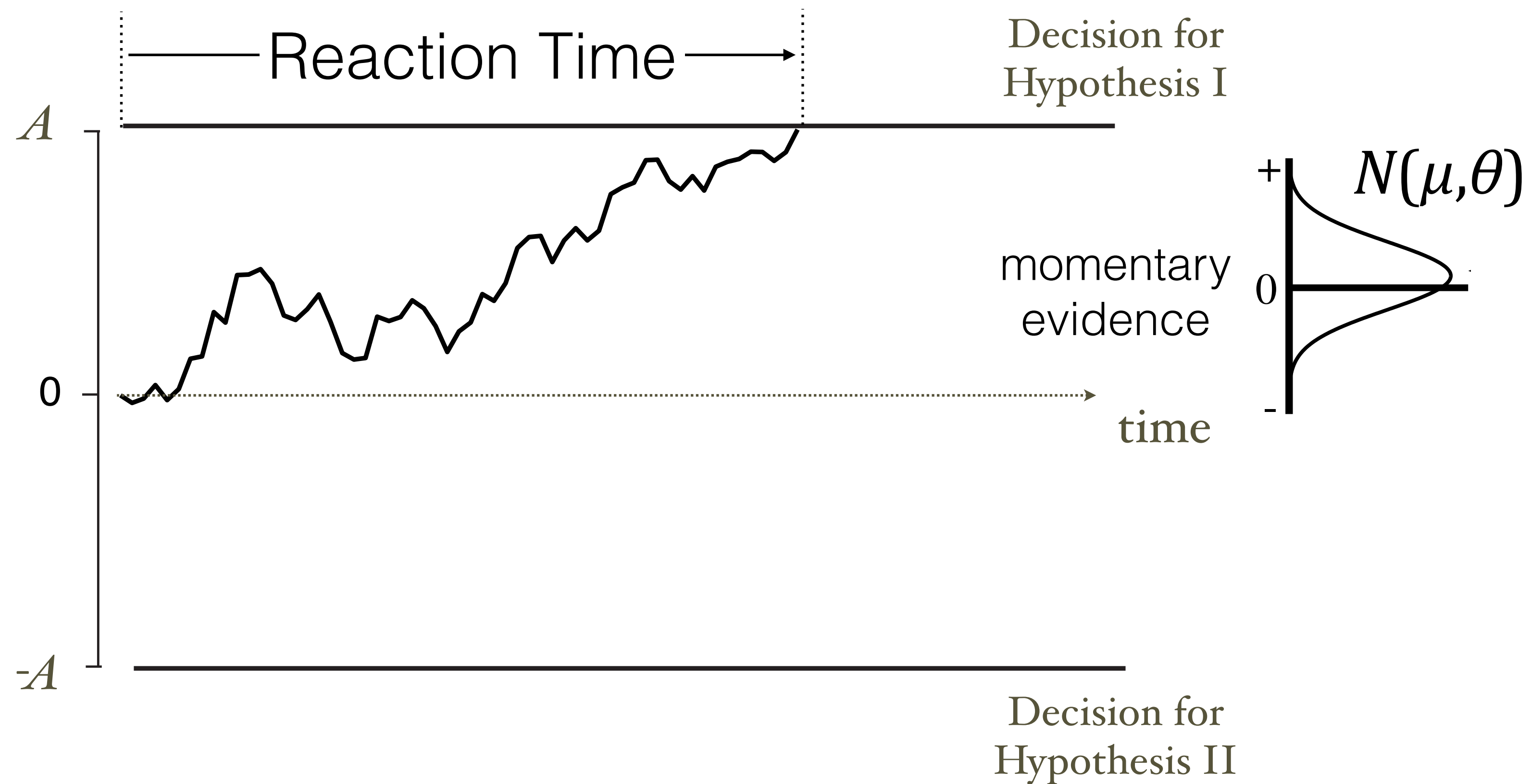
Psychometric and Chronometric Curves



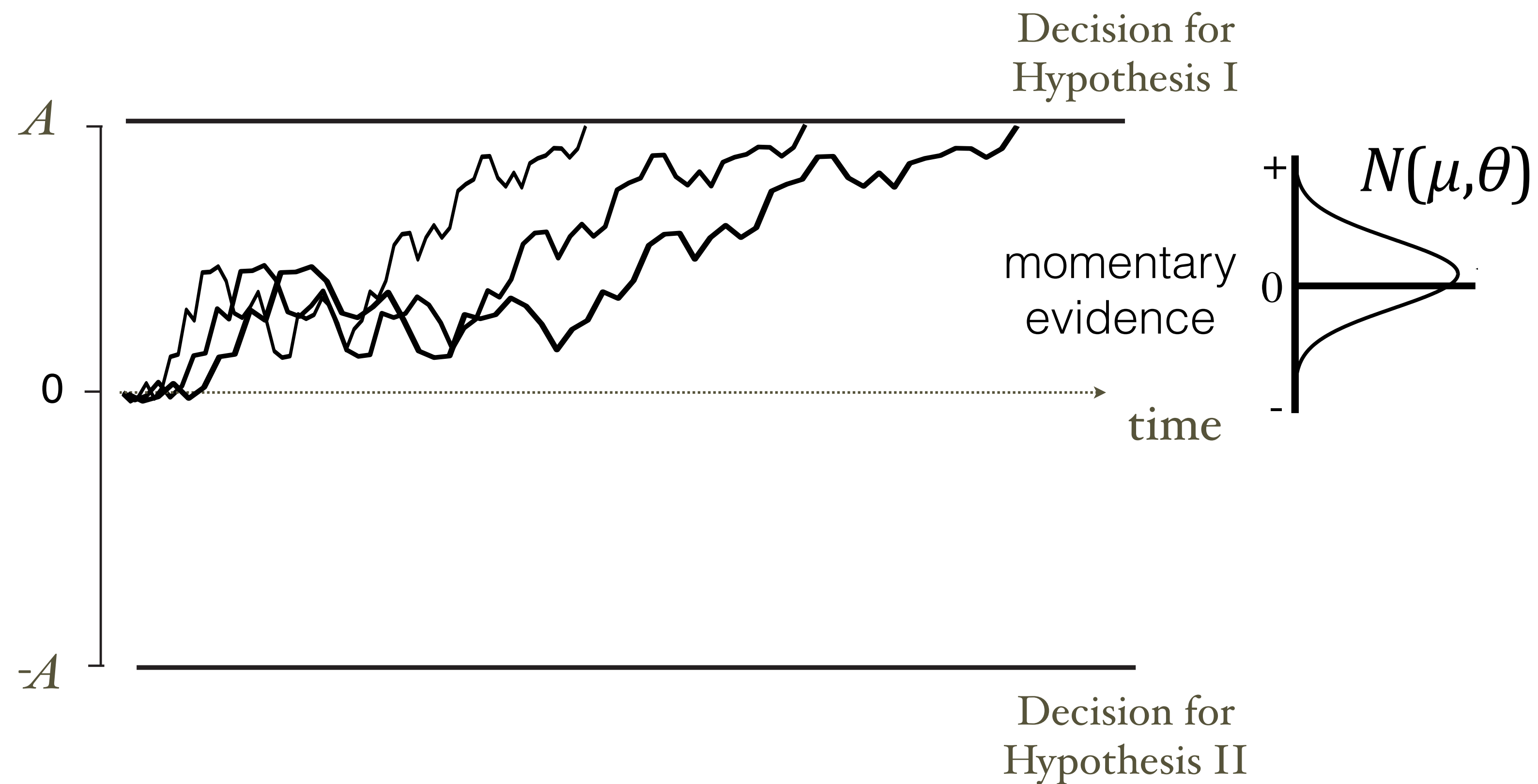
Drift Diffusion Model



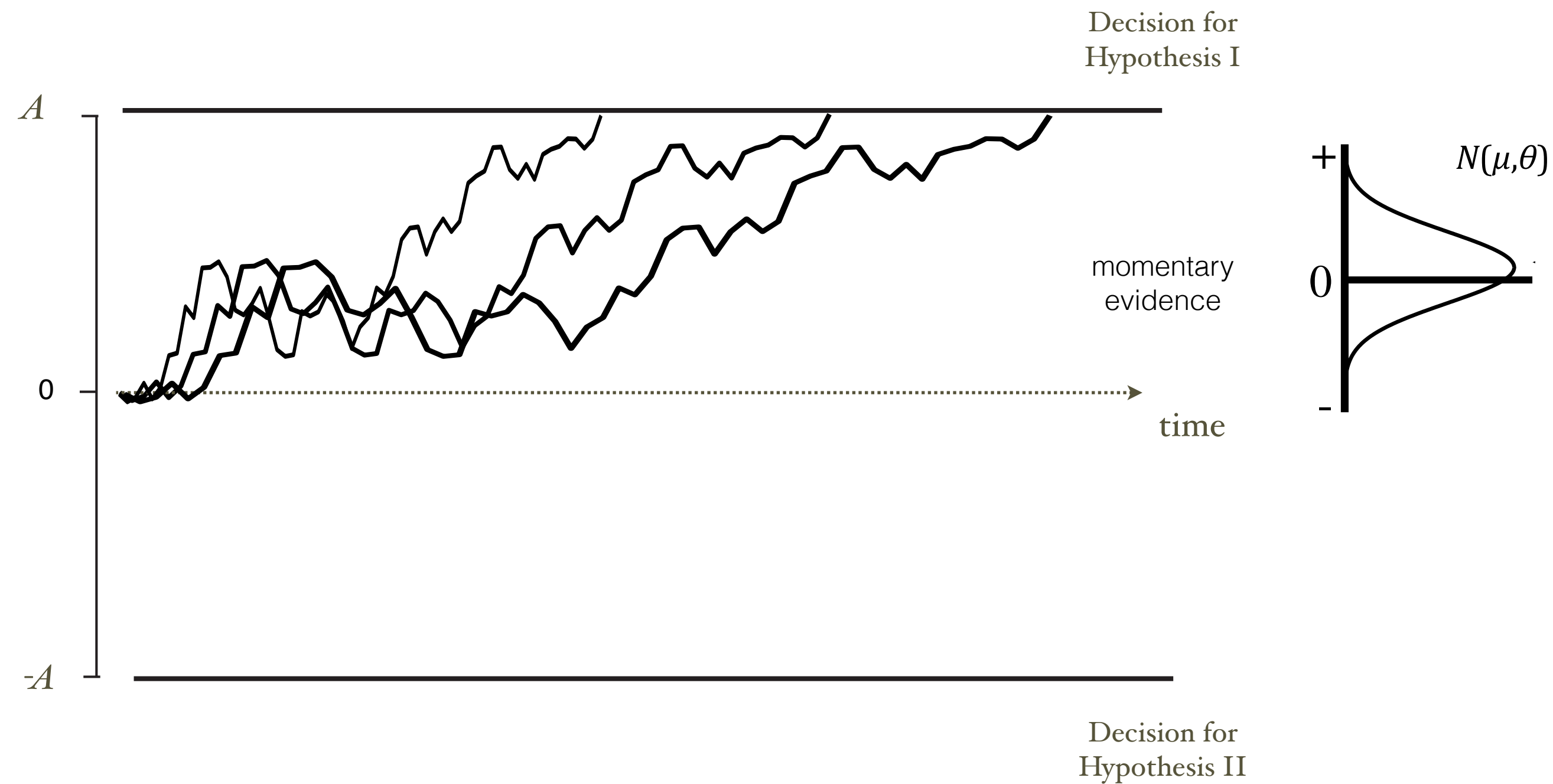
Drift Diffusion Model



Drift Diffusion Model



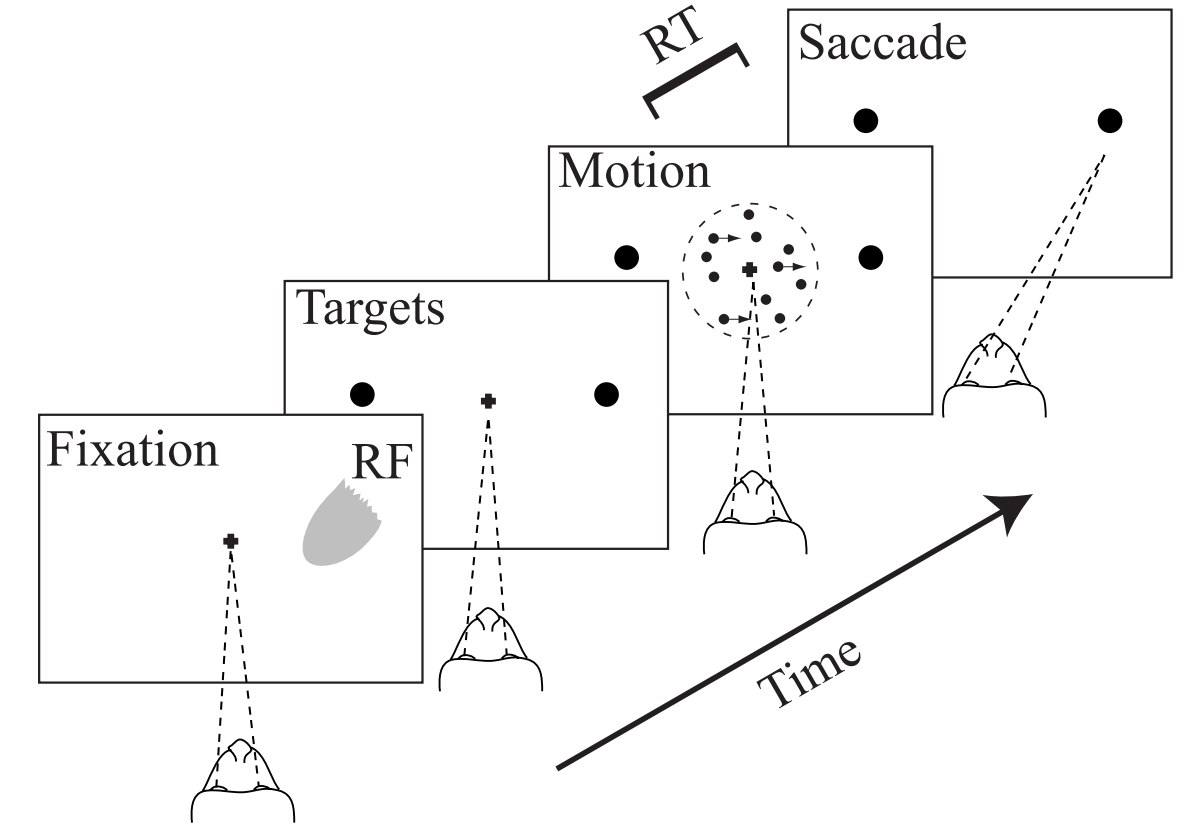
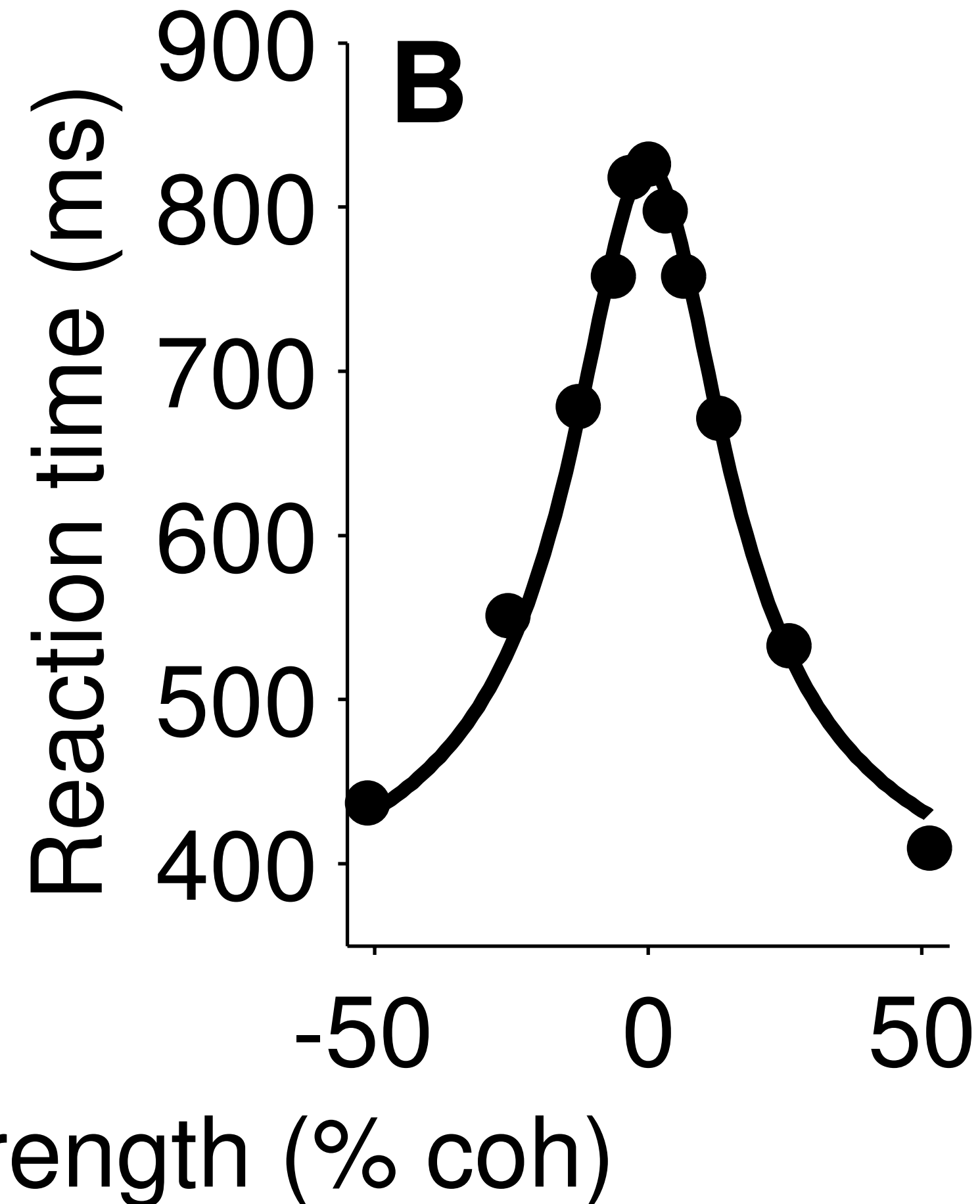
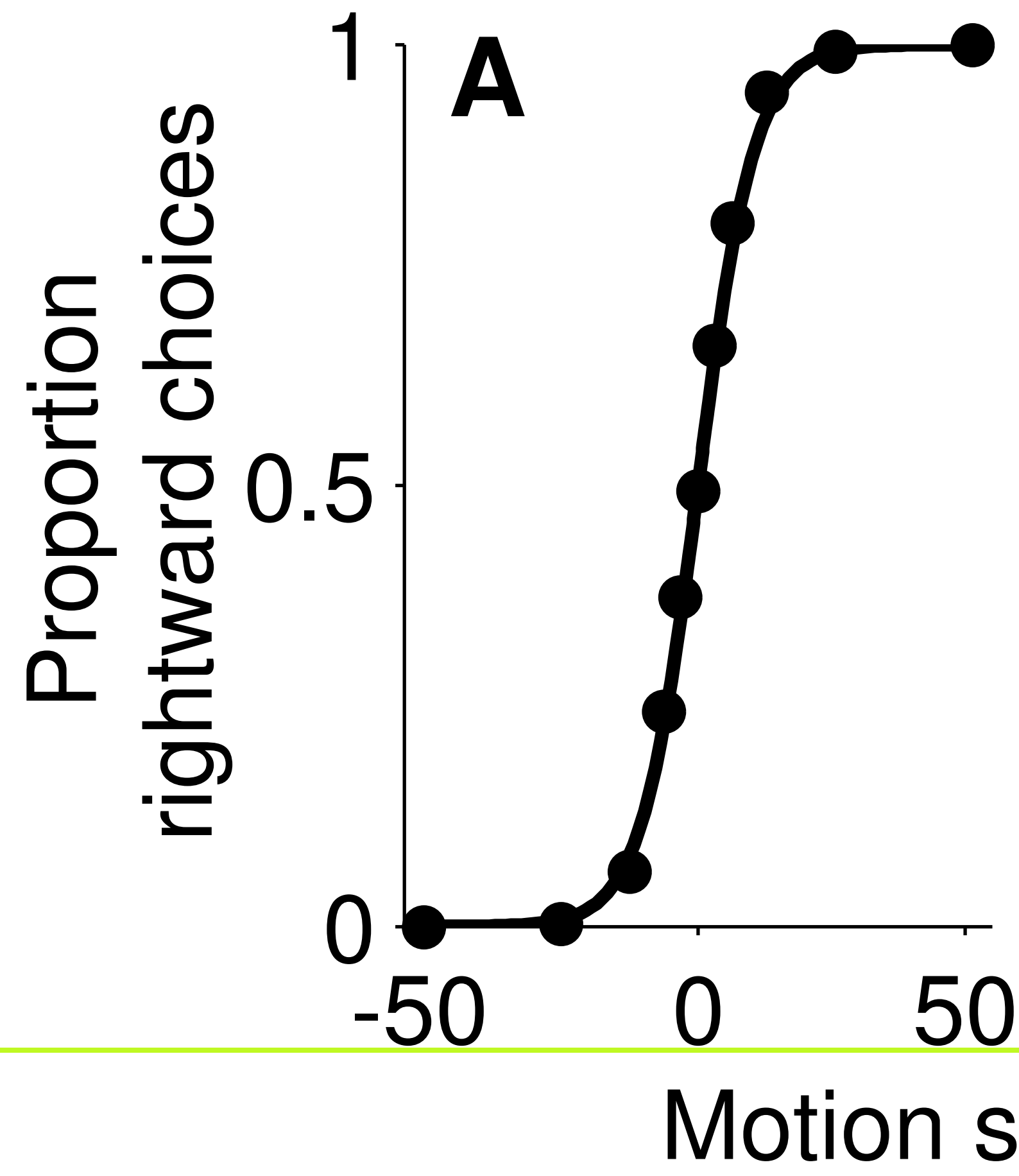
Drift Diffusion Model



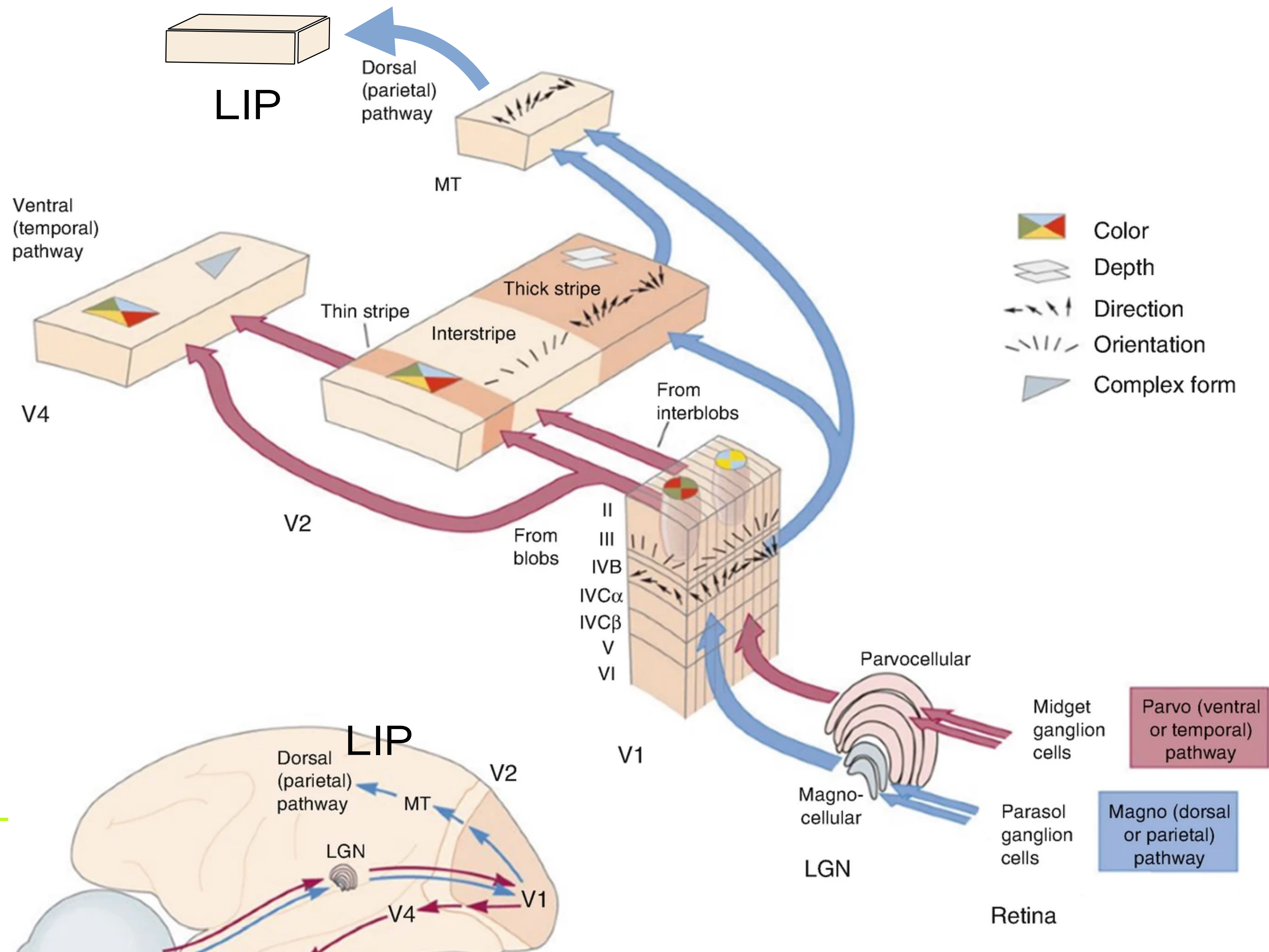
$$P_+ = \frac{1}{1 + e^{\frac{-2\mu}{\sigma^2} \cdot A}}$$

$$E[t] = \frac{A}{\mu} \tanh\left(\frac{\mu A}{\sigma^2}\right)$$

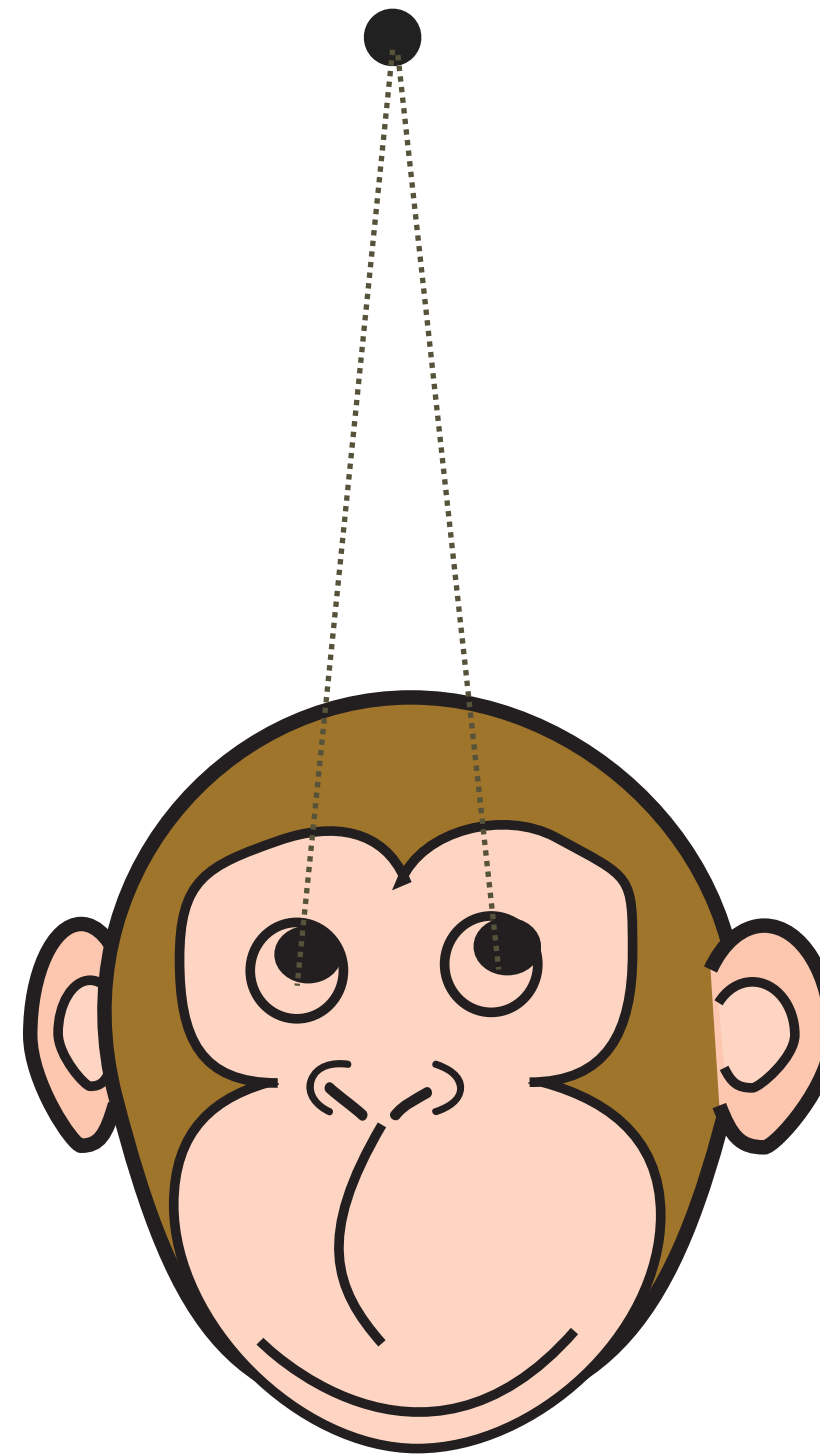
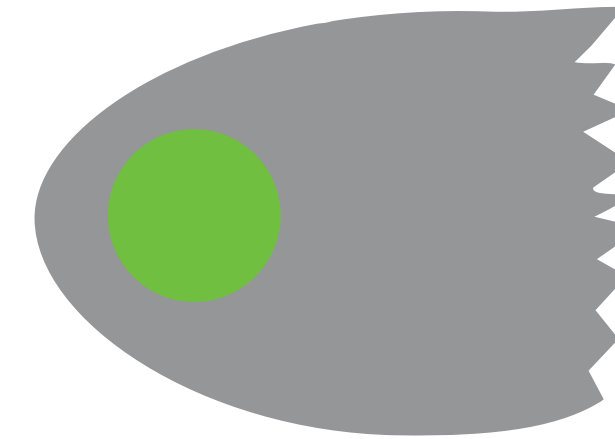
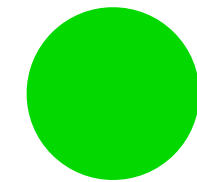
Reaction Time Task



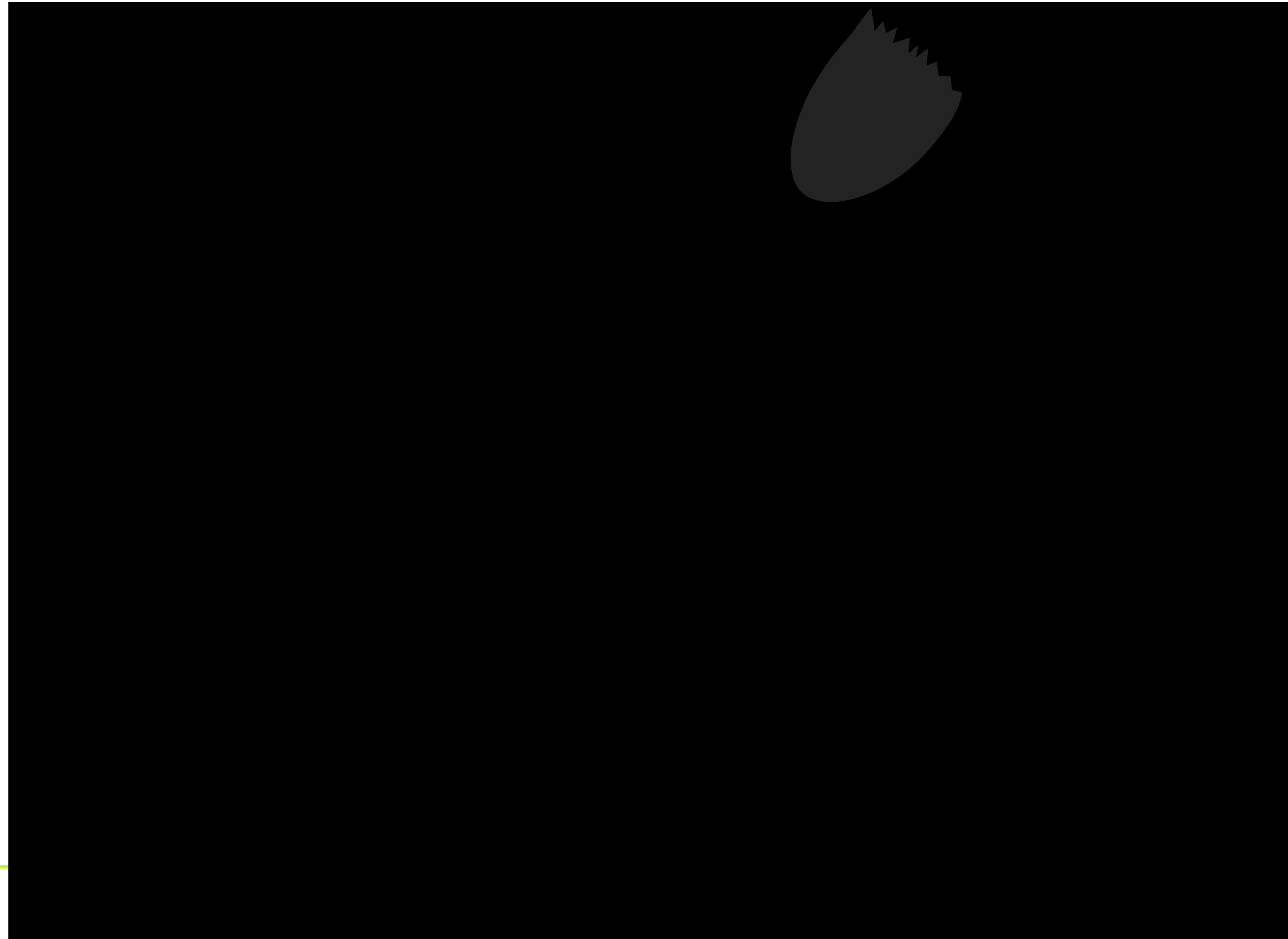
Lateral Intraparietal Area



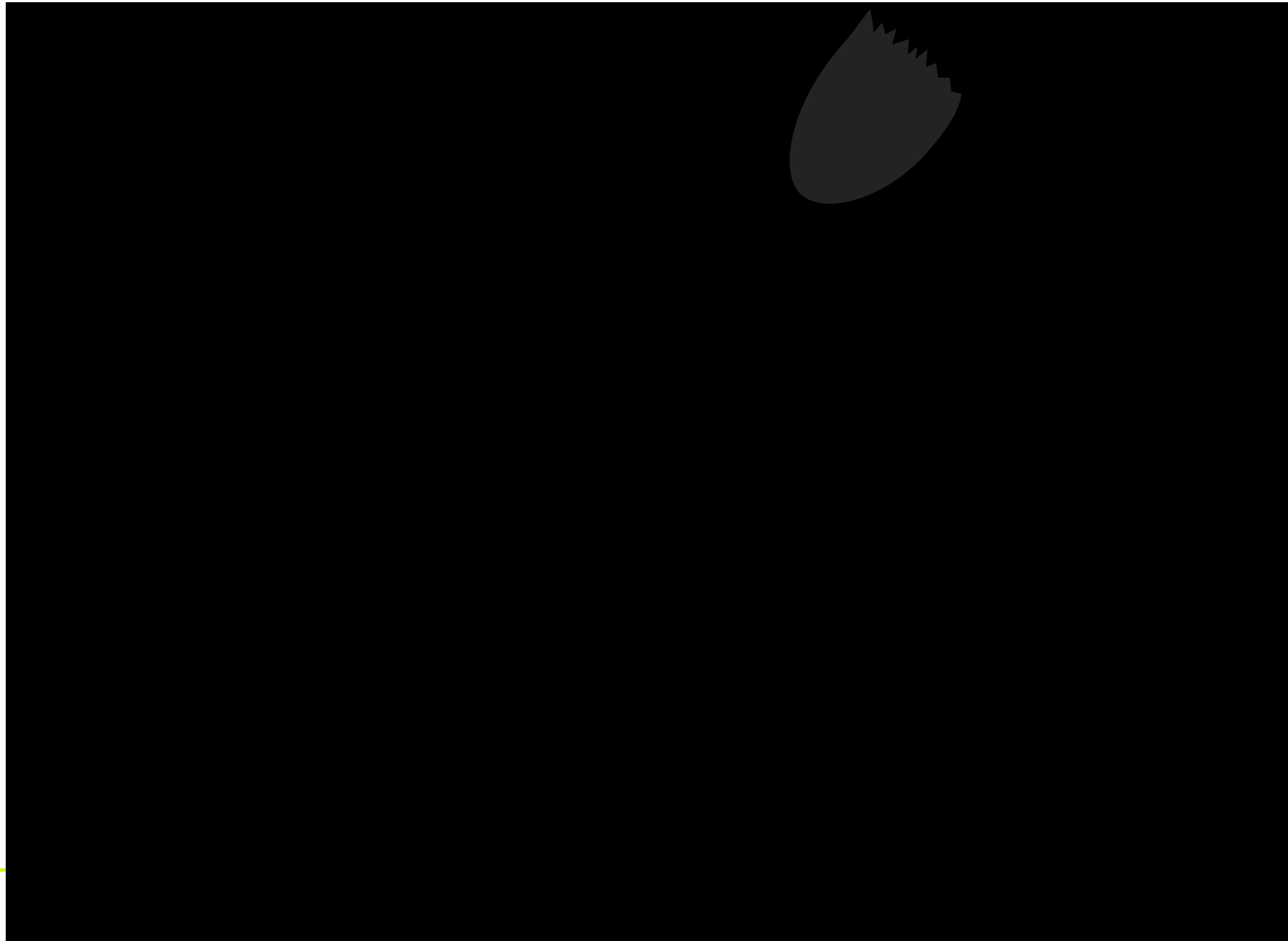
Response Field



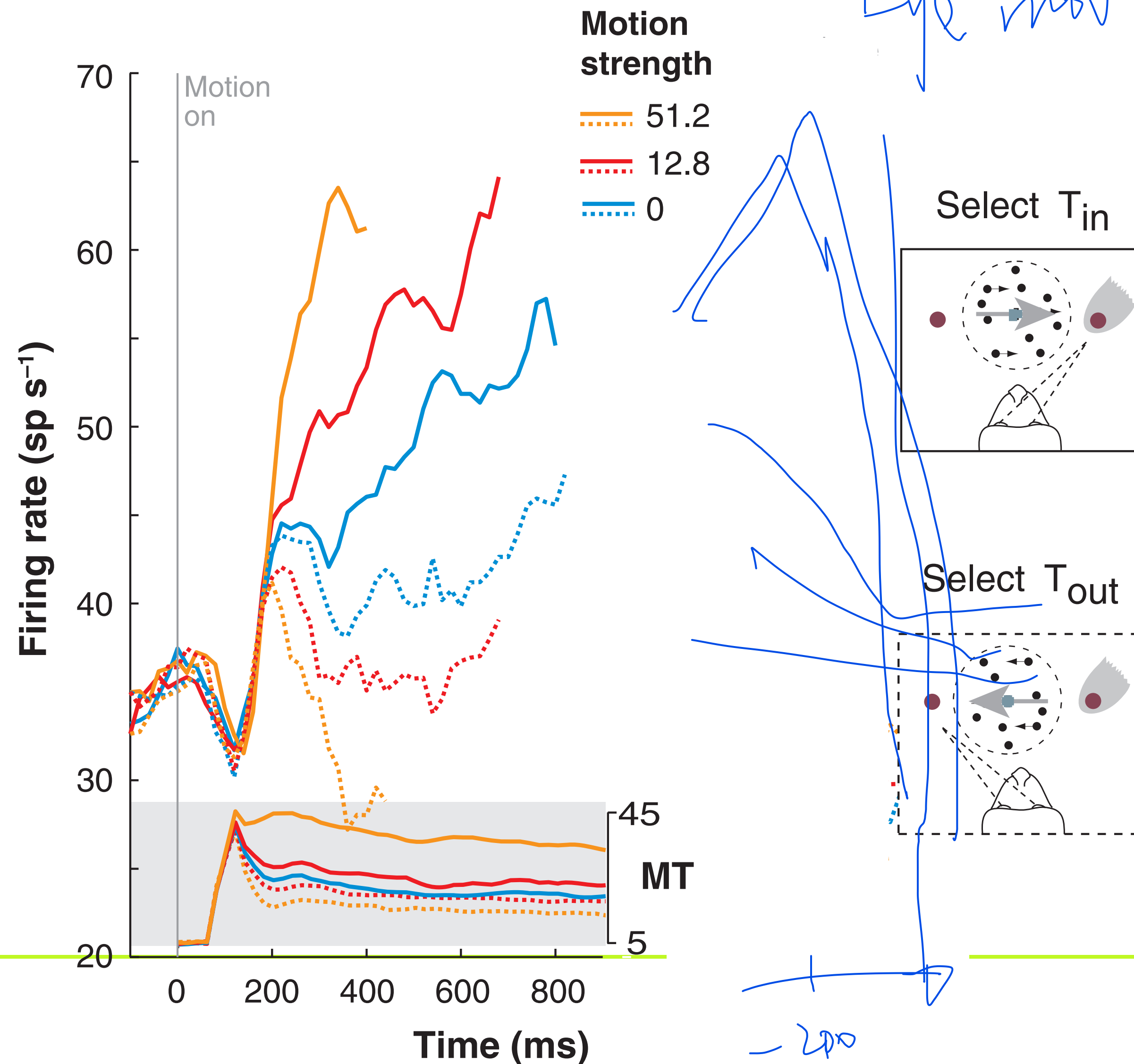
Example Neuron Movie



Example Neuron Movie



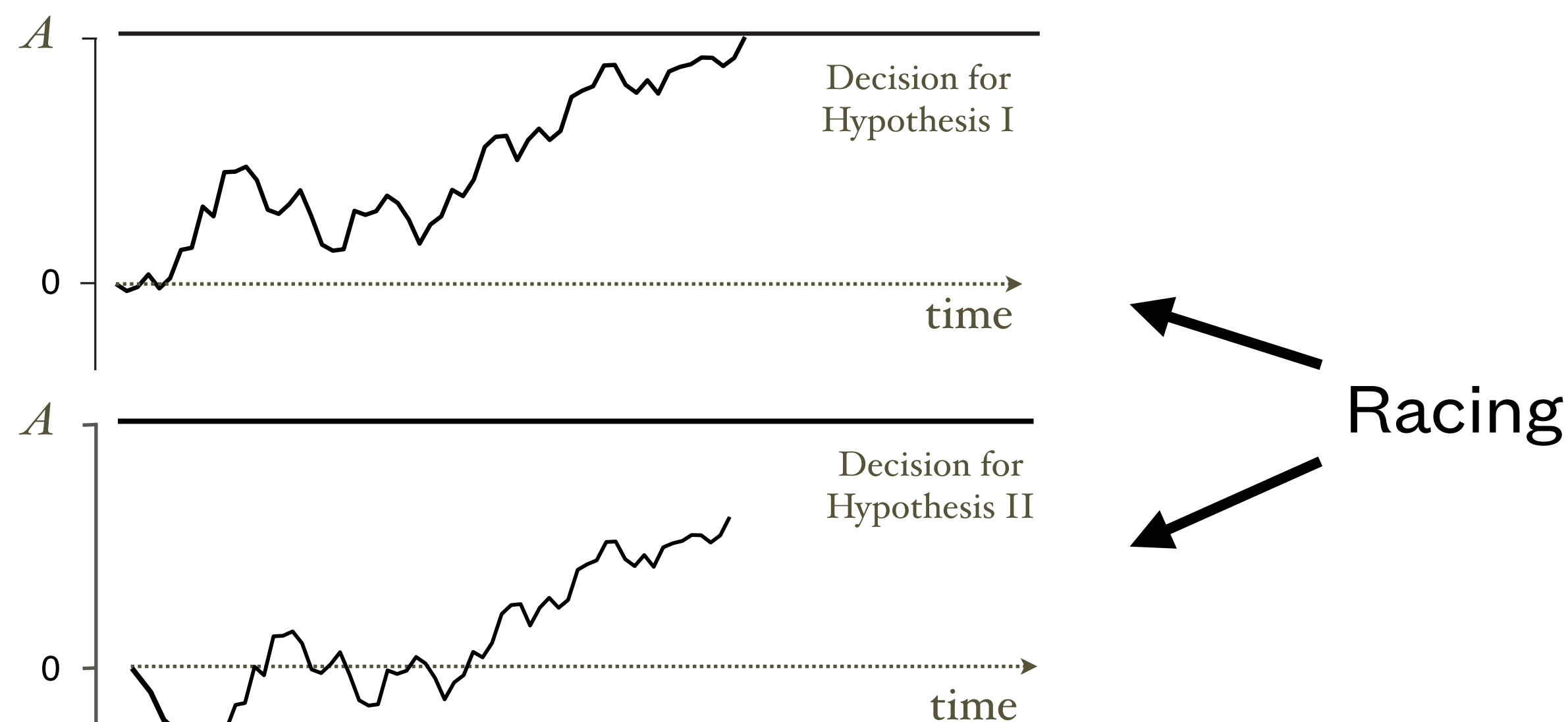
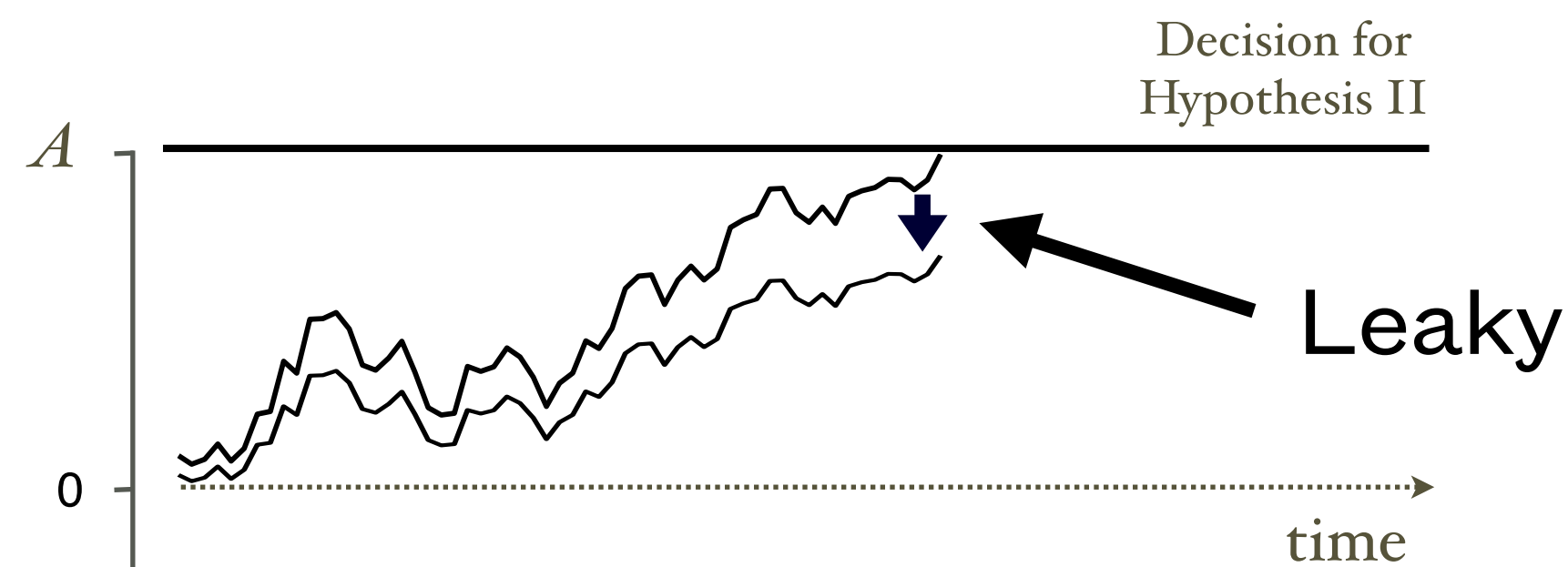
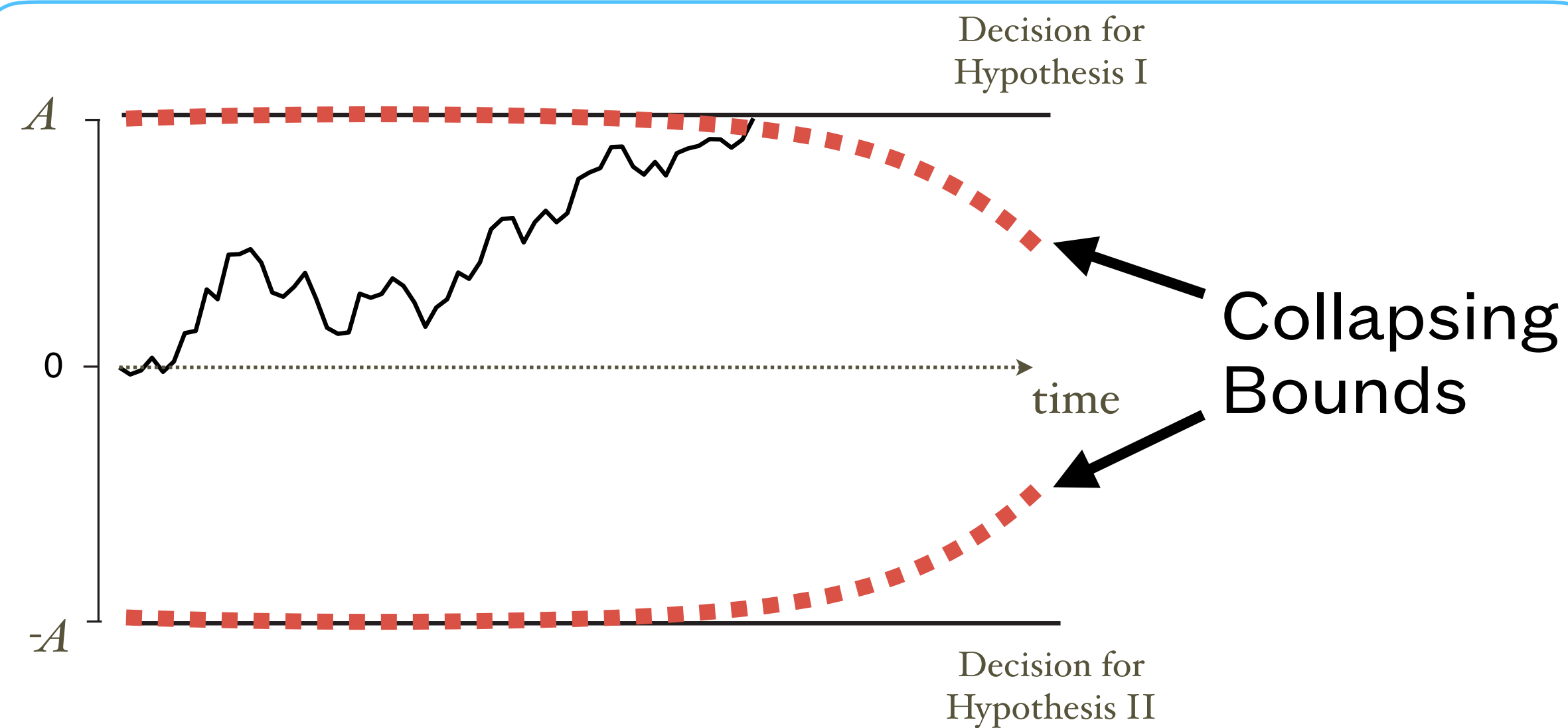
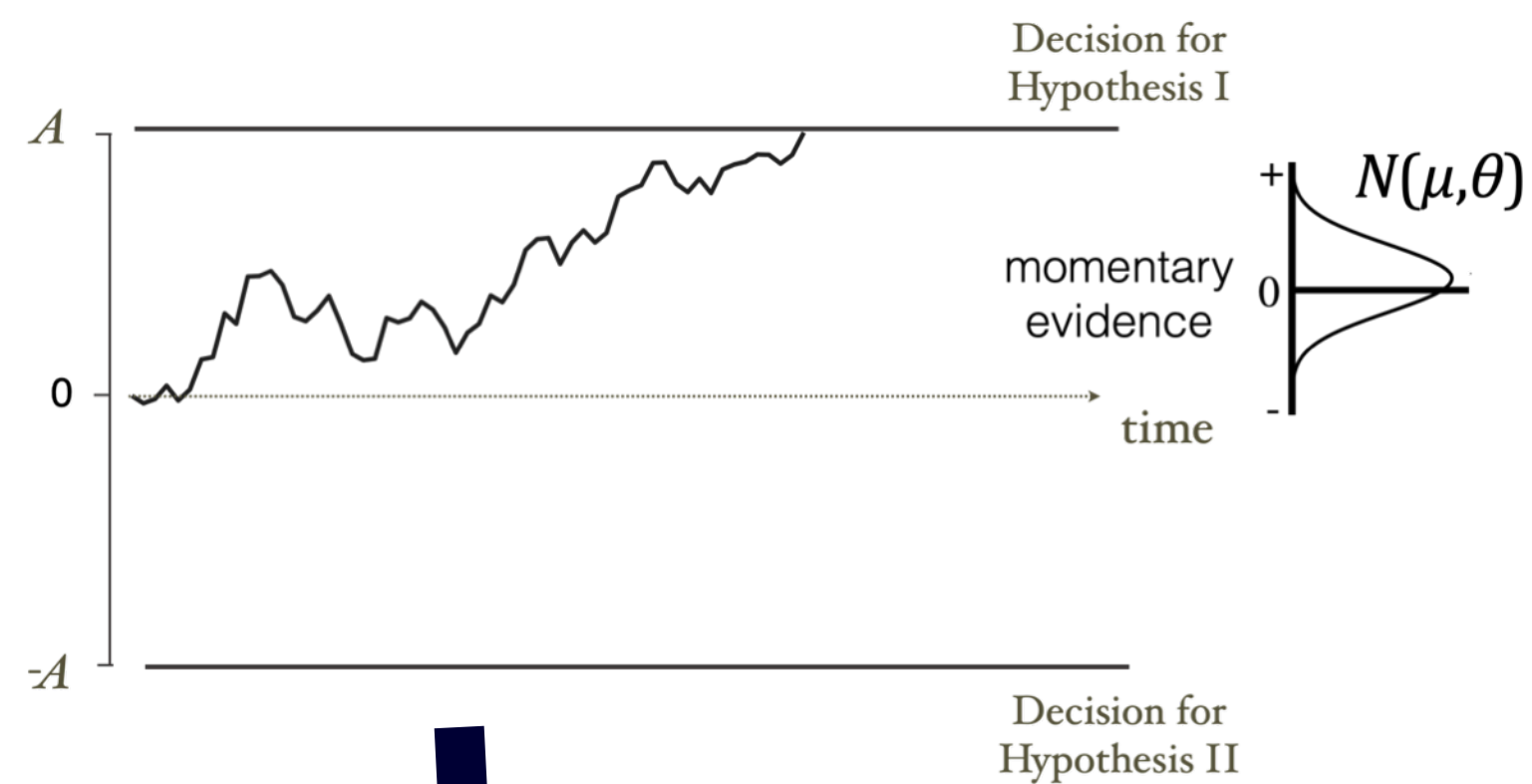
Ramping Activity



DDM Everywhere

- Perceptual decision making (eg. Roitman and Shadlen, 2002)
 - Value-based decision making (eg. Krajbich et al., 2012)
 - Time (eg. Jazayeri and Shadlen, 2015)
 - Memory retrieval (eg. Ratcliff, 1978, Gluth et al, 2015)
 - Probabilistic reasoning (eg. Yang and Shadlen, 2007)
 - Confidence (eg. Kiani and Shadlen, 2013)
-

Variations of DDM



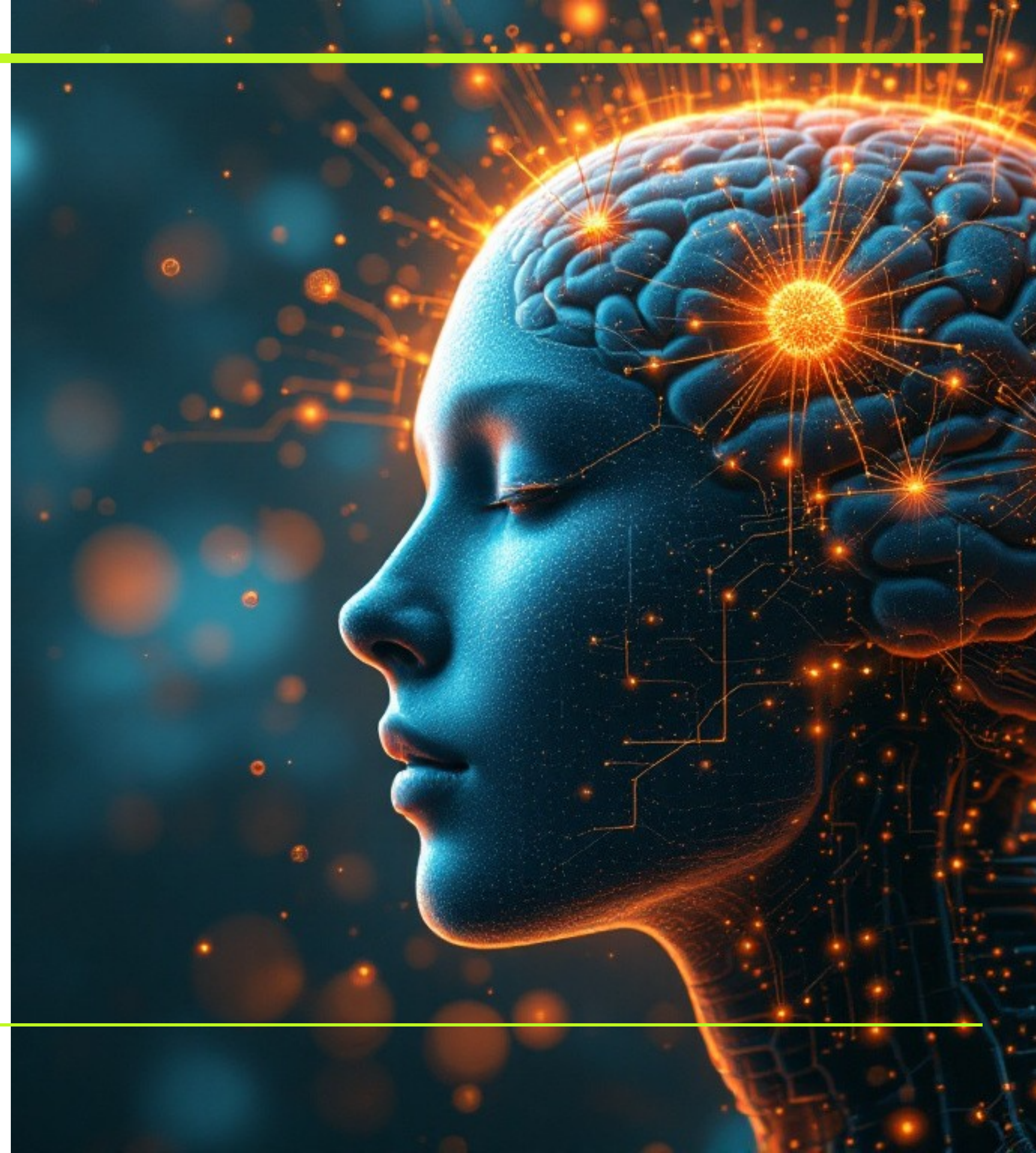
❖ Overview

❖ DDM

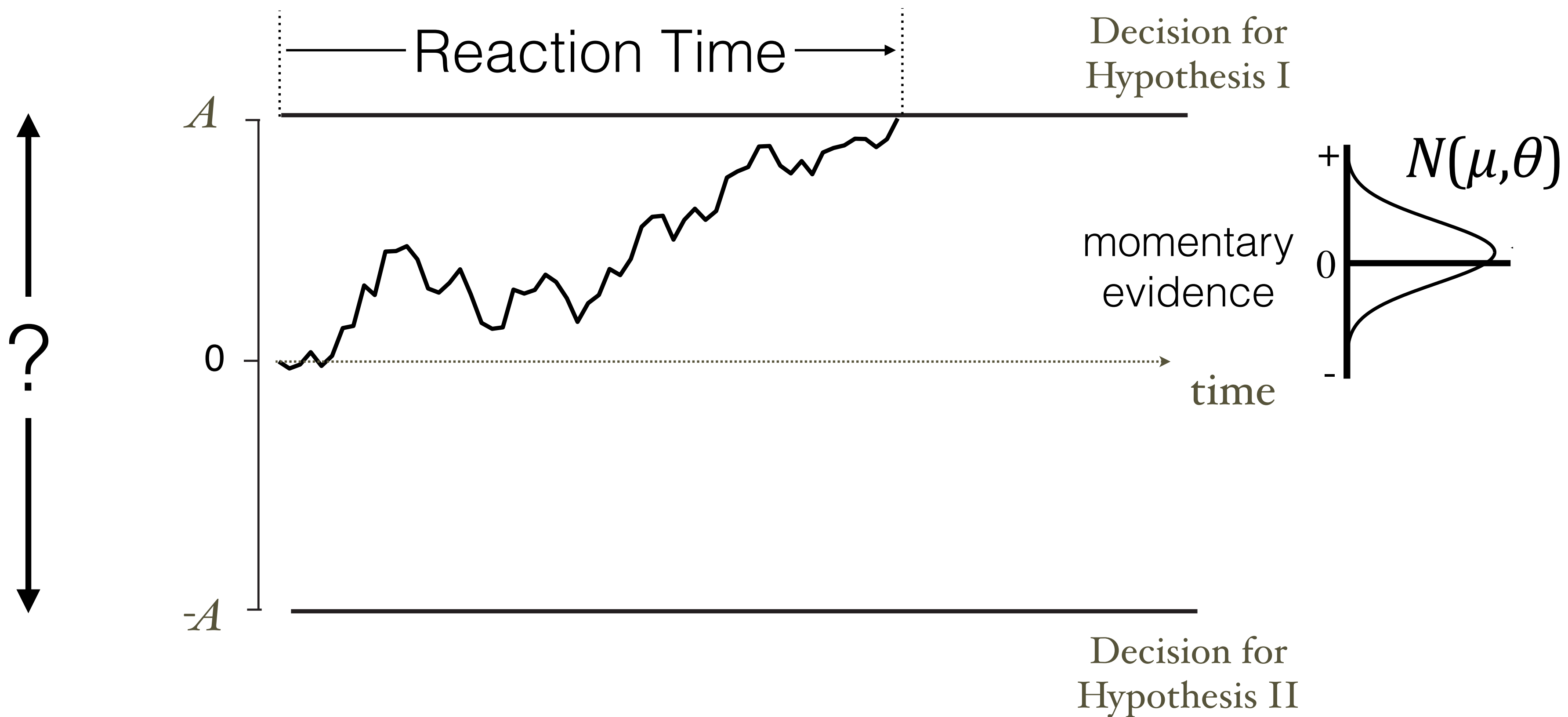
❖ **SPRT**

❖ Confidence

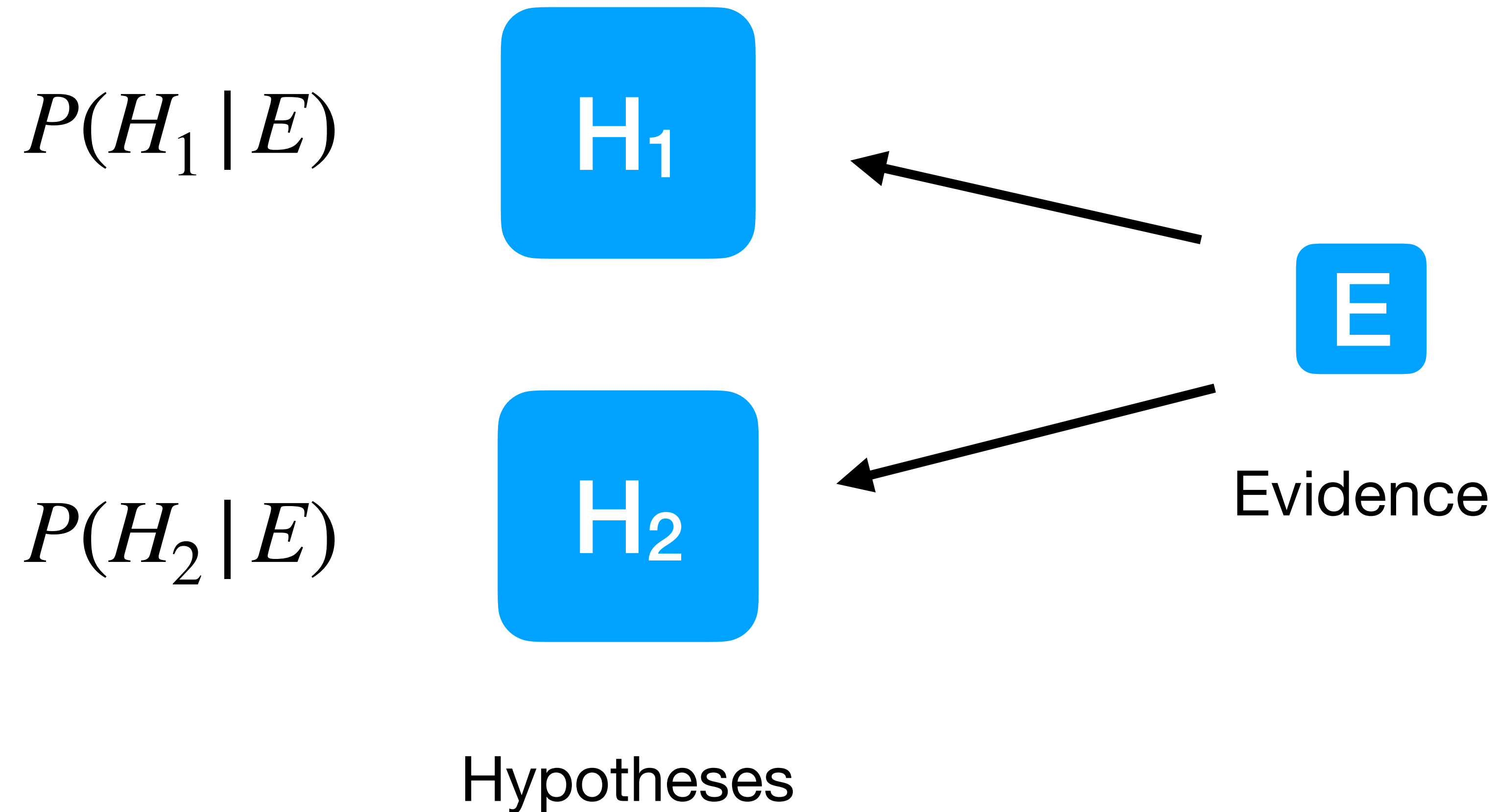
❖ Attractor Network



Drift Diffusion Model



Quantifying Evidence



Quantifying Evidence

$$P(H_1 | E) \propto P(E | H_1) * P(H_1)$$

$$P(H_2 | E) \propto P(E | H_2) * P(H_2)$$



Evidence

Quantifying Evidence

$$P(H_1 | E) \propto P(E | H_1) * P(H_1)$$

$$P(H_2 | E) \propto P(E | H_2) * P(H_2)$$



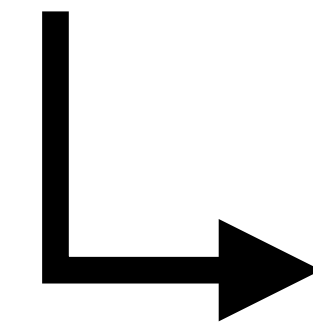
Evidence

Quantifying Evidence

$$\frac{P(H_1 | E)}{P(H_2 | E)} = \frac{P(E | H_1)}{P(E | H_2)} * \frac{P(H_1)}{P(H_2)}$$

Likelihood Ratio

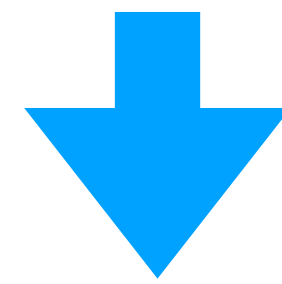
Bias



$$\frac{P(E_1 | H_1) * \dots * P(E_n | H_1)}{P(E_1 | H_2) * \dots * P(E_n | H_2)}$$

Quantifying Evidence

$$\frac{P(E_1 | H_1) * \dots * P(E_n | H_1)}{P(E_1 | H_2) * \dots * P(E_n | H_2)}$$

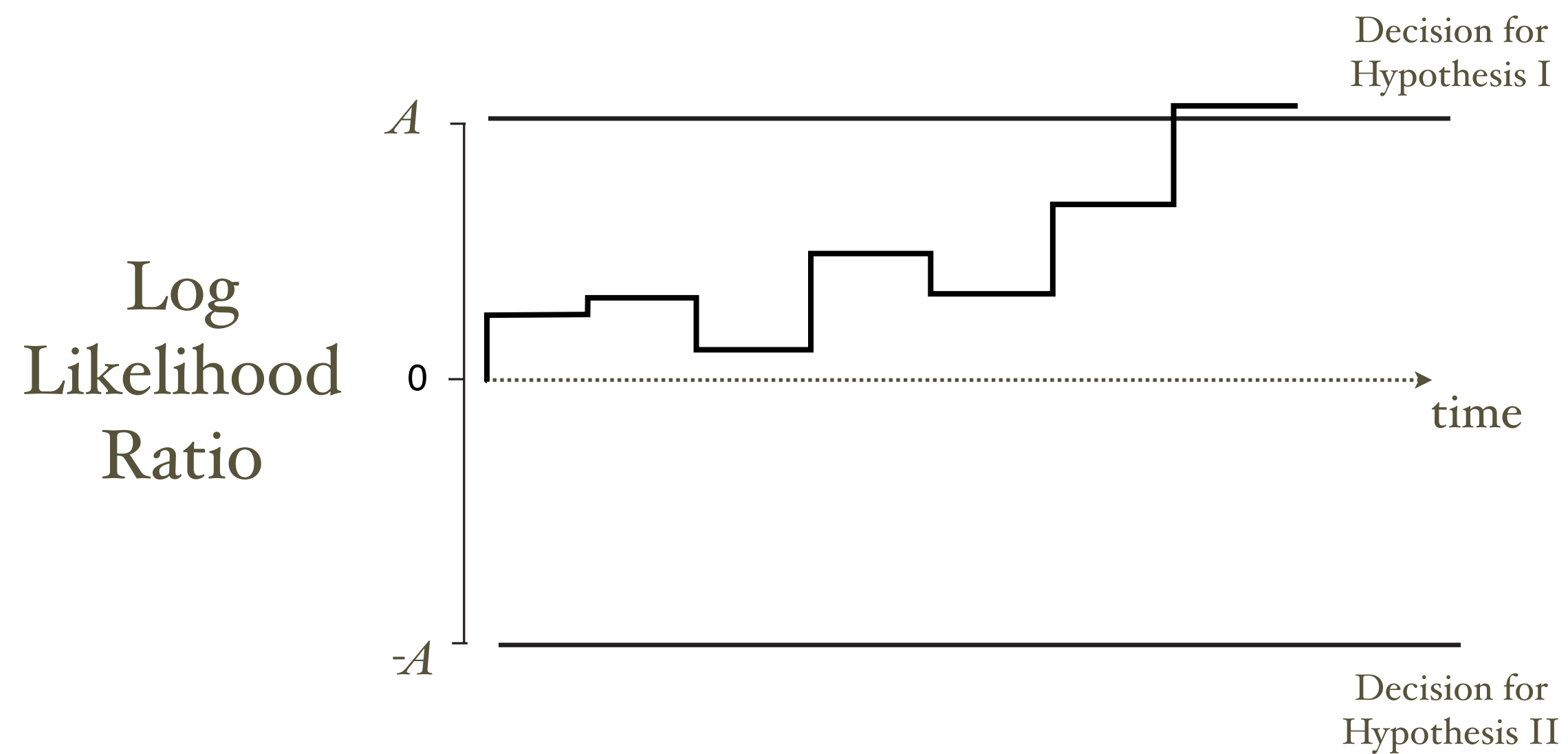


$$\log \frac{P(E_1 | H_1)}{P(E_1 | H_2)} + \dots + \log \frac{P(E_n | H_1)}{P(E_n | H_2)}$$

Log Likelihood Ratio

Sequential Probability Ratio Test

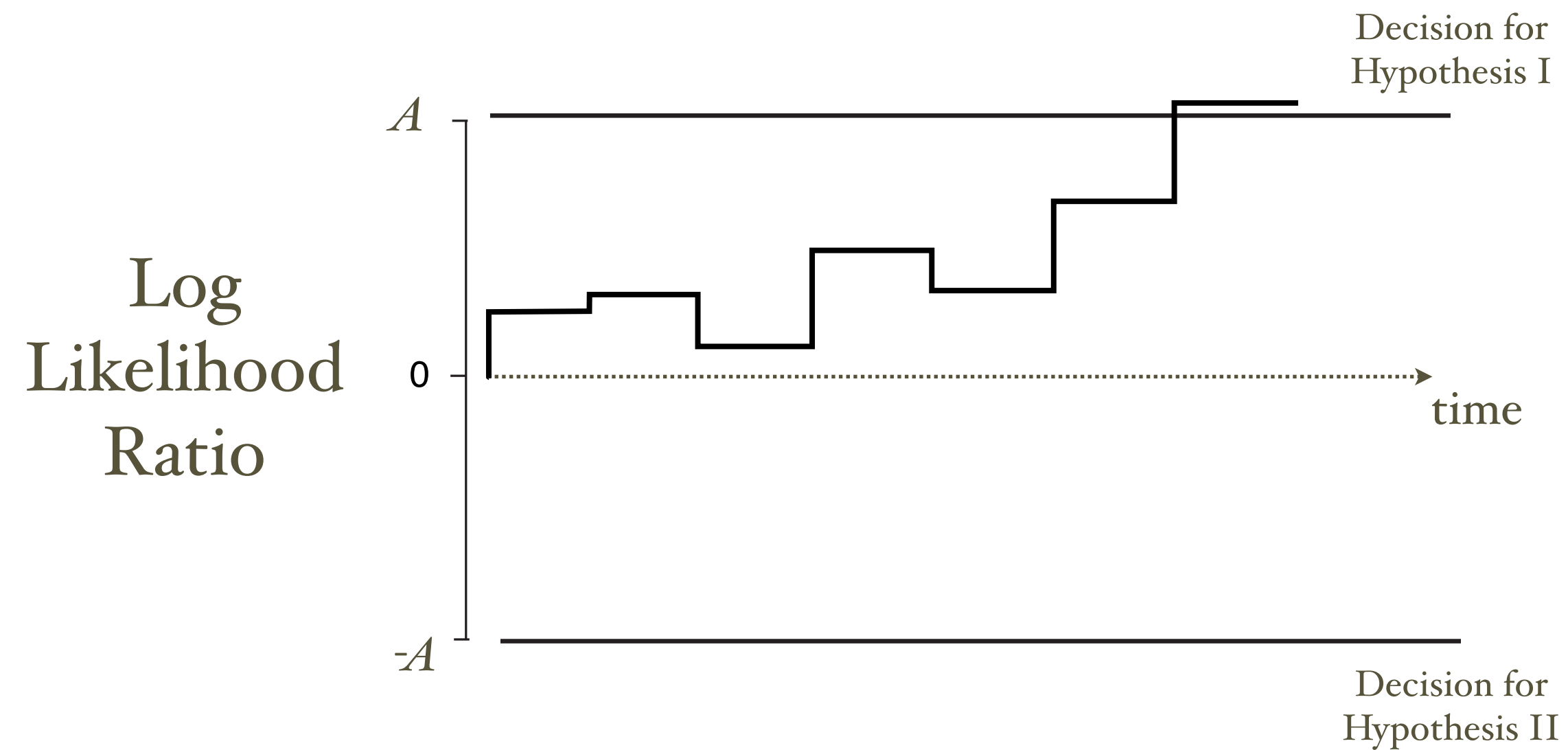
SPRT



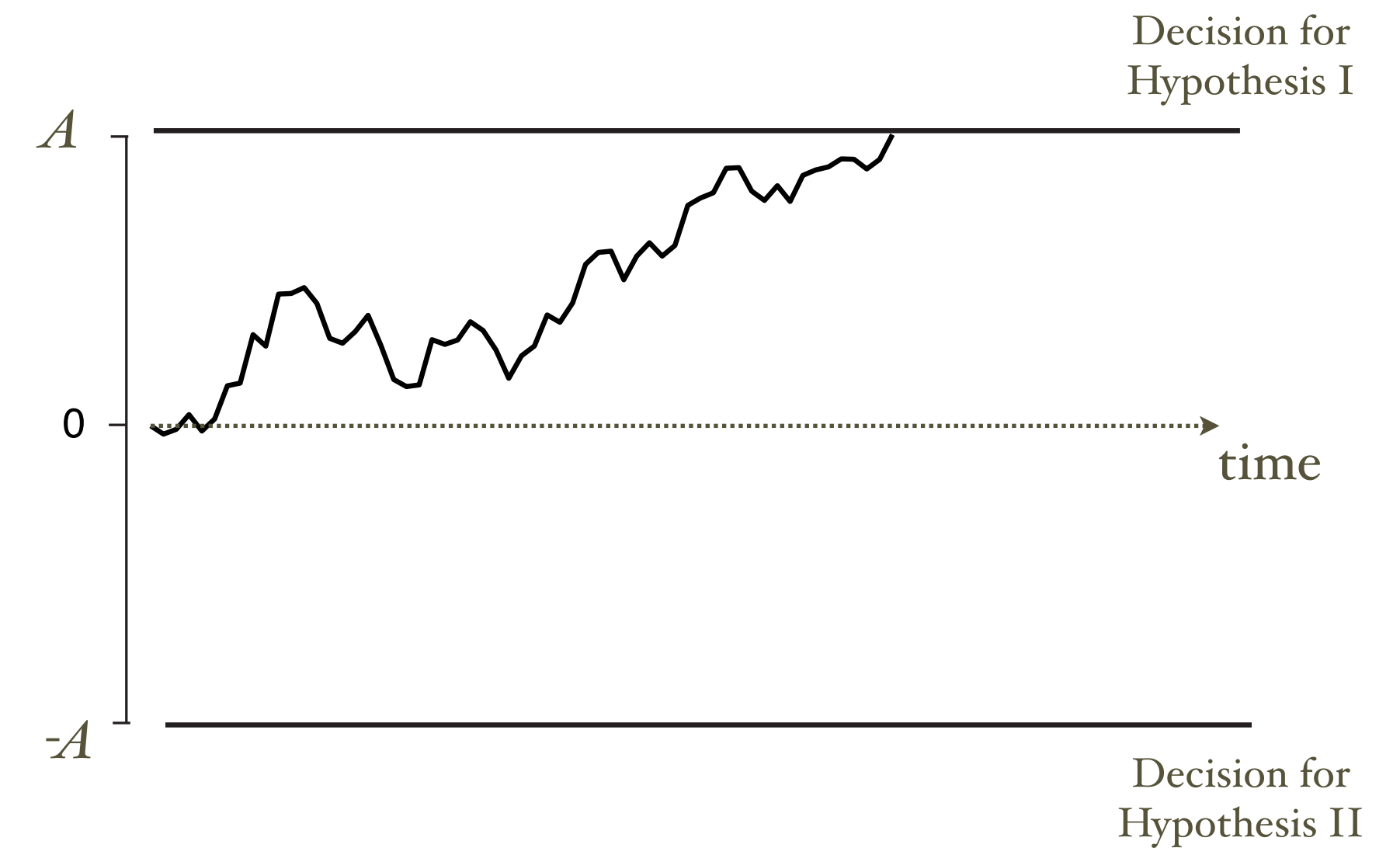
A. Wald, 1945

Sequential Probability Ratio Test $\leftrightarrow$ DDM

SPRT

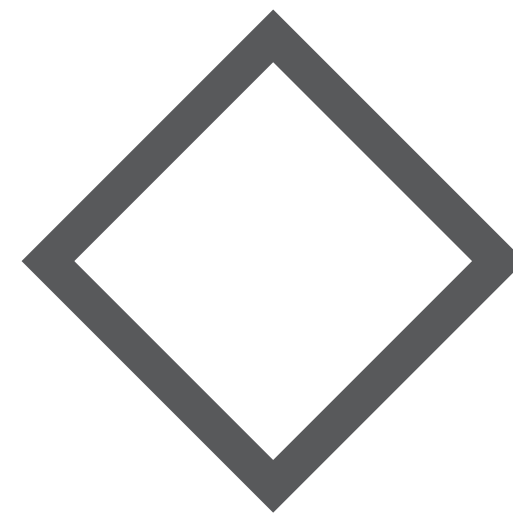
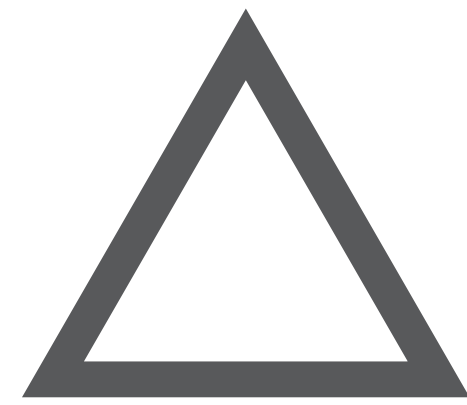
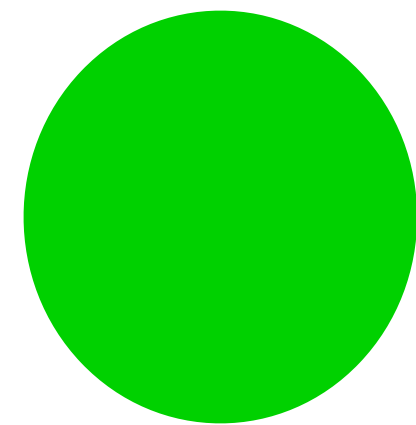


DDM

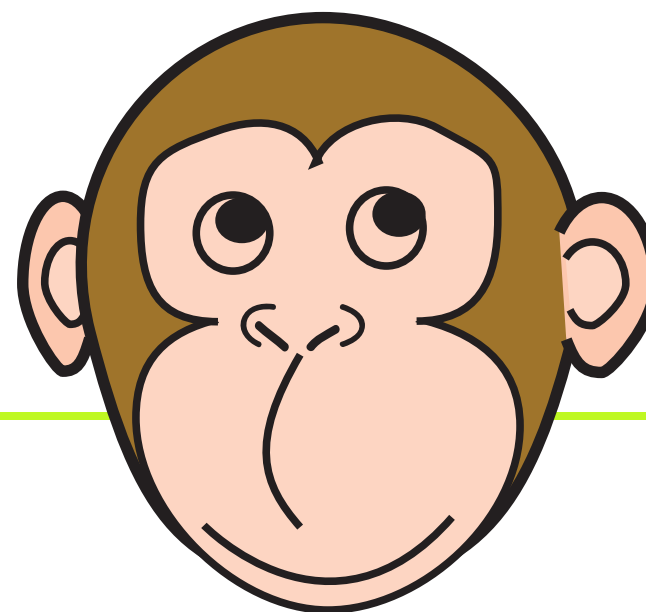
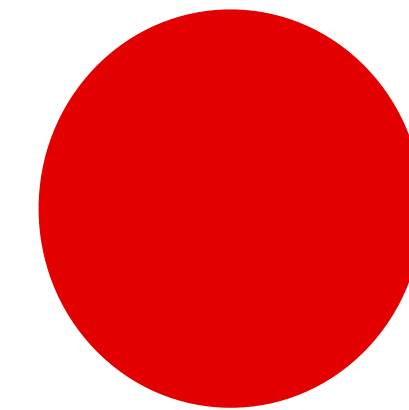


A. Wald, 1945

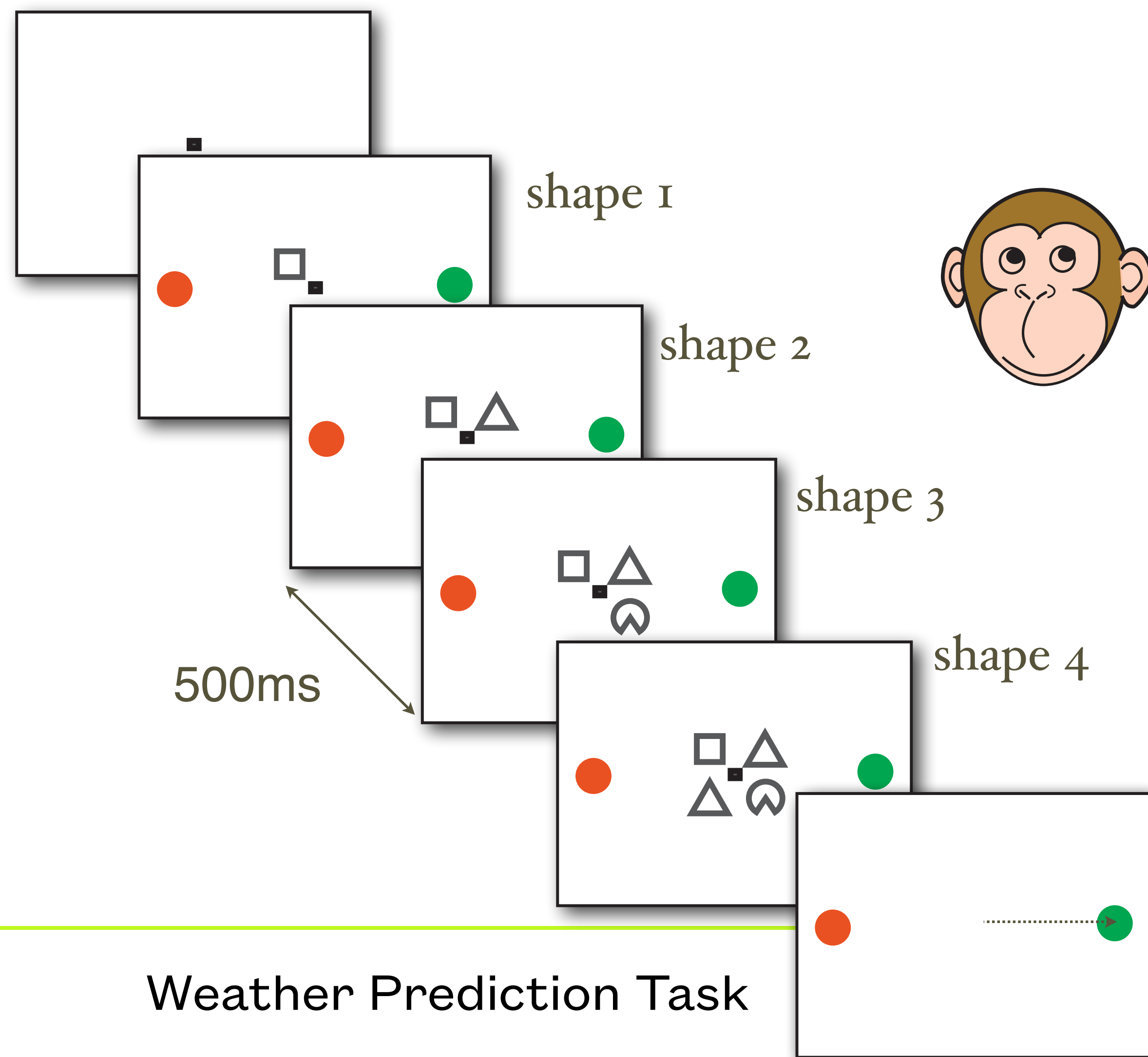
Probabilistic Reasoning



Weather Prediction Task



Probabilistic Reasoning



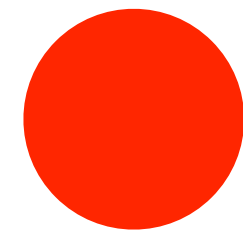
Weather Prediction Task

Yang and Shadlen, 2007

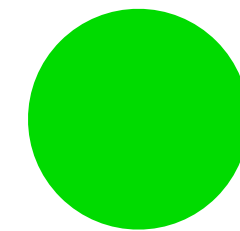


$$\sum \text{weights} = \log_{10} \frac{P(\text{shapes} \mid \text{red})}{P(\text{shapes} \mid \text{green})}$$

61%



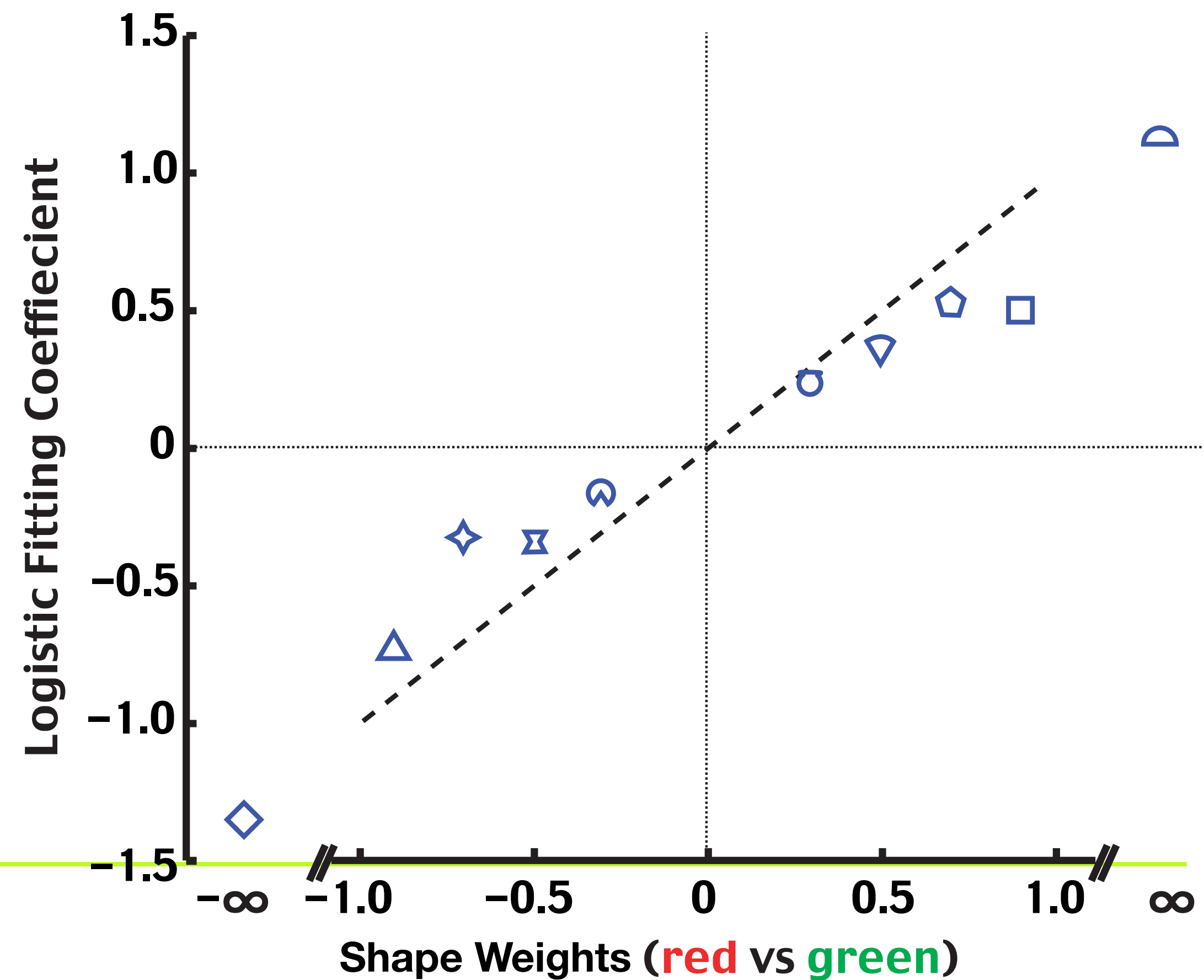
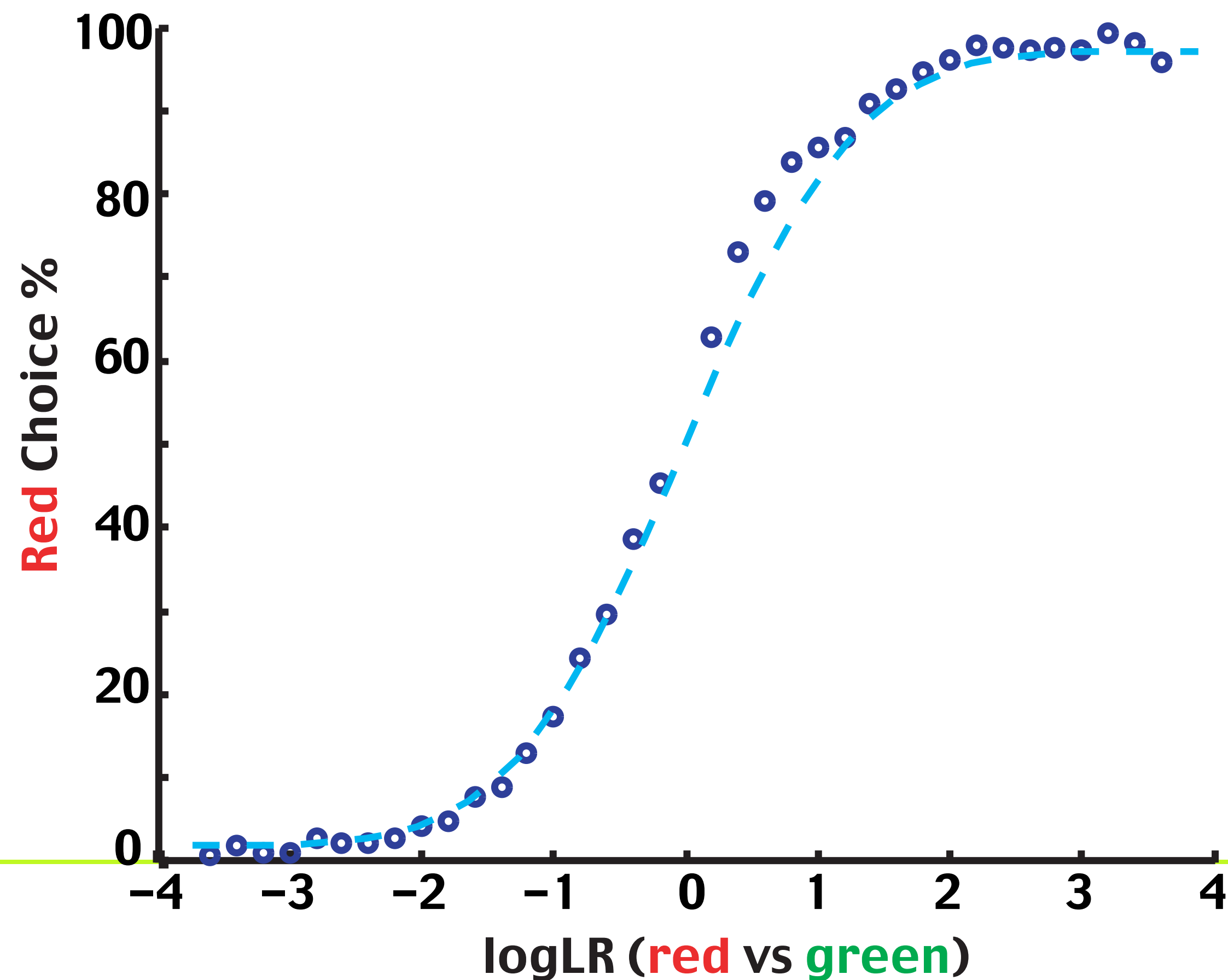
39%



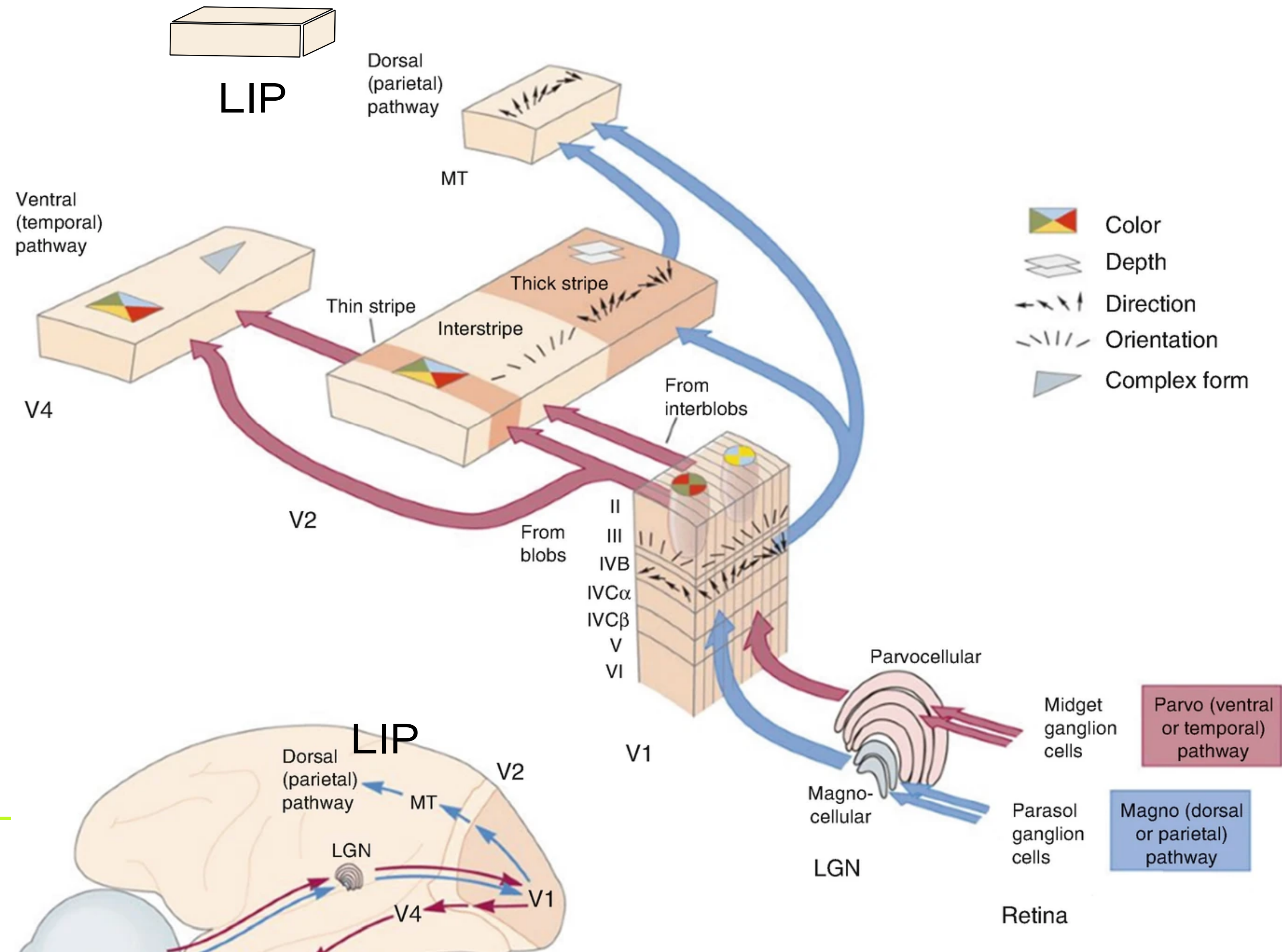
$-\infty$ -0.9 -0.7 -0.5 -0.3 0.3 0.5 0.7 0.9 ∞

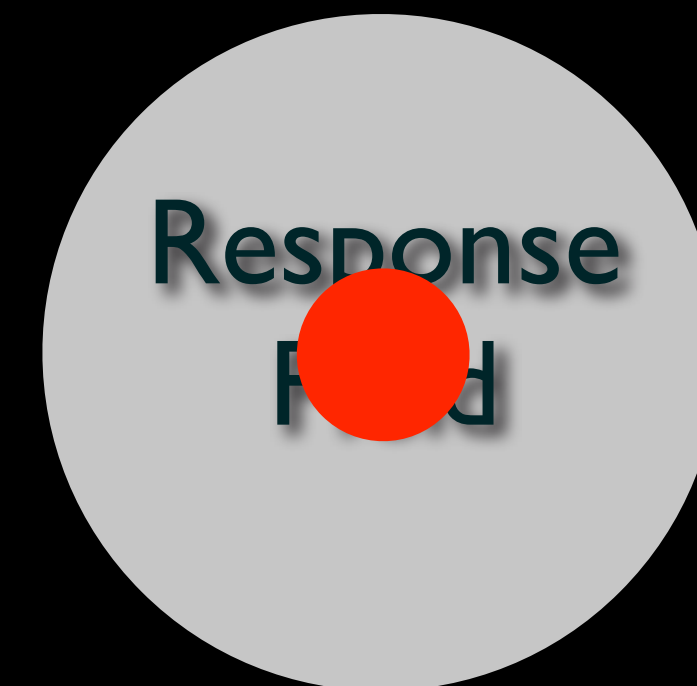
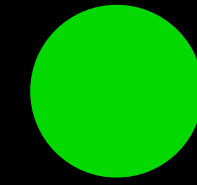
Monkey Performance

$$\text{logit}(\text{Choice}) \sim \sum_{i=1..10} \beta_i I_i$$

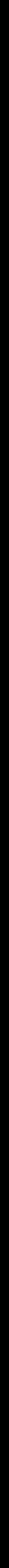


Target Brain Area: LIP





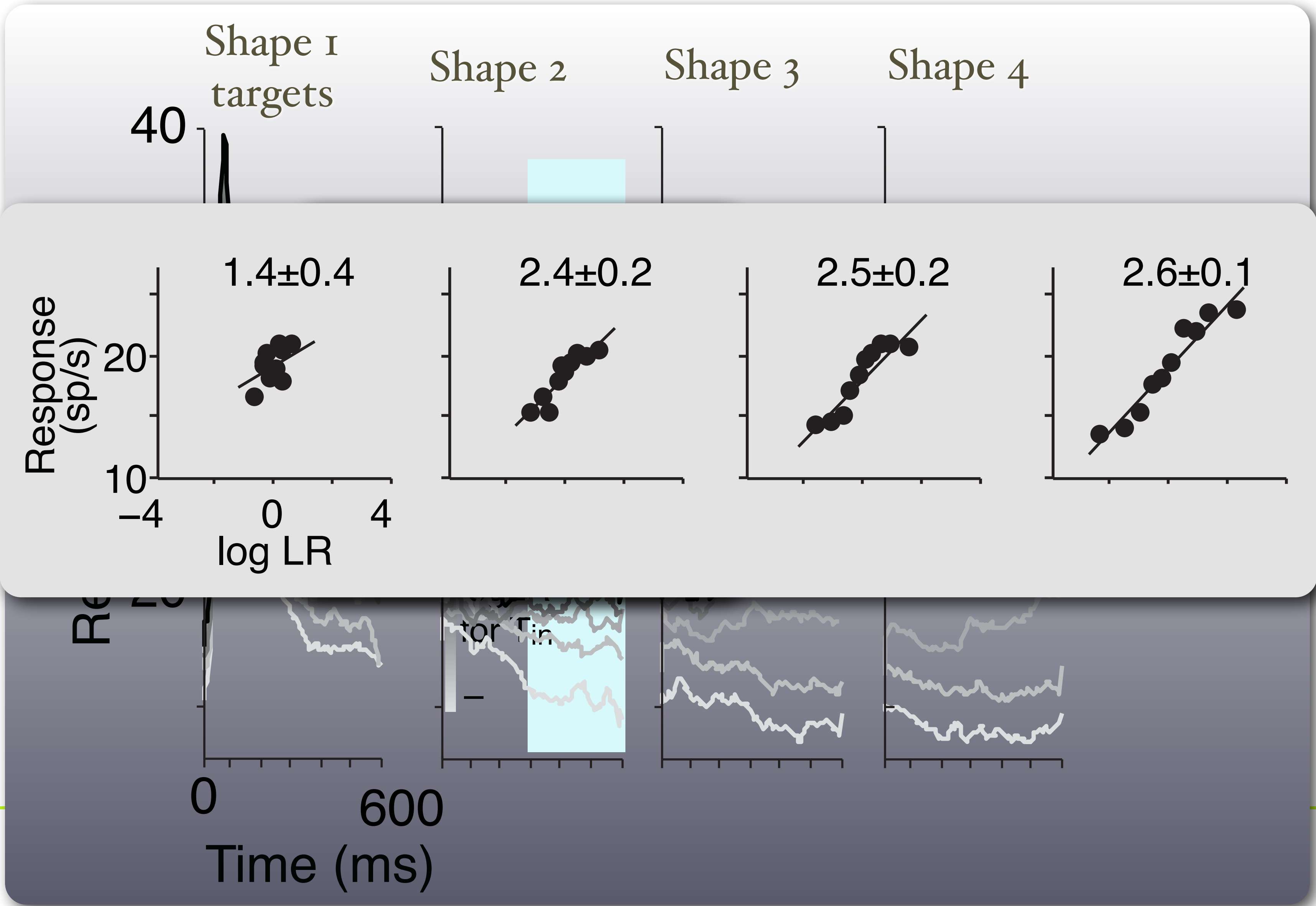
logLR
for T_{in}



$\log LR$
for T_{in}

•

LIP Neurons Encode logLR



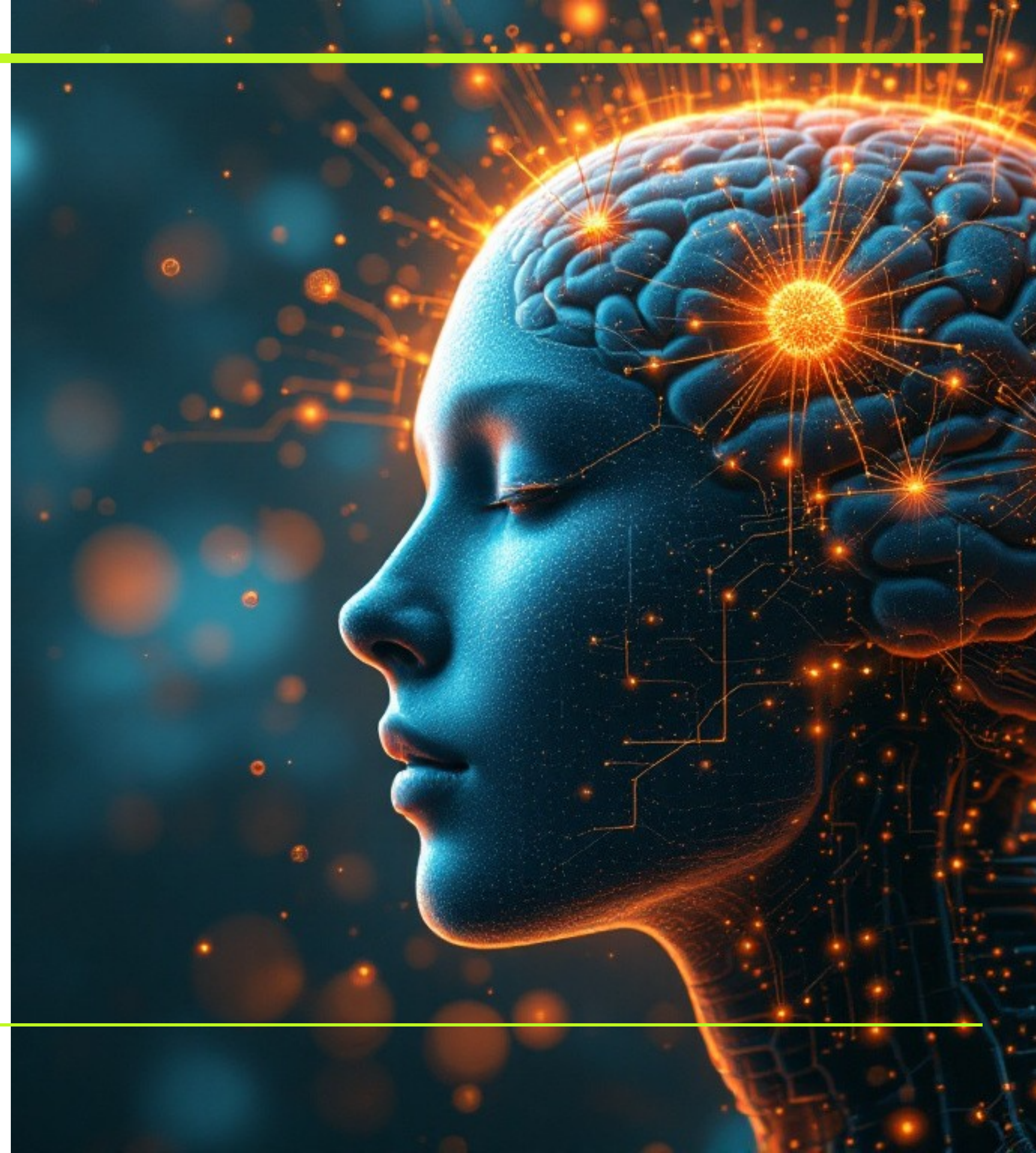
❖ Overview

❖ DDM

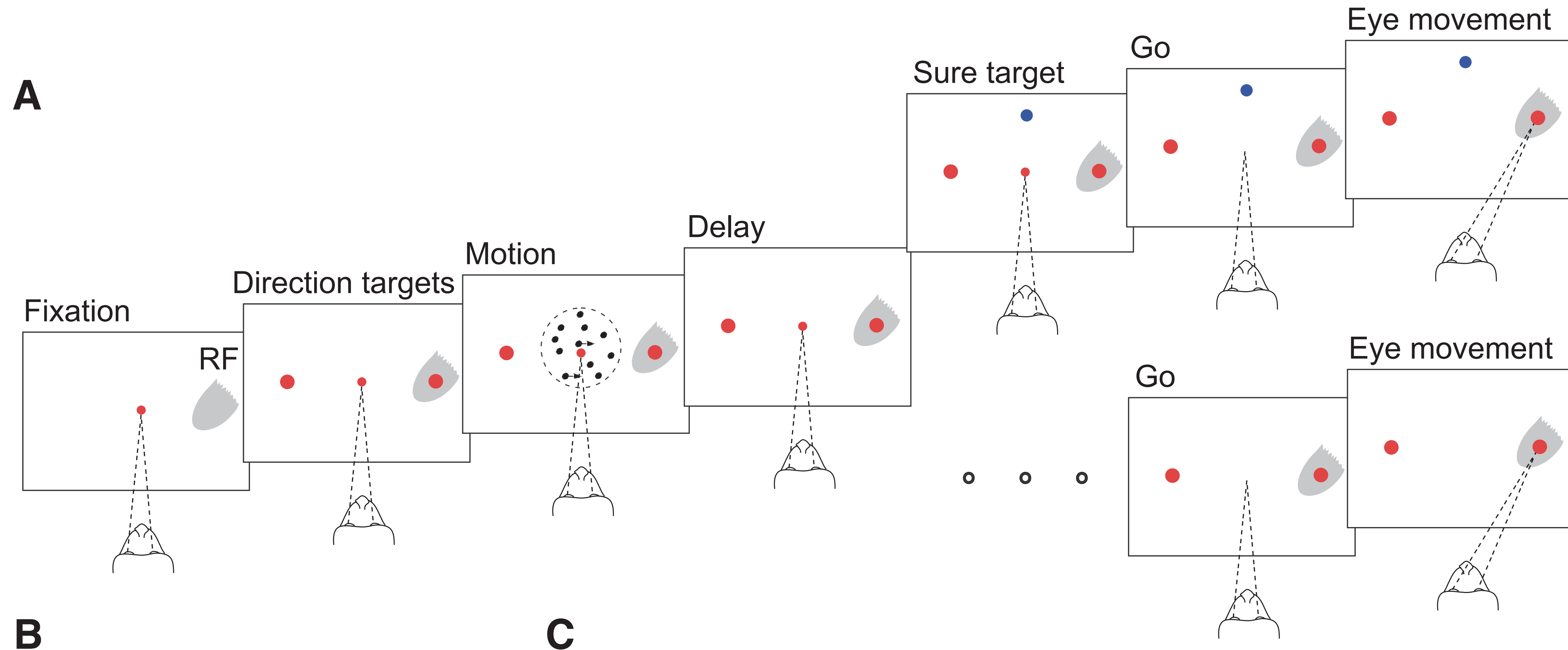
❖ SPRT

❖ **Confidence**

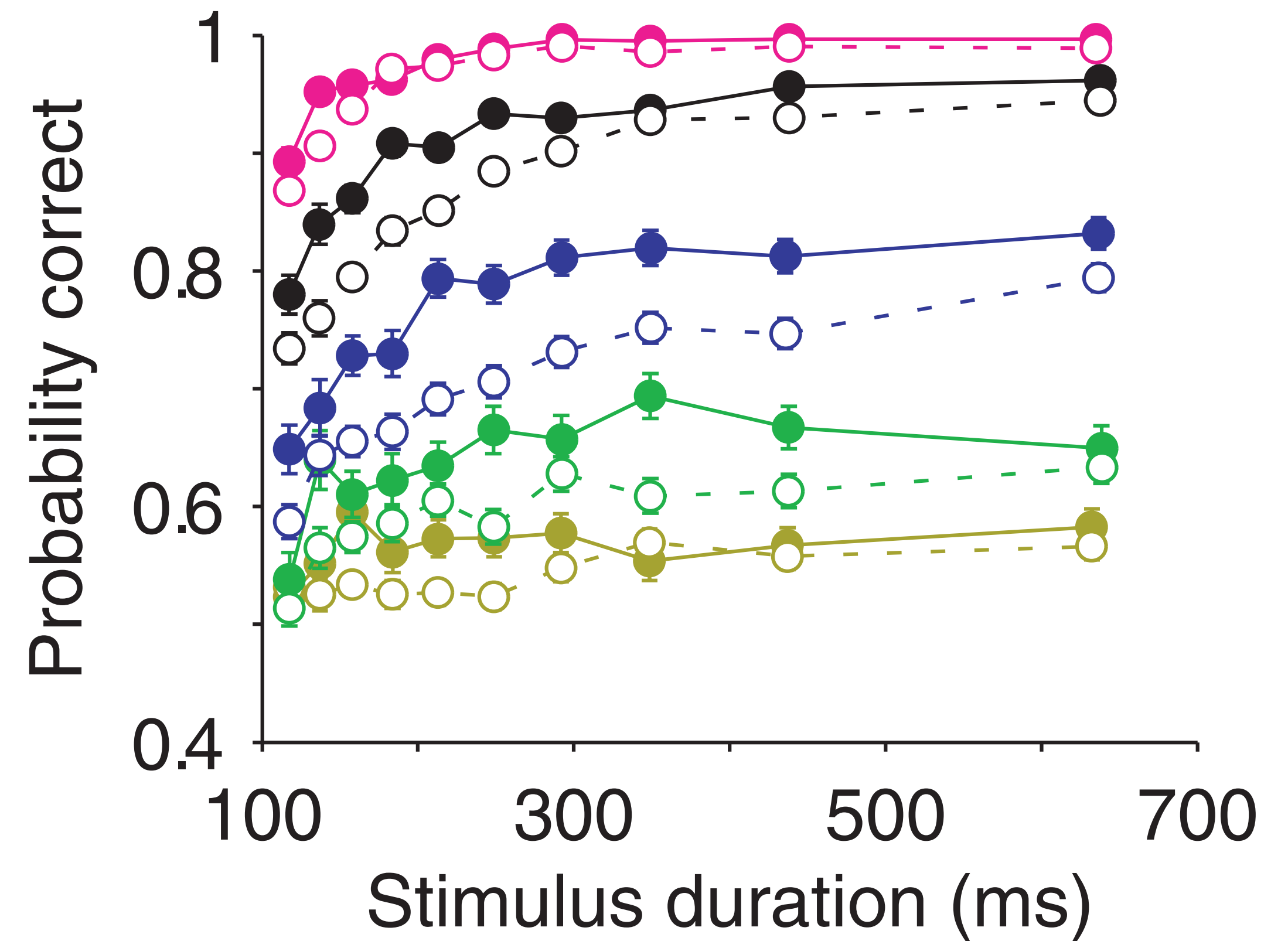
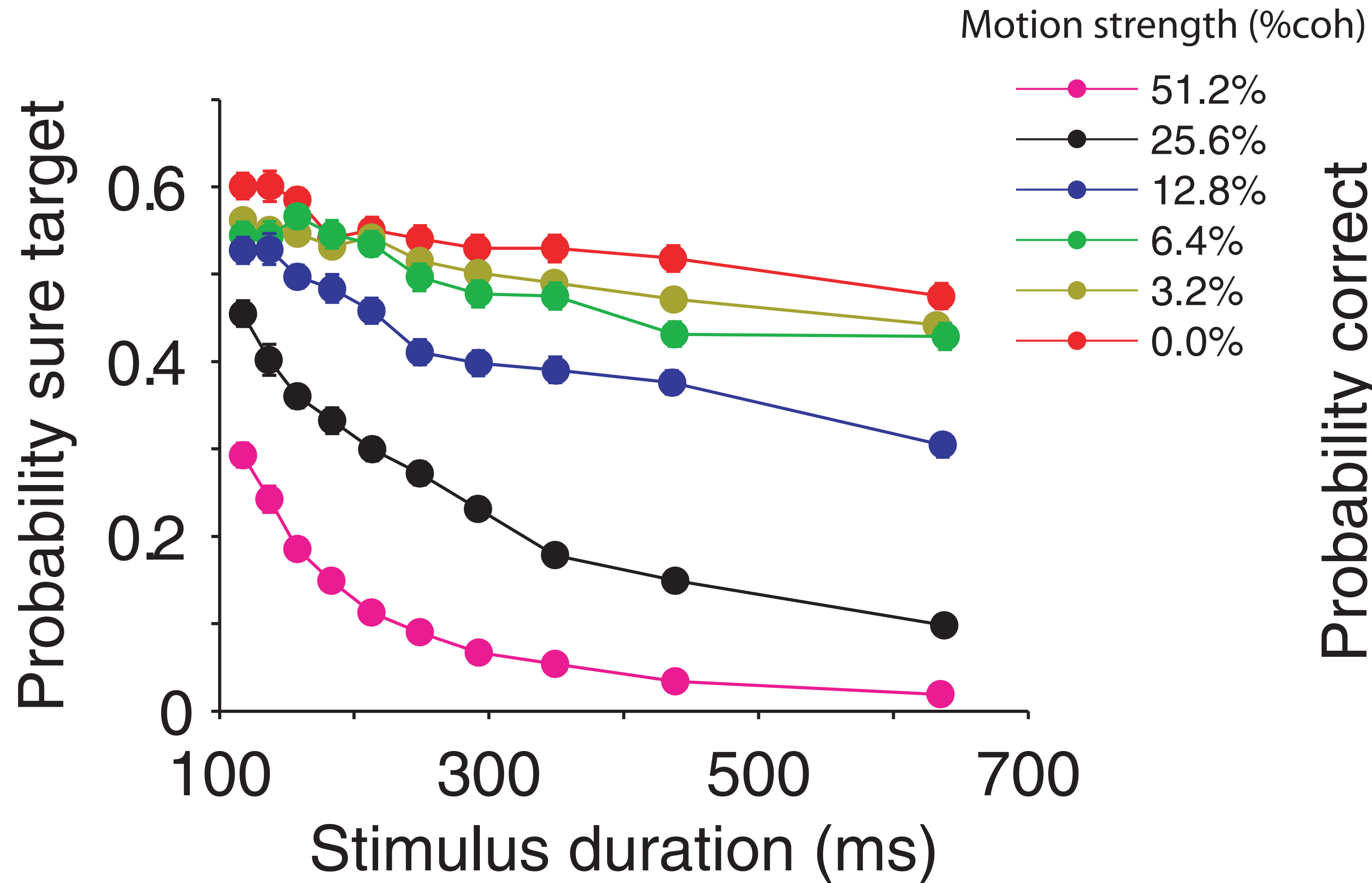
❖ Neural network Modeling



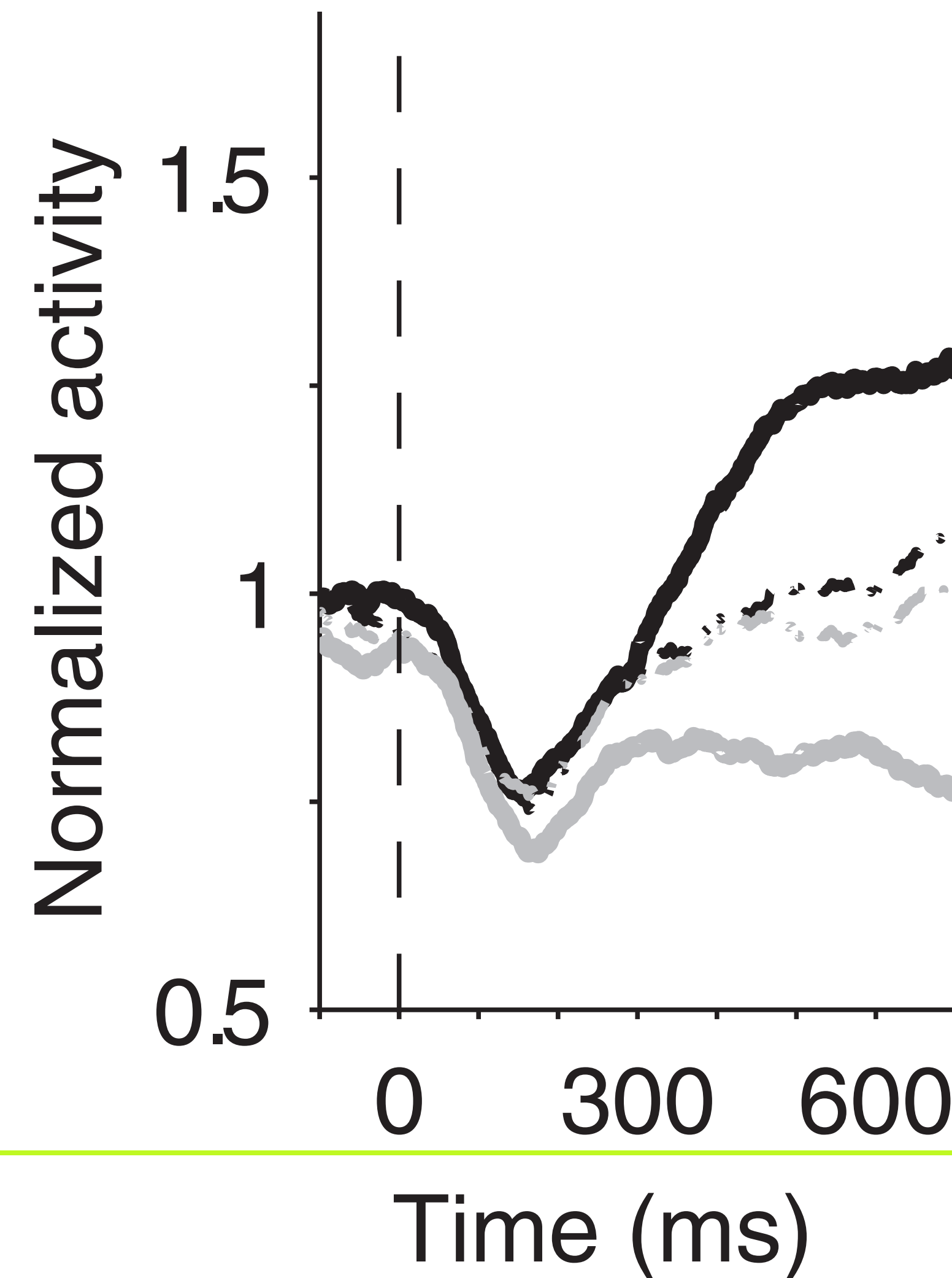
Confidence



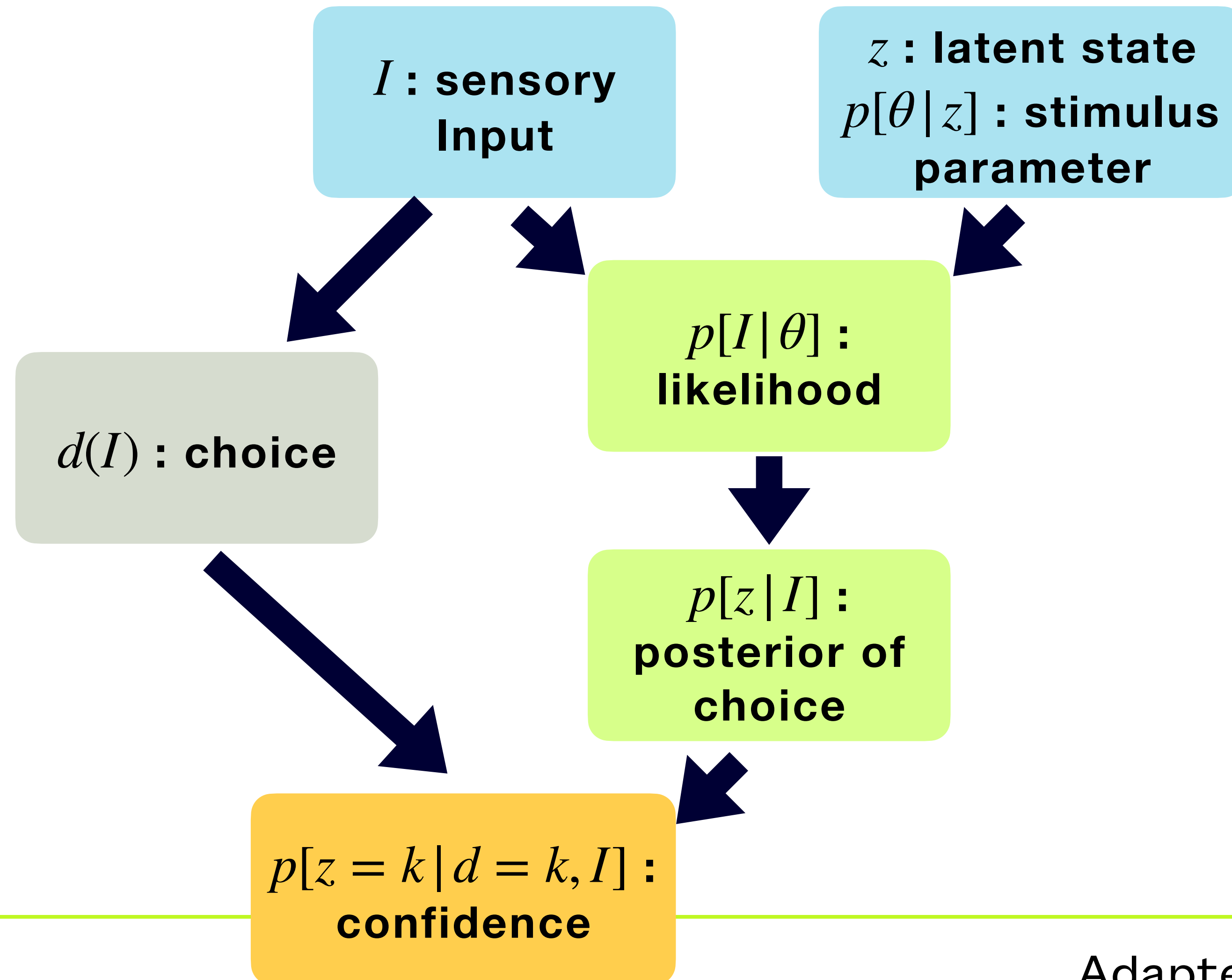
Measuring Confidence



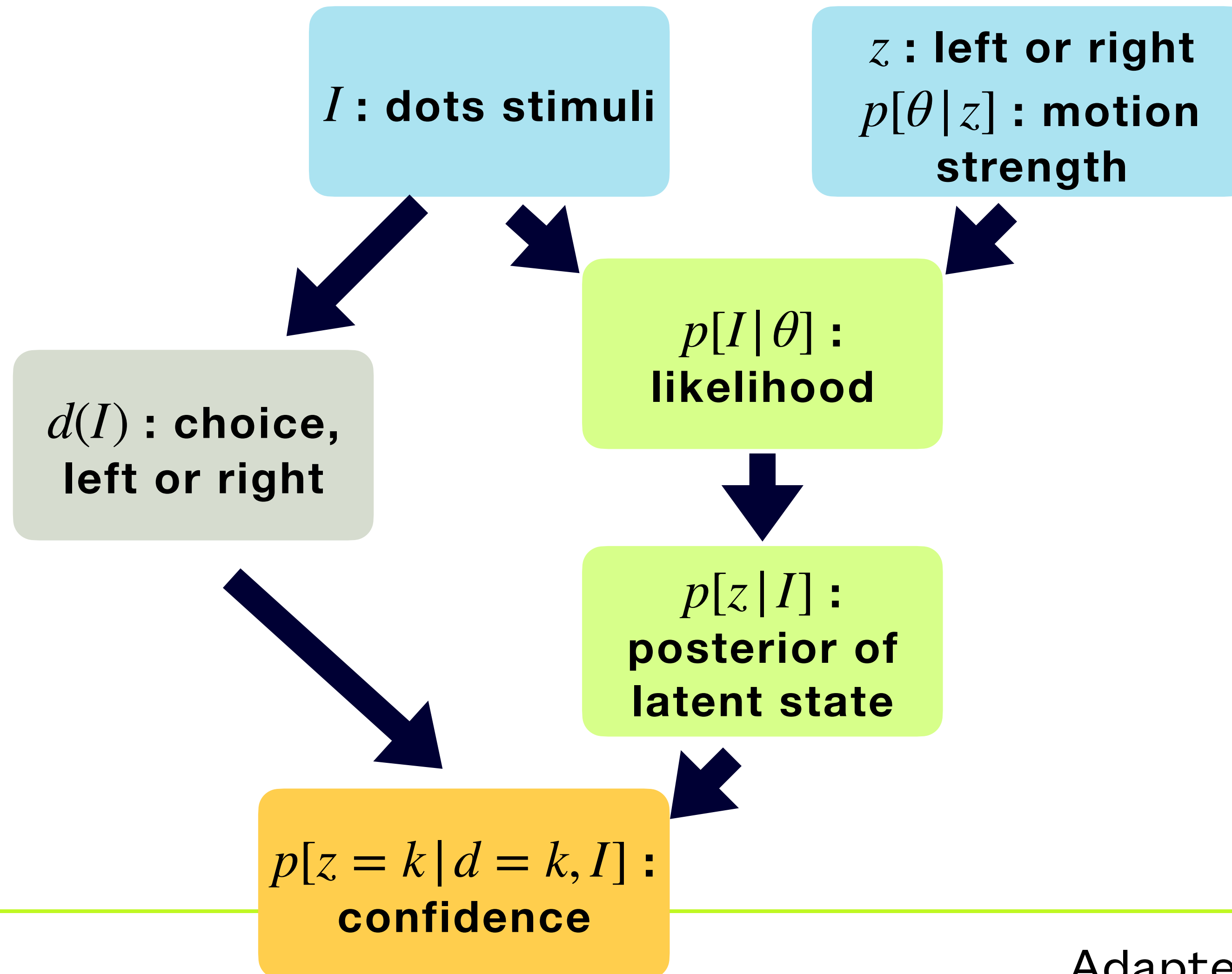
Confidence Represented by LIP



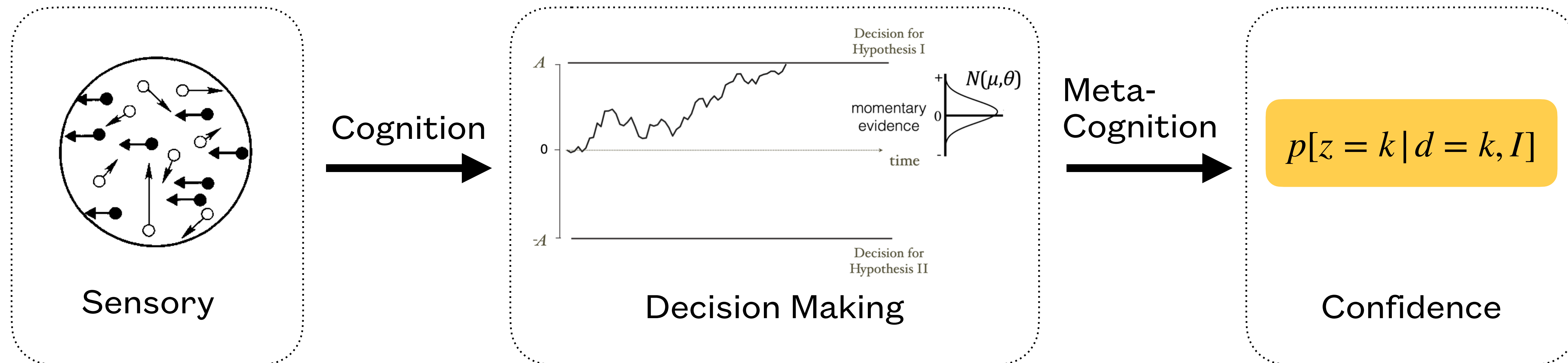
Computing Confidence



Computing Confidence



Meta-Cognition



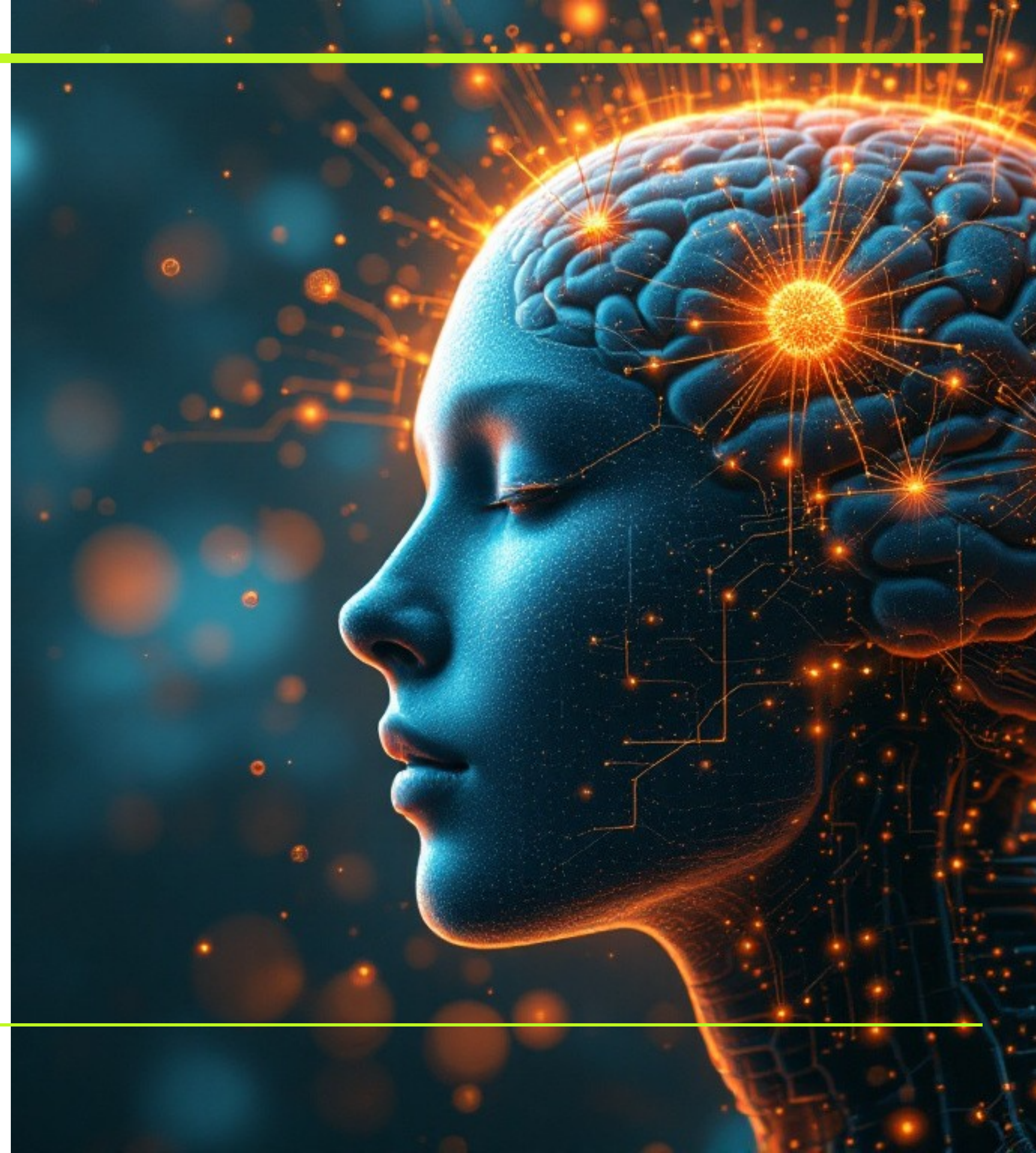
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❖ DDM

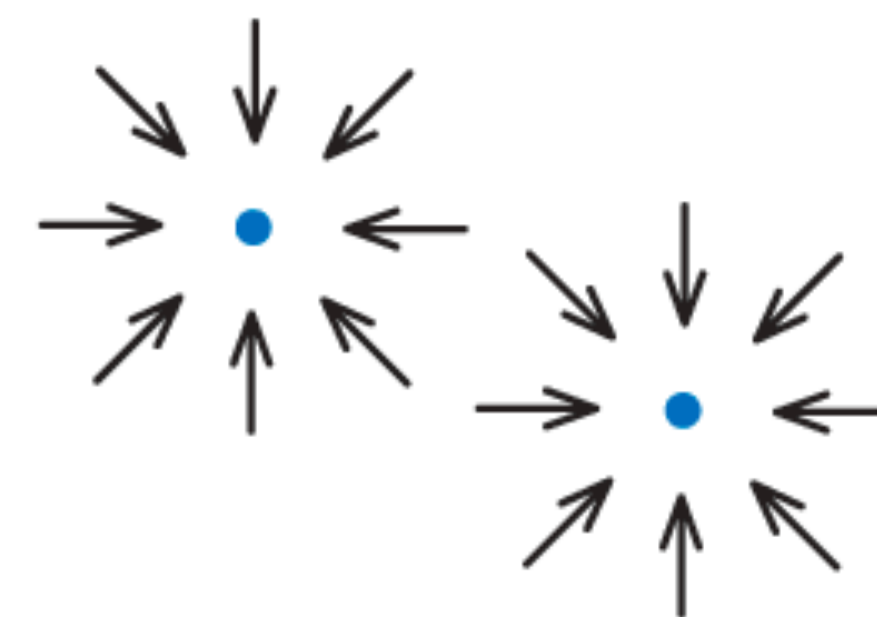
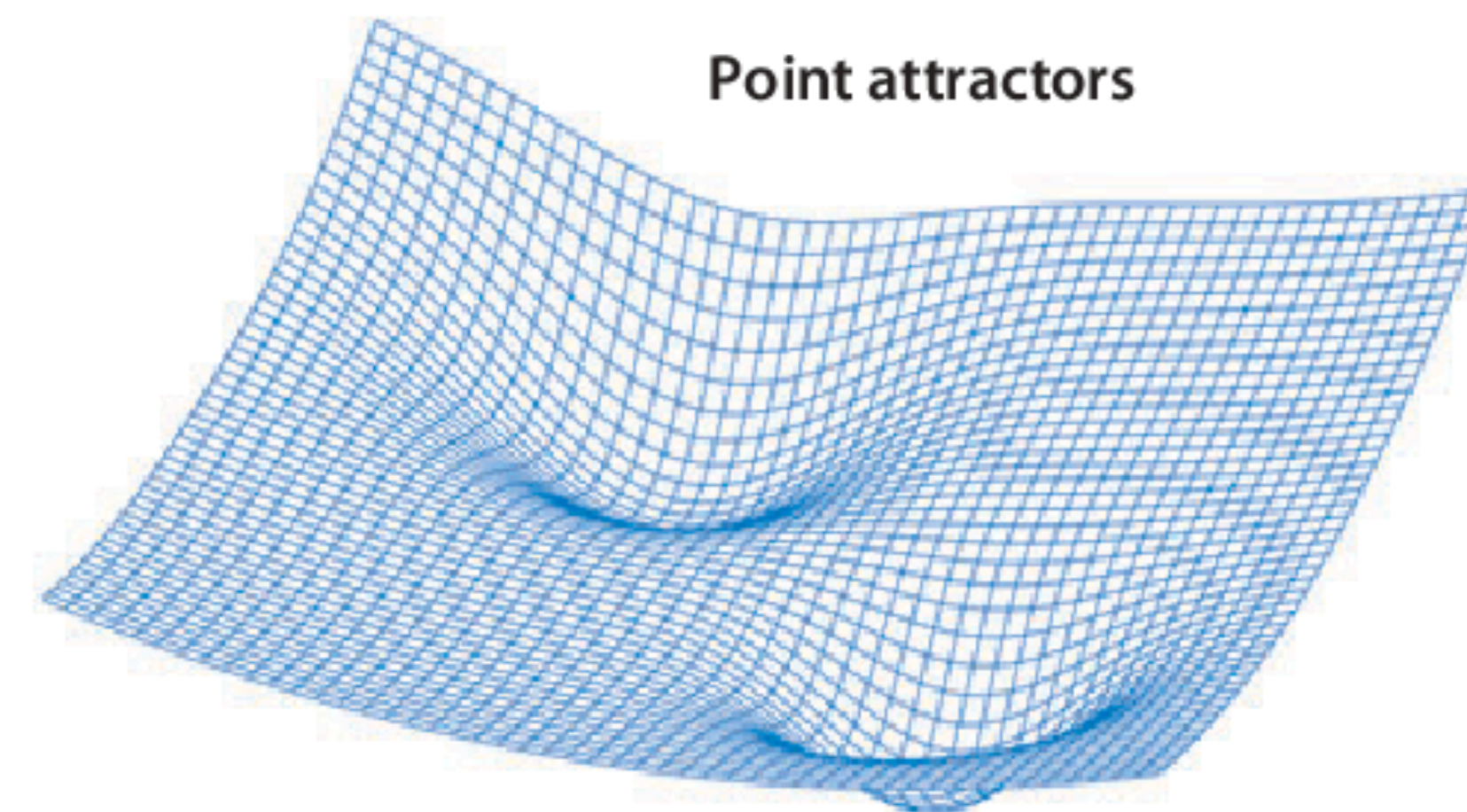
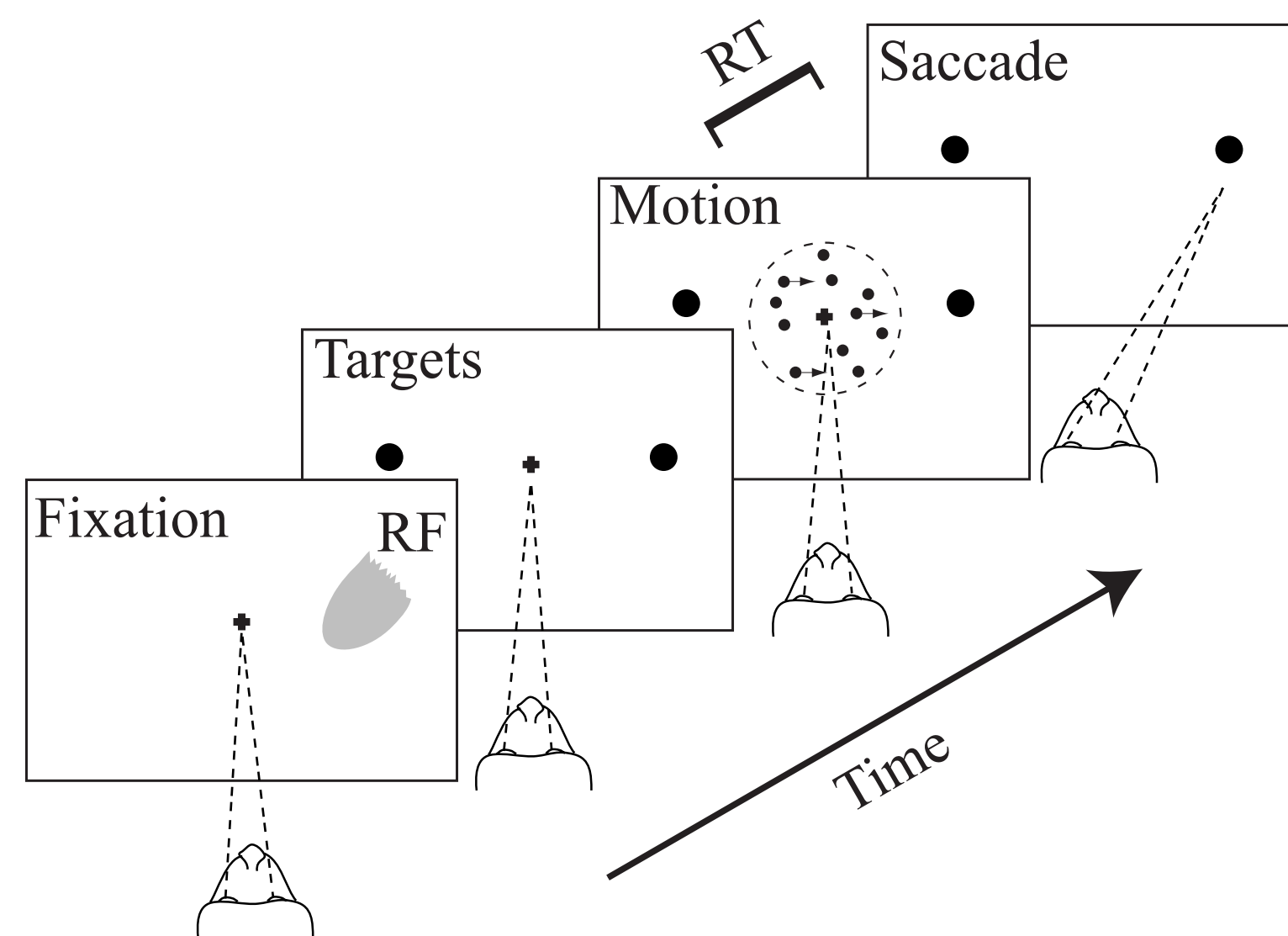
❖ SPRT

❖ Confidence

❖ **Attractor Network**

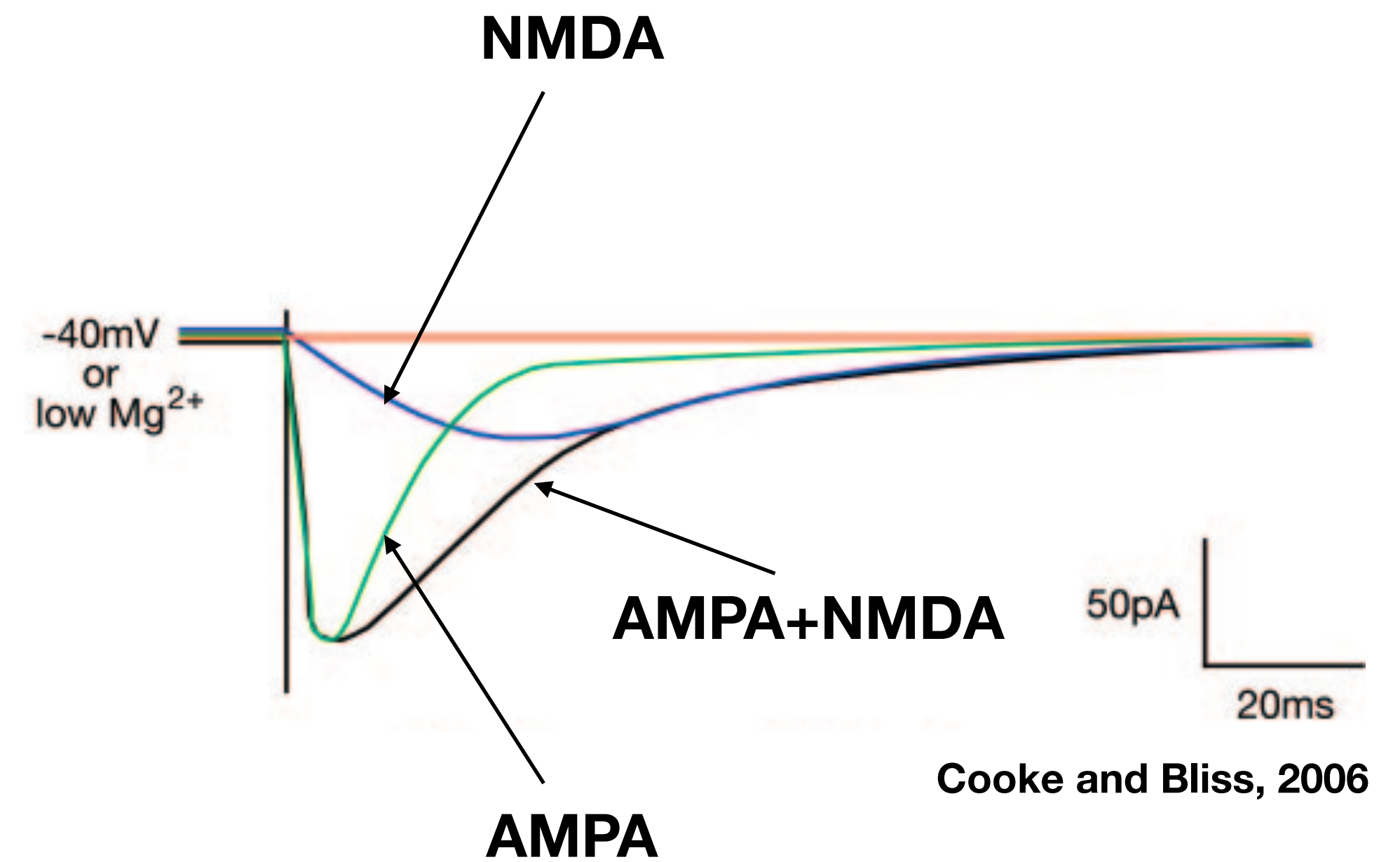
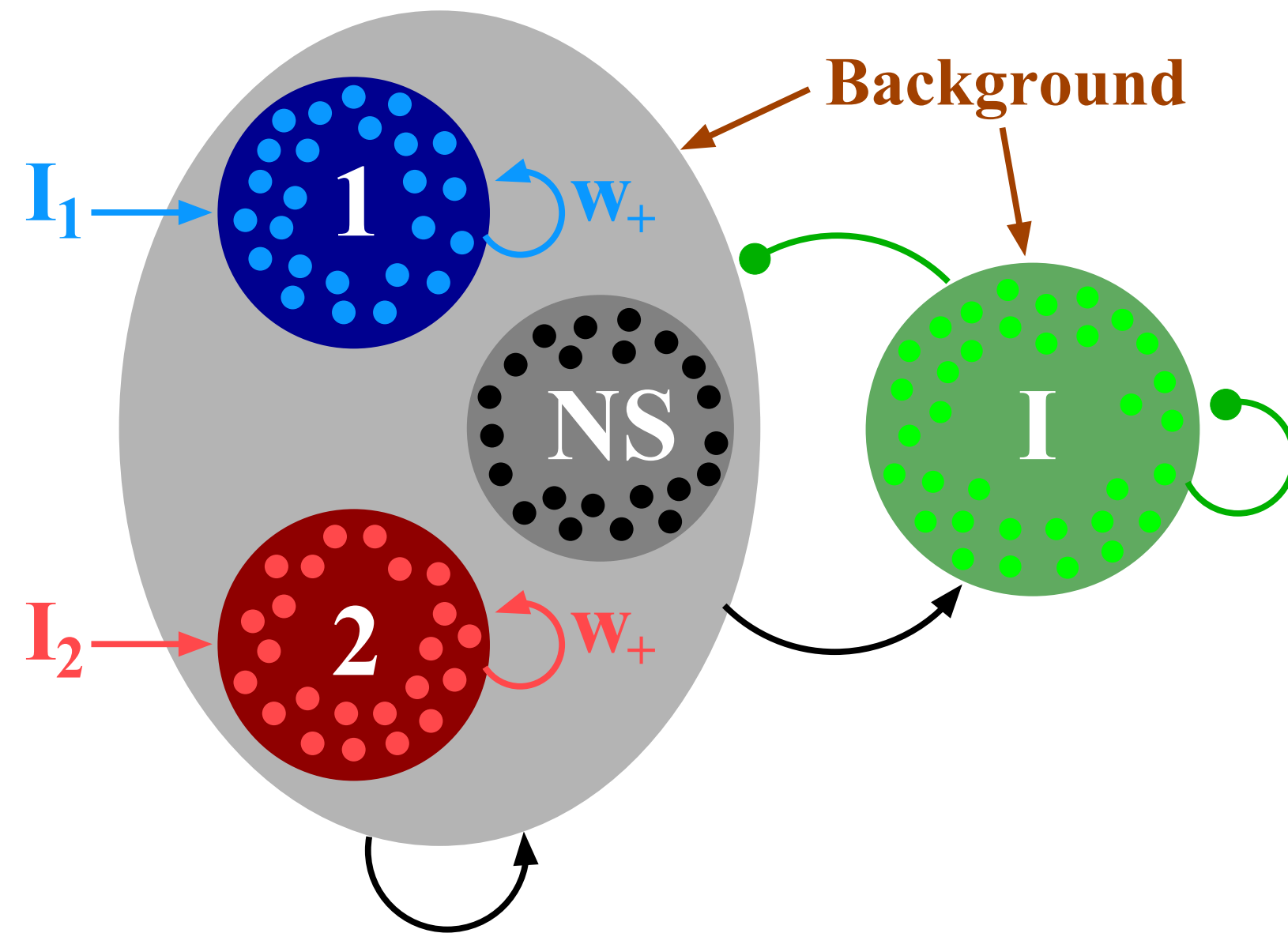


Attractor Networks

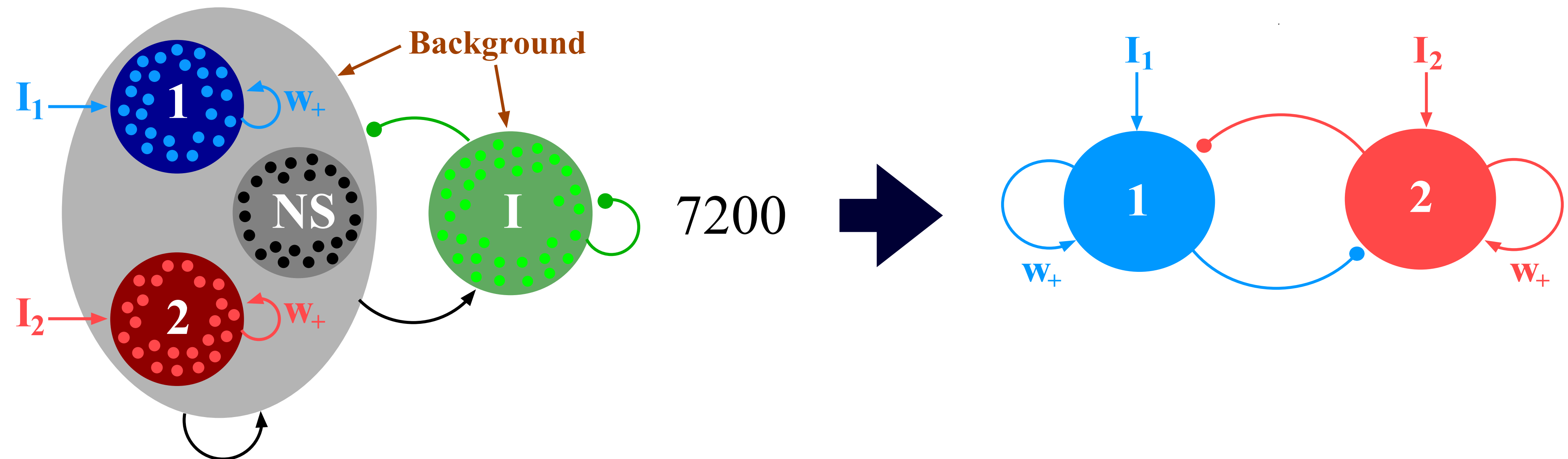


Network state space

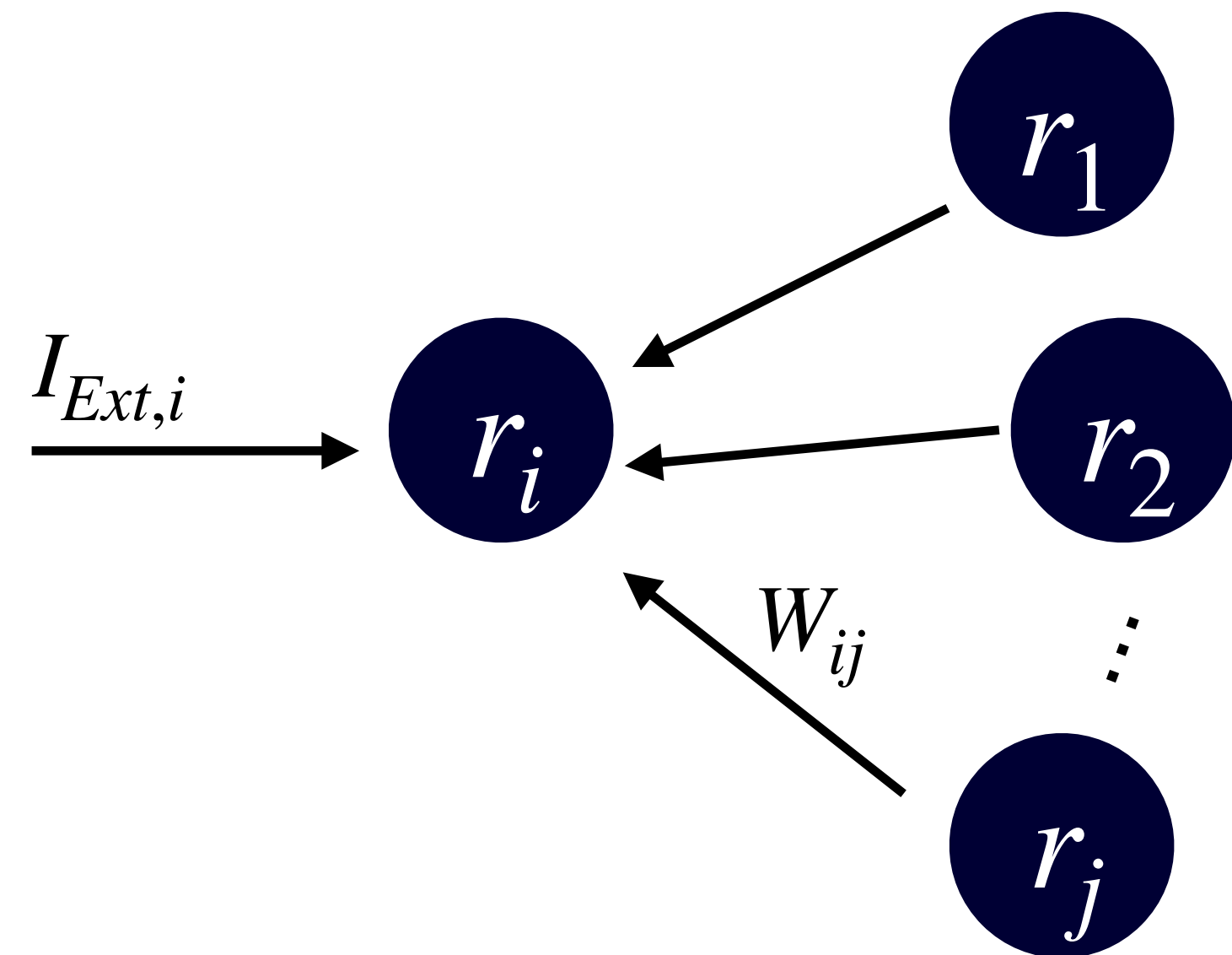
Attractor Neural Network



Simplified Model



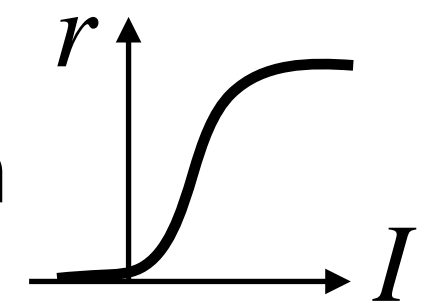
Rate Model of Neural Network Modeling



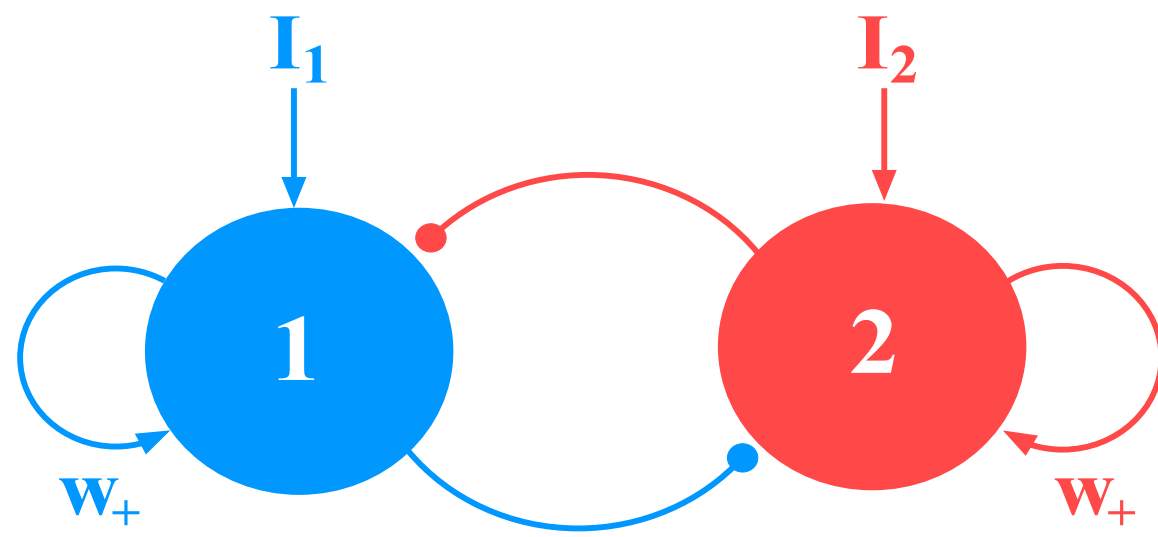
Time Constant

$$\tau_i \frac{dr_i}{dt} = -r_i + \Phi_i \left(\sum_j W_{ij} r_j + I_{Ext,i} \right)$$

Input-Output Transfer Function



Simplified Model



$$\frac{dS_{1,NMDA}}{dt} = -\frac{S_{1,NMDA}}{\tau_{NMDA}} + (1 - S_{1,NMDA})\gamma r_1$$

$$I_{syn,1} = J_{NMDA,11}S_{1,NMDA} - J_{NMDA,12}S_{2,NMDA} + J_{AMPA,11}r_1 - J_{AMPA,12}r_2 + I_0 + I_{input,1} + I_{noise,1}$$

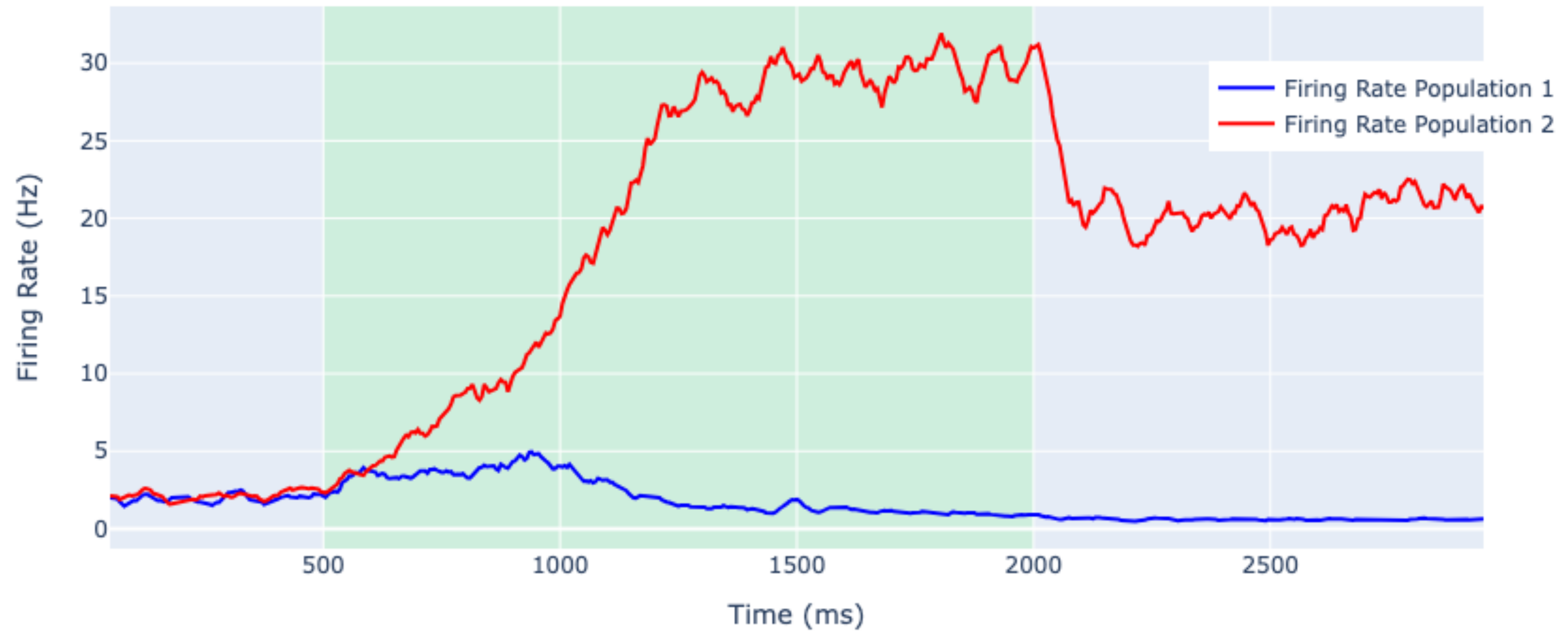
$$r_1 = \phi(I_{syn,1})$$

$$\frac{dS_{2,NMDA}}{dt} = -\frac{S_{2,NMDA}}{\tau_{NMDA}} + (1 - S_{2,NMDA})\gamma r_2$$

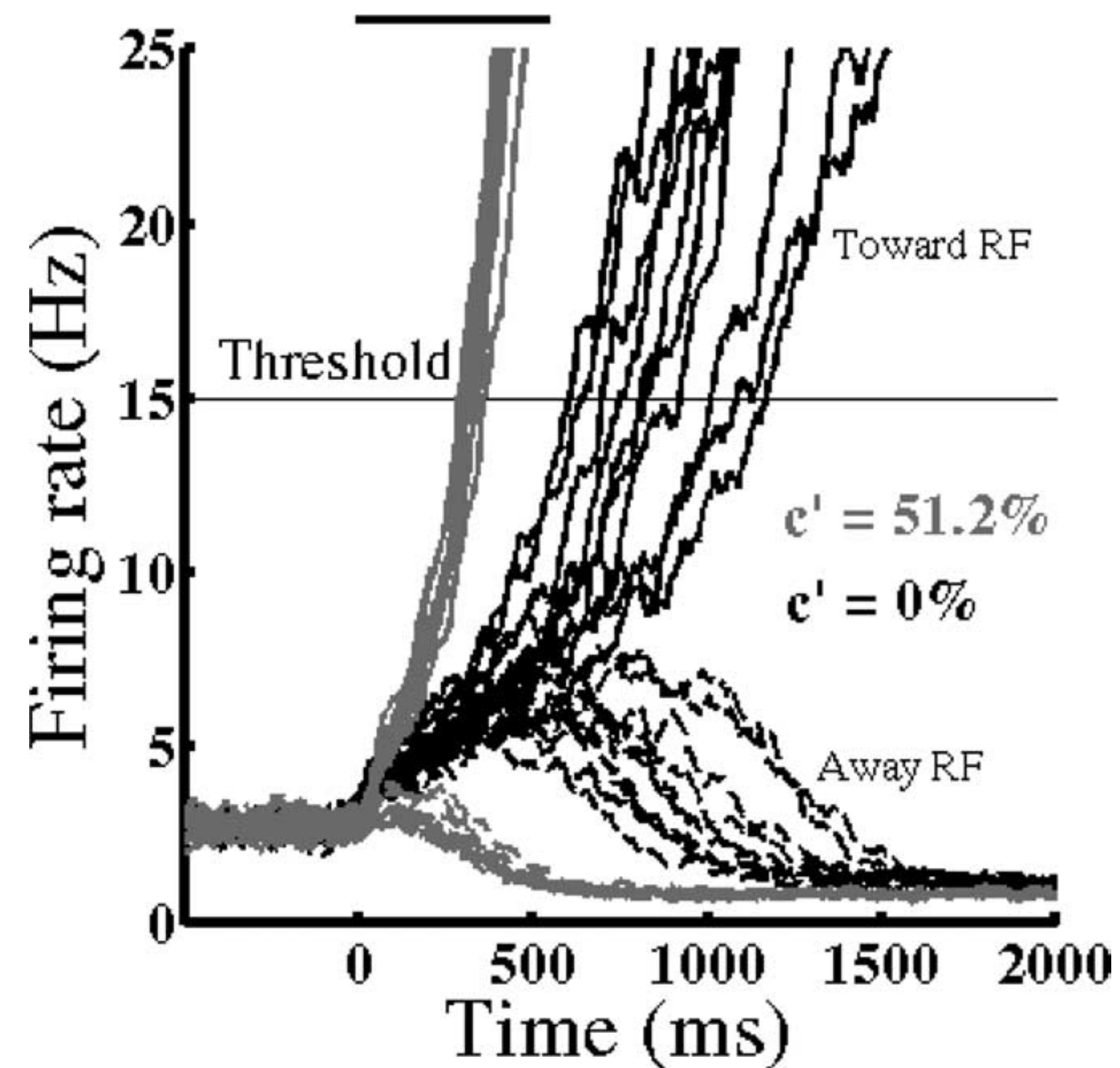
$$I_{syn,2} = J_{NMDA,22}S_{2,NMDA} - J_{NMDA,21}S_{1,NMDA} + J_{AMPA,22}r_2 - J_{AMPA,21}r_1 + I_0 + I_{input,2} + I_{noise,2}$$

$$r_2 = \phi(I_{syn,2})$$

Simplified Model



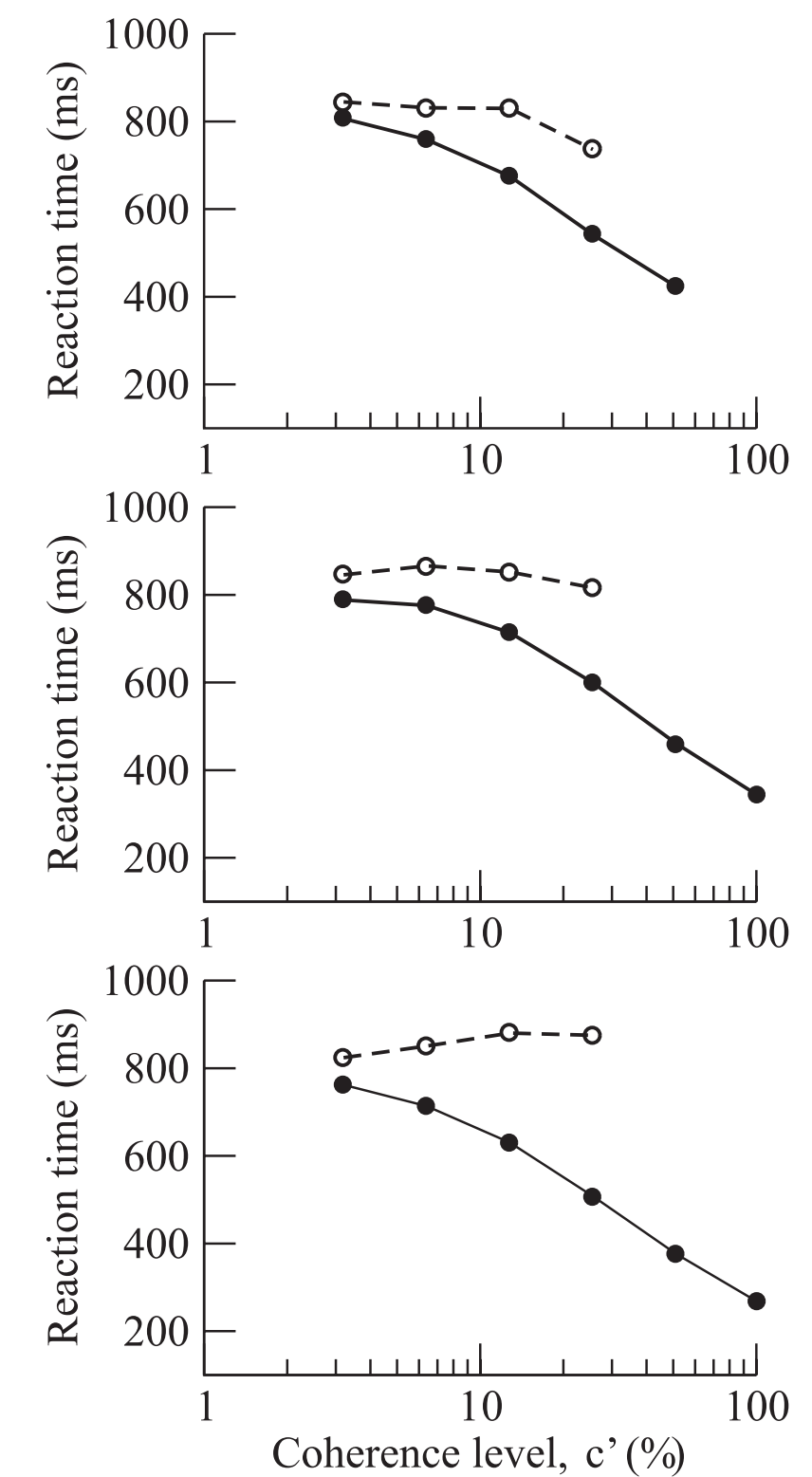
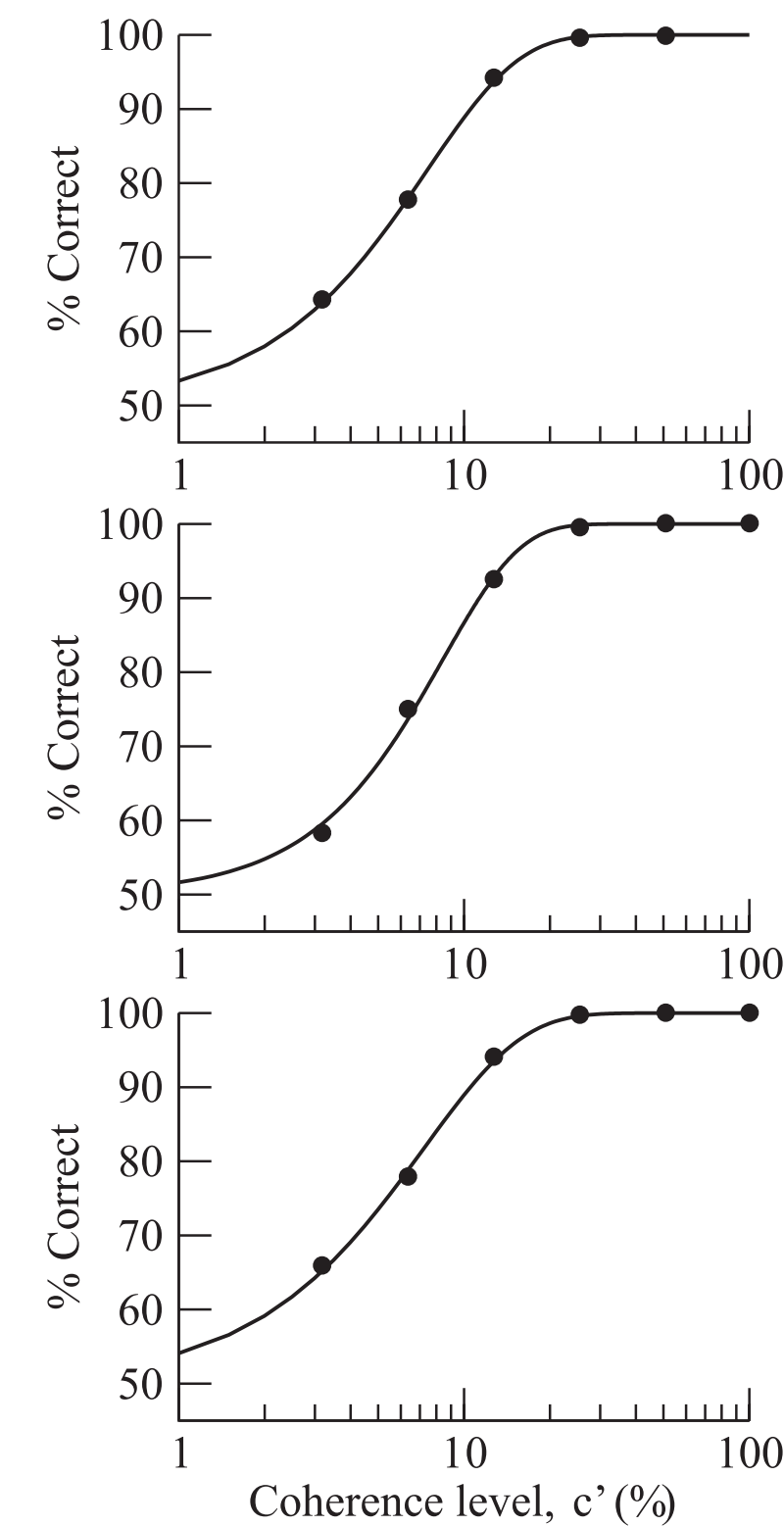
Model Performance



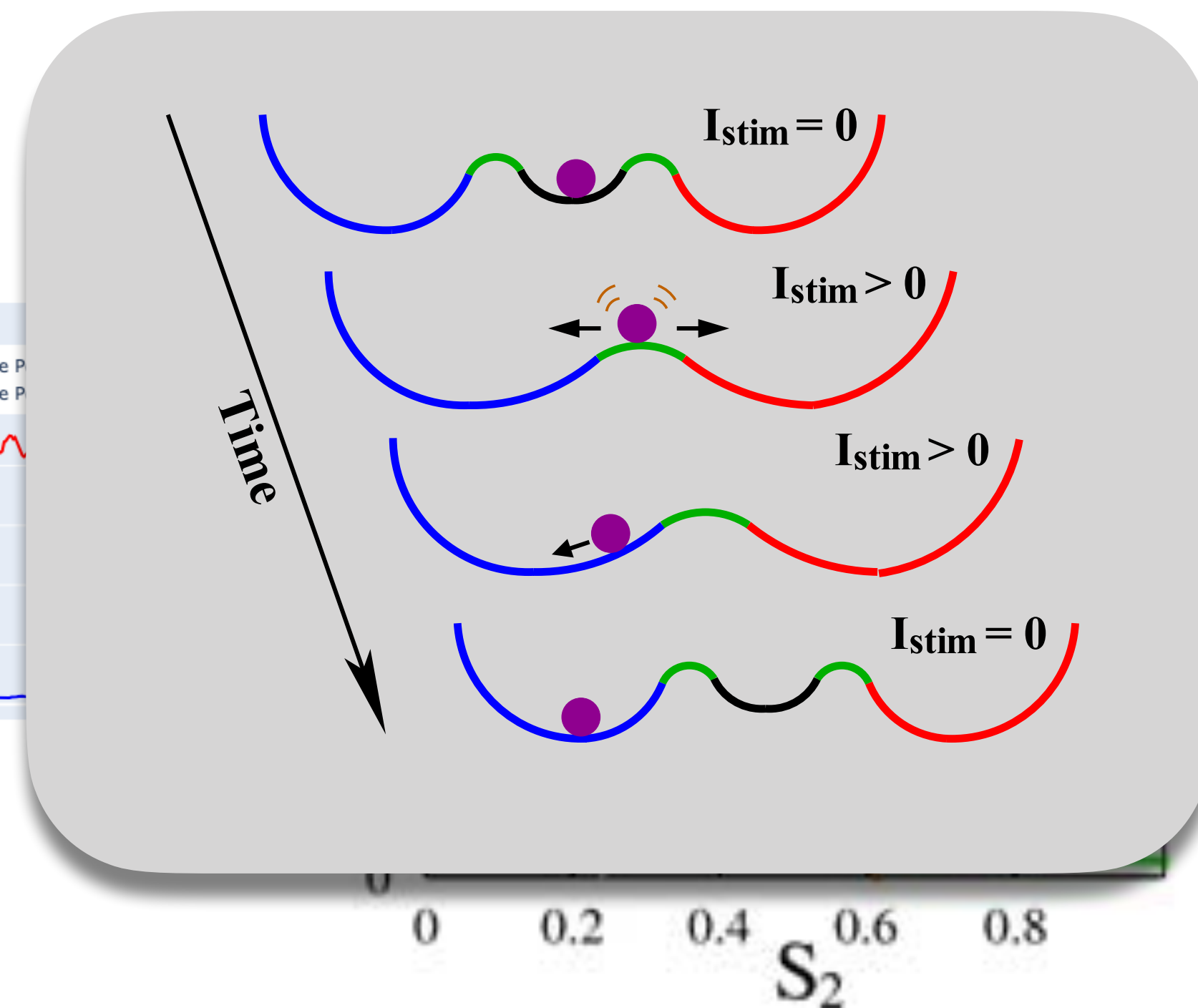
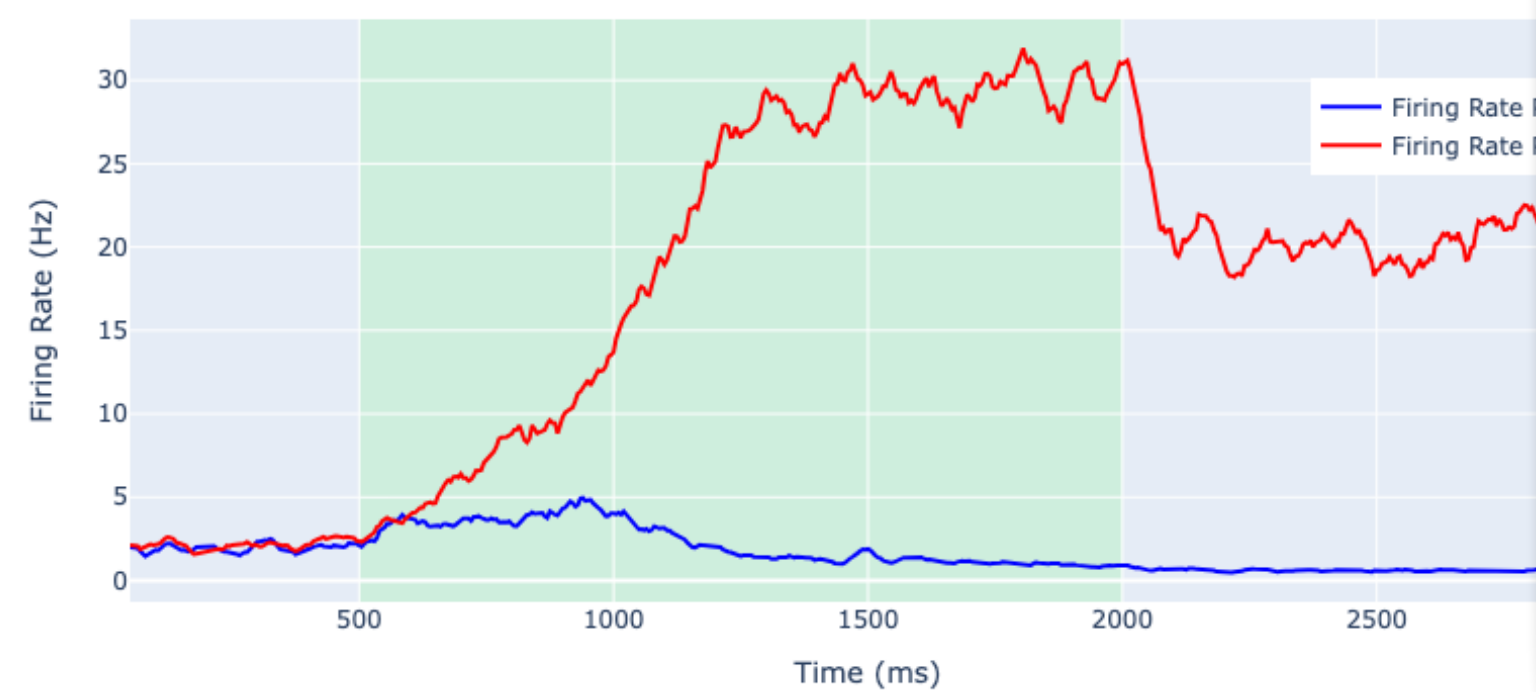
Experimental
data

Spiking
neuronal
network
model

Reduced
two-variable
model

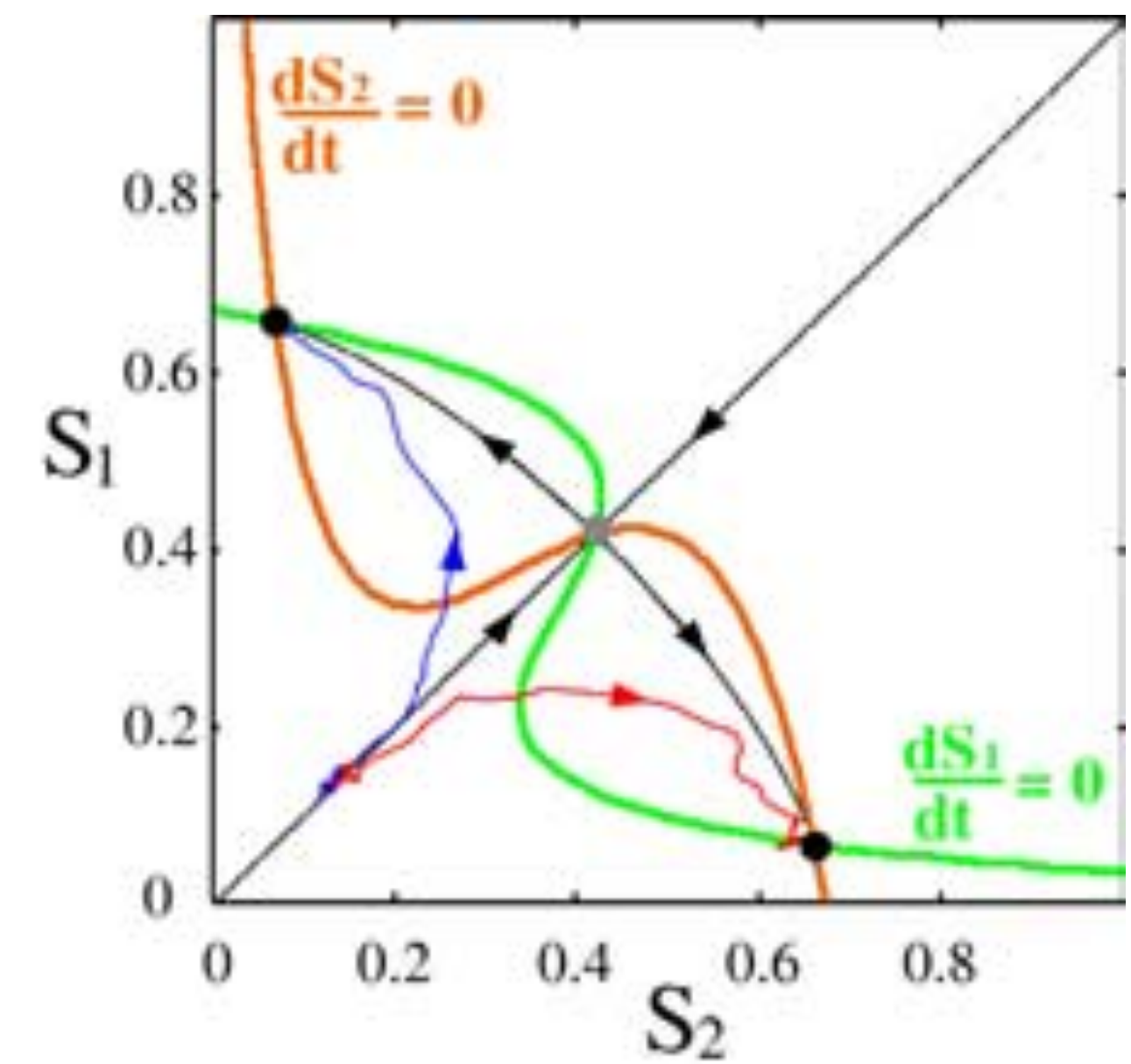


Phase Plane



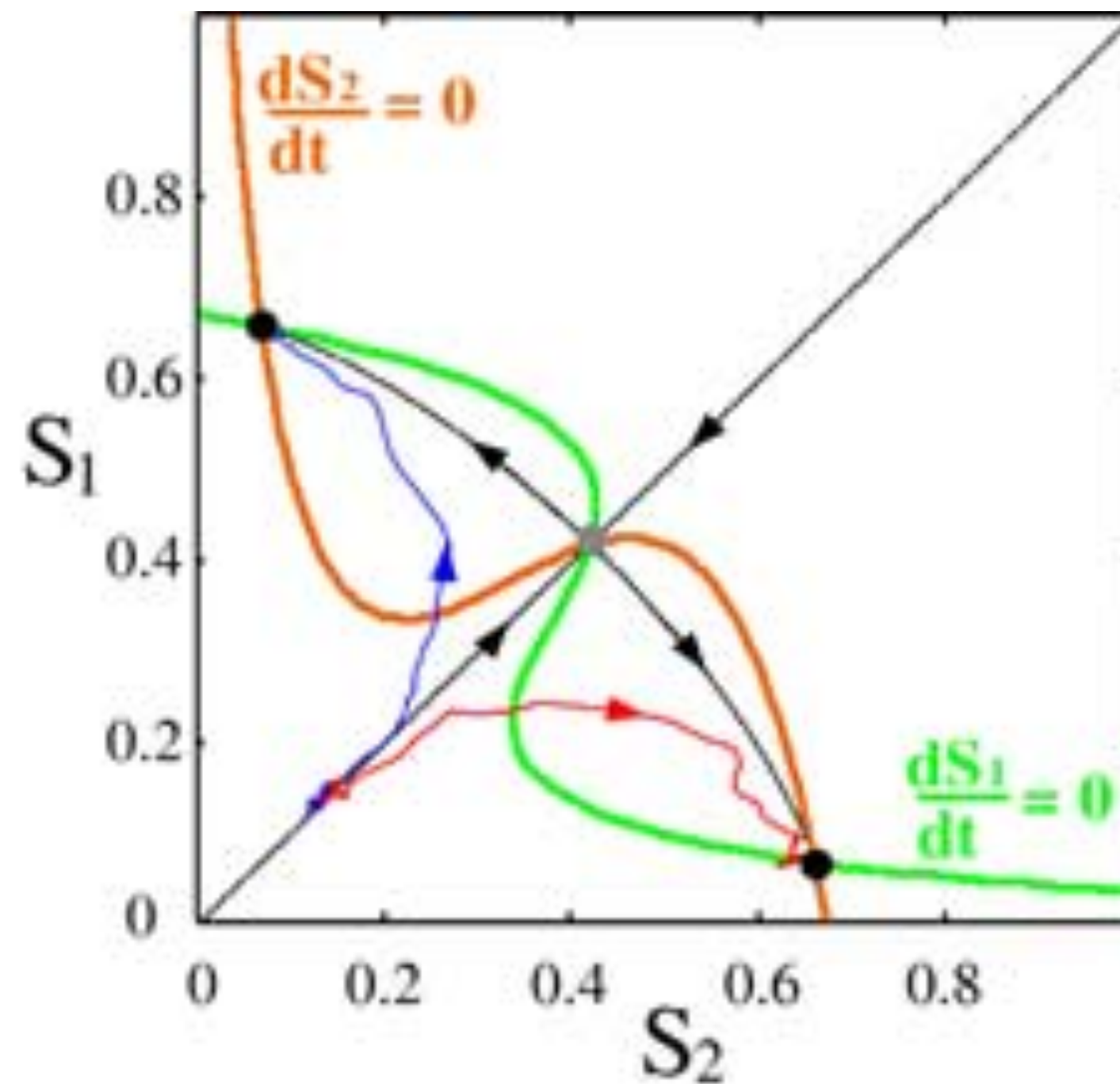
B

$c' = 0\%$

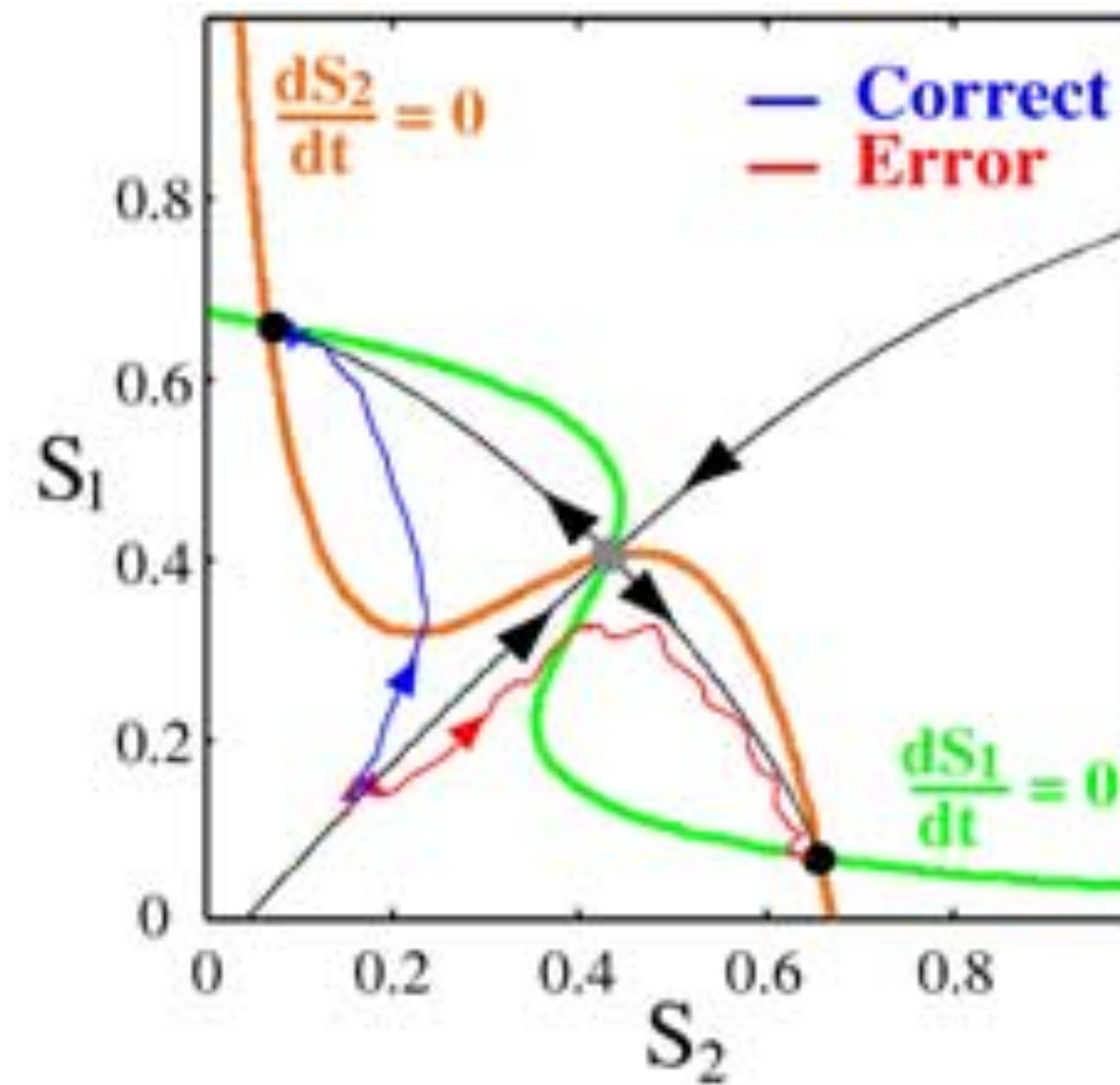


Biased inputs

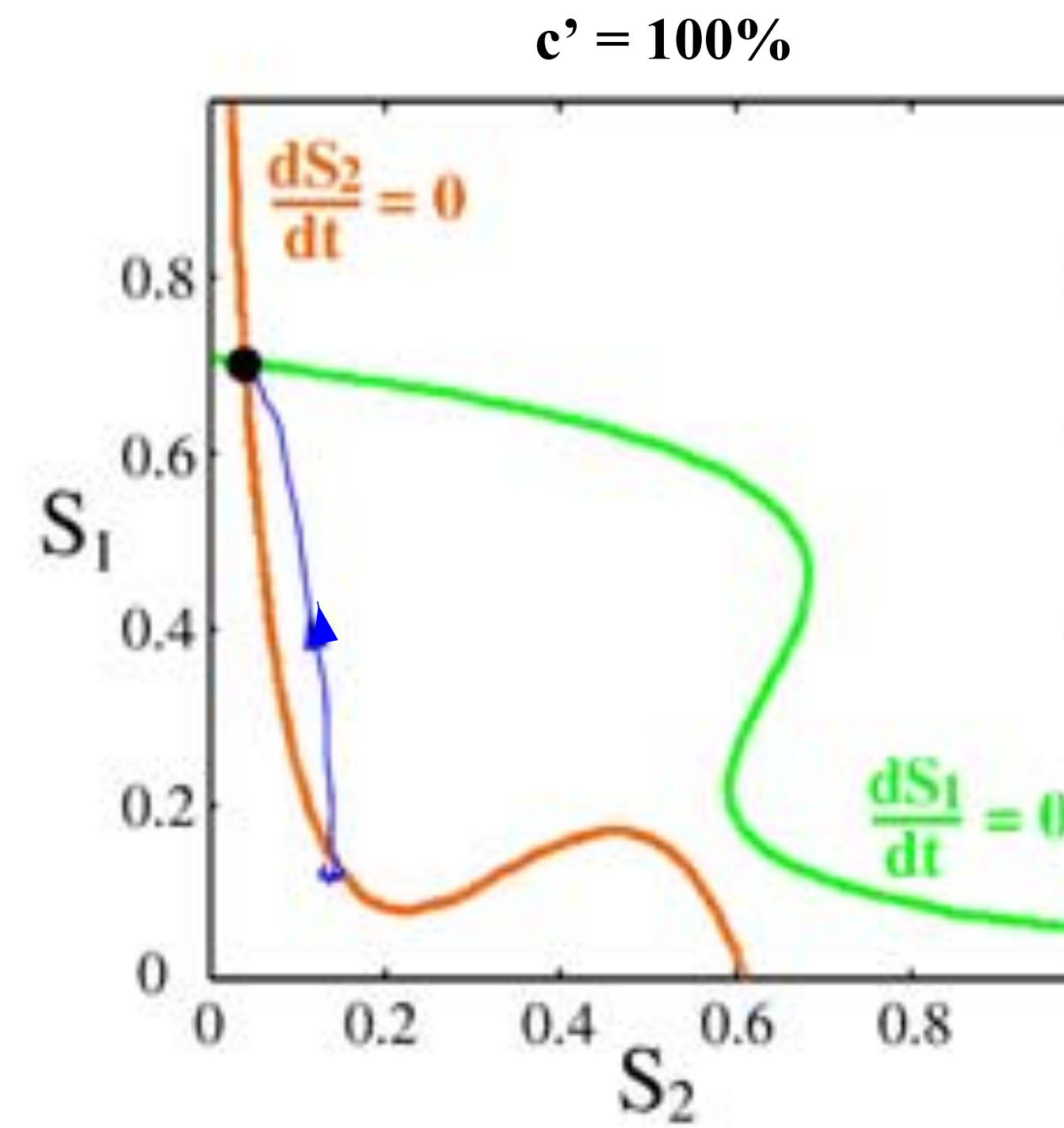
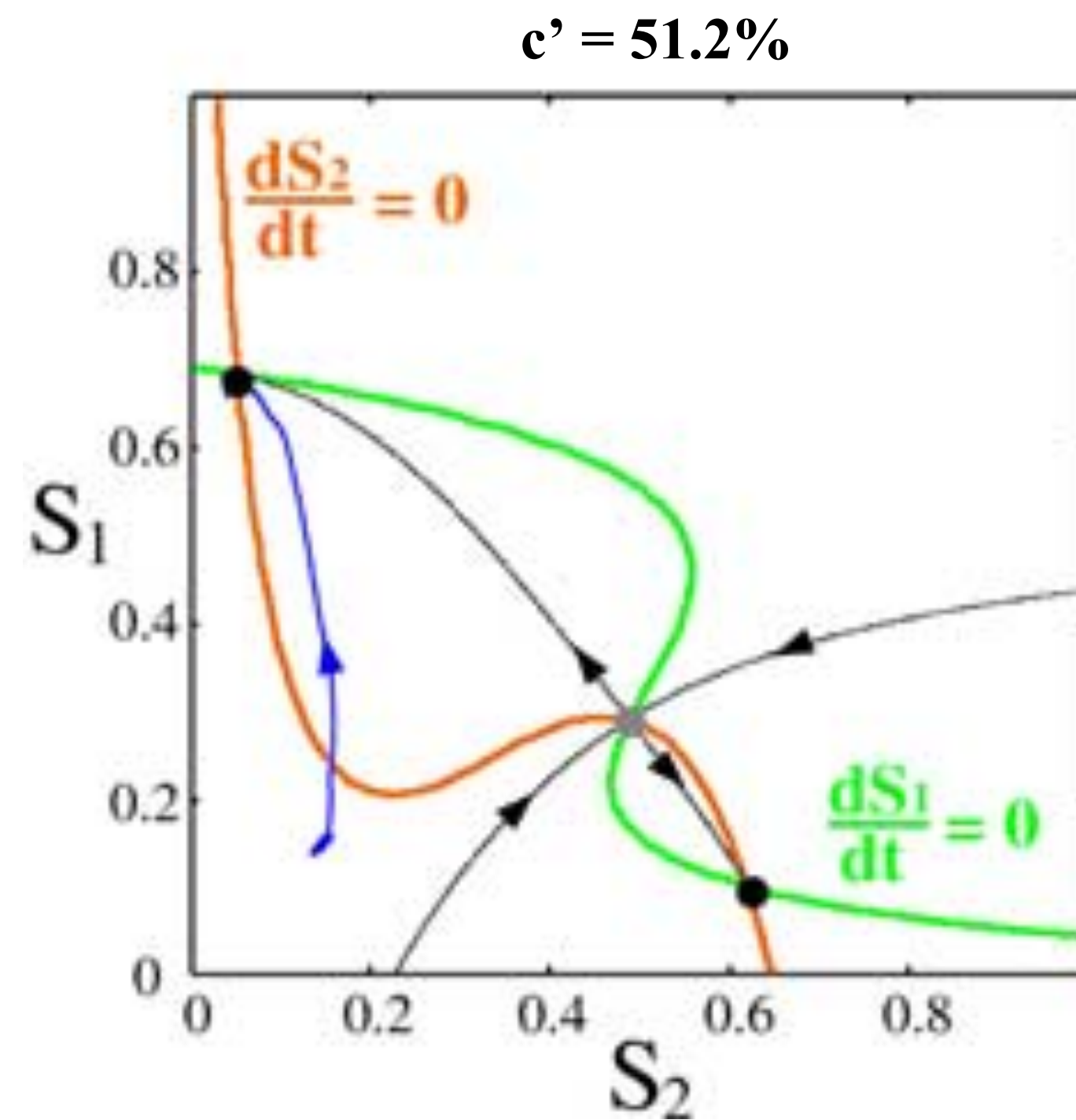
$c' = 0\%$



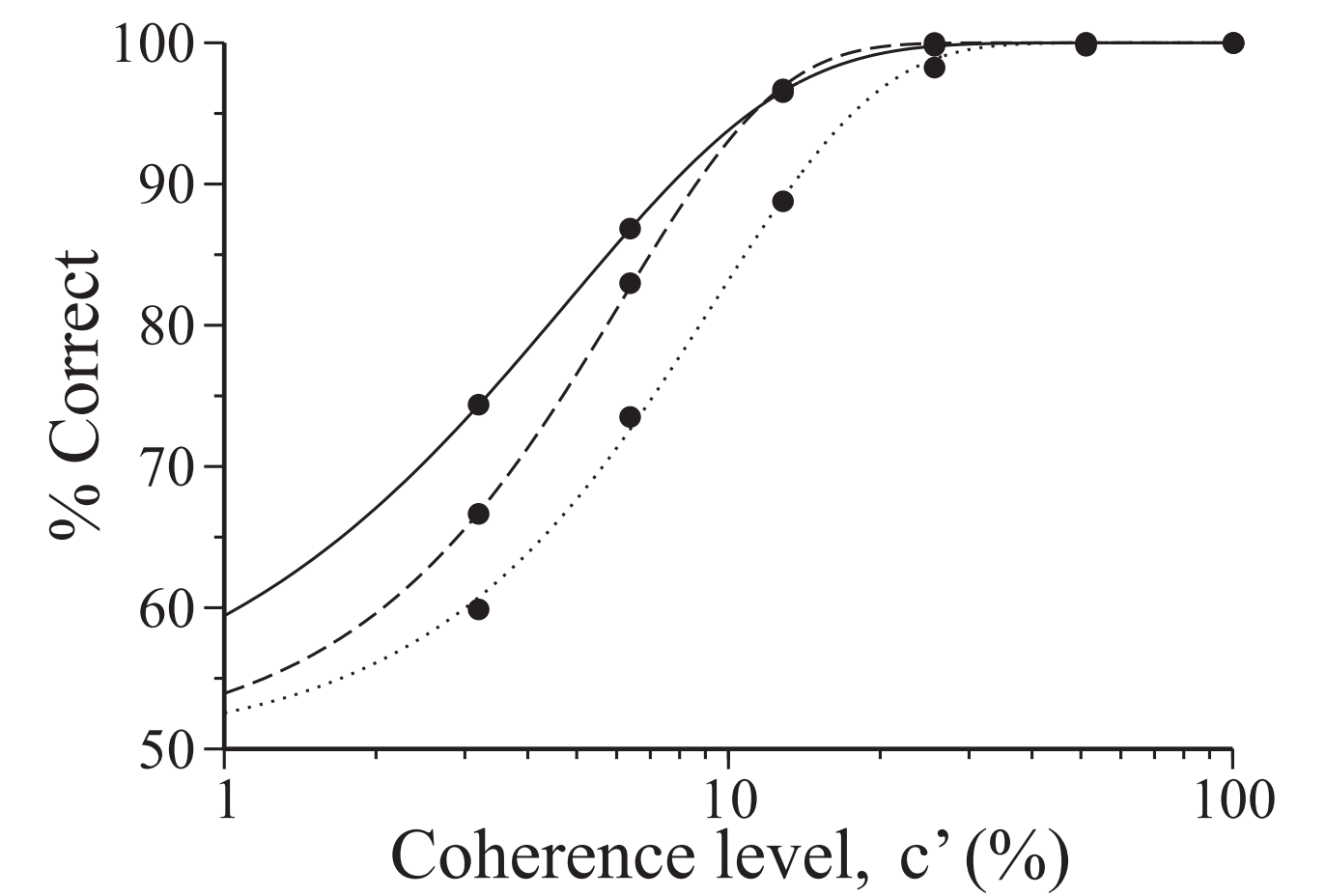
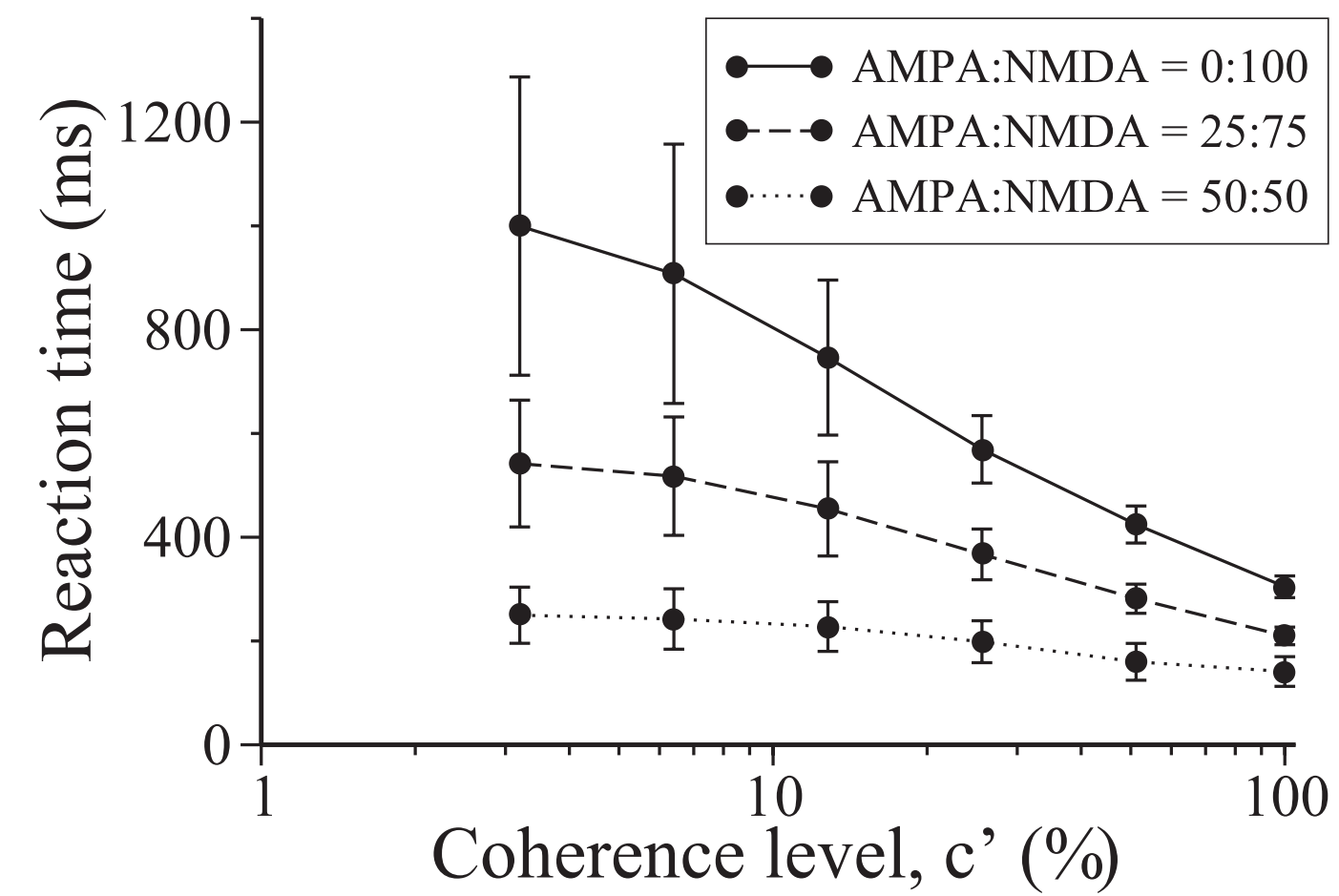
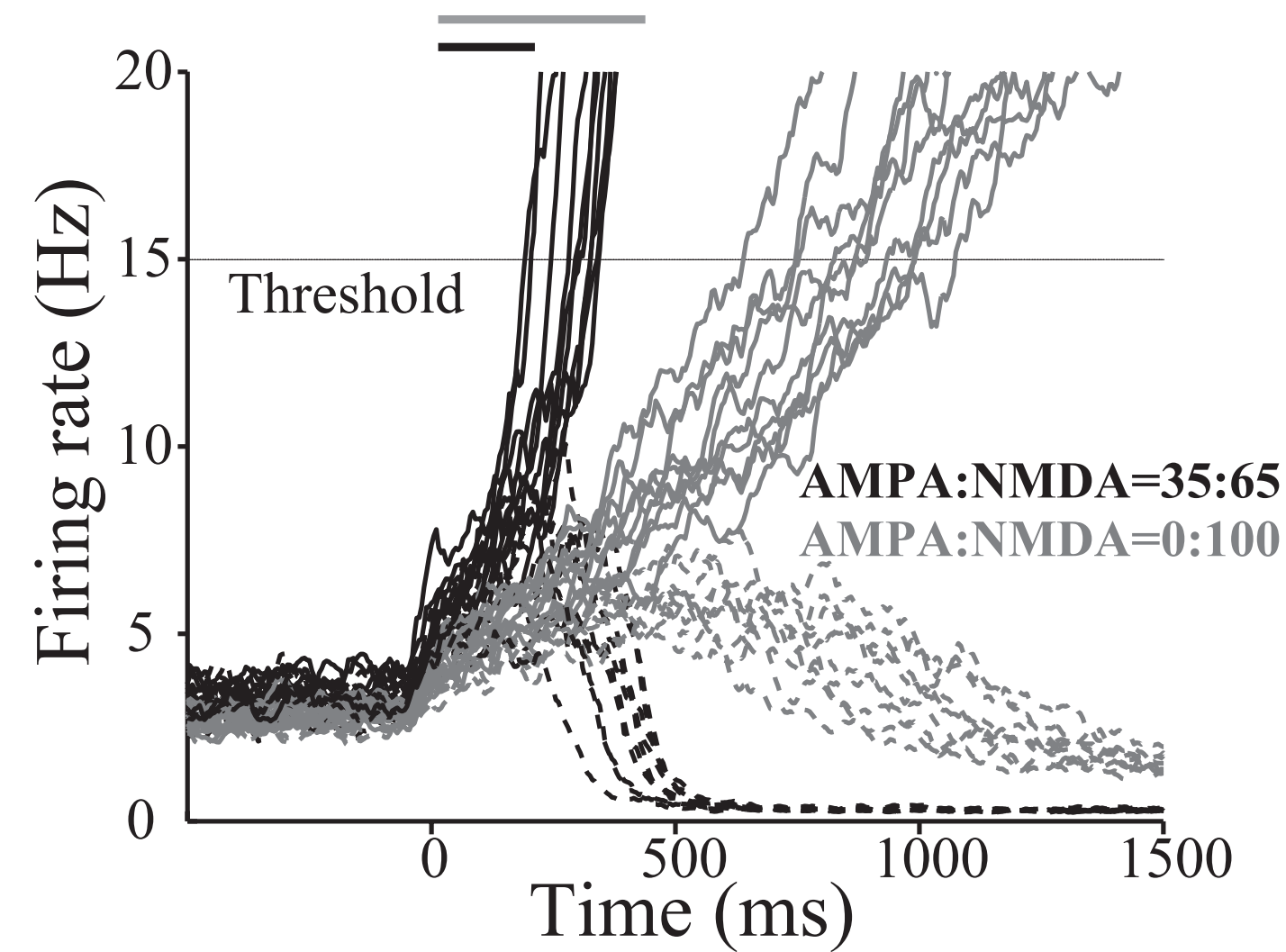
$c' = 6.4\%$



Extreme biases



NMDA is important



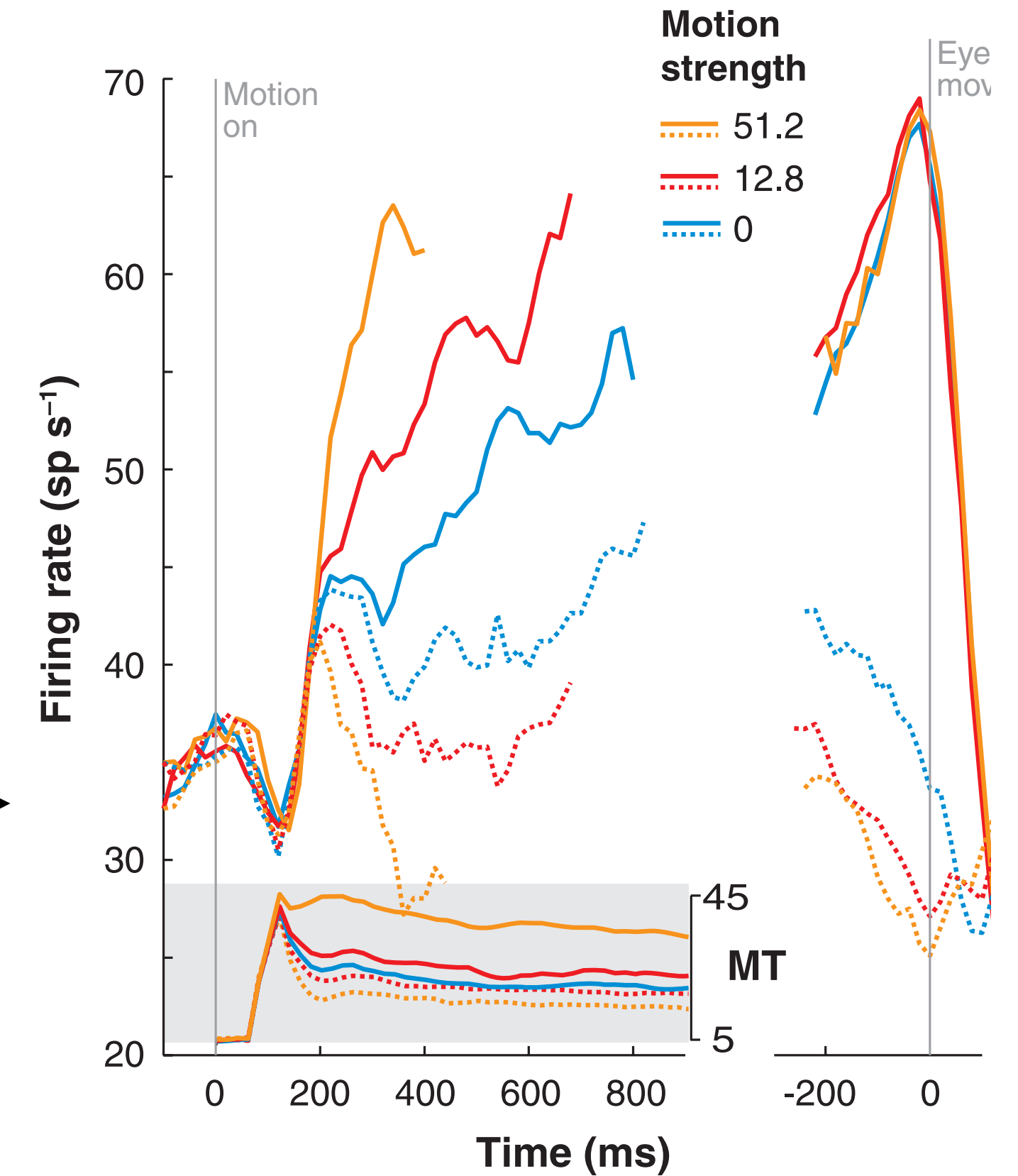
Summary

- DDM
 - RDM, LIP, Ramping
- SPRT
 - LogLR, Probabilistic reasoning
- Confidence
 - Definition, meta-cognition
- Attractor Network
 - Neural network modeling, Phase plane



Homework

- Understand and play with the two simulation codes I send out
- DDM
 - Implement a variation of DDM based on this code (collapsing bound, leaky integration...), and compare its behavior against the standard DDM
- Attractor Network
 - Modify the code to plot the simulation for the figure on the right →
 - Modify the code to add a decision mechanism (ref. Wong and Wang, 2006). Plot a psychometric and a chronometric curve.



Project Idea

- Design a decision-making experiment, measure the accuracy and reaction time, determine whether DDM is a good model for this experiment.